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BOOK II.

THE OPERATIONS OF THE CENSUS.

CHAPTER XI.
THE CENSUS OF THE PROVINCE.

PART I.—PRELIMINARY OPERATIONS.

834. Territorial sub-division into Census units.—The unit adopted for Census purposes was the area of which the Census was to be taken by a single *enumerator*. This was called the *block*. The blocks were grouped into *circles*, each of which was under a *supervisor*, and the circles were again grouped into *charges*, with a *superintendent* over each. These charges were arranged by tahsíls, a gazetted officer being placed, wherever possible, in charge of each tahsíl. A map of the tahsíl showing the village boundaries was taken, and the population of each village according to the Census of 1868 marked upon it. With the help of this map it was easy to fix all blocks which consisted of integral villages, to decide roughly how many blocks each of the larger villages should be divided into, and to mark out the rural circles and charges. In fixing these, established boundaries, such as those of police and patwári's circles, zaildár's circuits, wards in towns and the like, were followed as far as possible. Patwáris' circles were found especially convenient, the patwári being often appointed supervisor of his circle.

The sub-division was strictly one of *area*; that is to say, wherever a village was divided into two or more blocks, the lands attached to it were also divided. Thus the collective blocks of a tahsíl included the whole of its area; and it was carefully explained to each enumerator that he was responsible for the enumeration, not only of persons present in the portion of the village site that was under his charge, but also of all such as might be present in the fields or other area attached to it. The directions laid down that in making this division, well-defined lines, such as roads, canals, streams, and the like, should be followed. This was one of the points to which too little attention was paid, the boundary not unfrequently consisting of an imaginary line drawn across the area, which had no tangible existence.

835. The index-register and circle-map.—As the blocks were roughly marked off, an index-register (Census Form A, General Letter A, Appendix D) was drawn up which served as an index to the Census sub-divisions of the tahsíl, showing the relation which they bore to the various villages, the number of houses, the population by last Census for each, the names of the Census staff, and so forth. Perhaps the most important columns in this form, and the ones which needed most insistance to ensure their being properly filled up, were columns 5 and 6 headed "Subsidiary parts." In these columns were shown all the minor hamlets and groups of houses or temporary huts which are so often scattered about the village area, and which, without some such record, were so likely to escape enumeration. The Deputy Commissioner of Ambála took upon himself to omit these columns as useless. But it seems to me that they are absolutely essential; and especially so in the Multán and Deraját divisions and the hill tracts, where tiny hamlets are dotted about all over the district. Some suggestions for facilitating the preparation of this form will be found in the section on Difficulties and Suggestions.

The register, when the rough entry of blocks was complete, was compared with the list of survey villages which is kept up in the tahsíls, to ensure that no portion of the area had escaped inclusion in some one block or other. Each supervisor was then furnished with a copy of so much of the register as referred to his circle, together with a rough sketch-map of the circle showing the village boundaries and the rough division into blocks; and he was then sent into his circle to conduct the house numbering, and to finally demarcate the blocks where these latter consisted of portions of villages or towns.

836. House-lists and block-maps.—The next step was to prepare a rough sketch-map of each block, showing its boundaries and the position of every house included within them. Together with this was prepared a house-list (Census Form B, Appendix D), the two standing to each other in the relation of the settlement field map and register. In this house-list was entered a detail, not only of the houses of the block, but also of the families dwelling in each house. This detail enabled the supervisor to complete the demarcation of blocks, finally fixing their limits in towns and in those villages which were too large for a single enumerator to deal with. Each house then had a separate number allotted to it, the numbering being consecutive within each block, and following the order in which it would be most convenient for the enumerator to visit the houses on the night of the Census. In Kúlu the houses were numbered downhill beginning from the highest point, to which the enumerator climbed up during the day, and so had no uphill work after he had once begun his enumeration. This was a capital device which should be followed in all hilly tracts.

It was laid down that each homestead (*ábádi*) should have a separate series of house numbers, except in the case of towns, in which each *muhallah* was to be numbered separately. These directions did not work well. The small scattered hamlets of which in some parts of the Province the village consists often include only four or five houses; while in many towns the *muhallahs* are not wards, but streets or quarters consisting of dwelling-houses as opposed to shops, and do not include the bazaars at all; and even where there are true *muhallahs*, they are often too large for the purpose, as only one numbering party can work at a time in the unit which is to bear a continuous series of numbers. The best plan would probably be to keep to one series of numbers throughout the whole of a village or Census circle, whichever might in each case be smaller.

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Part I.—Preliminary Operations.

The number allotted to each house was marked on it in a conspicuous manner, so as to be easily visible by the enumerator when he made his rounds on the Census nights. The method of affixing the numbers varied from place to place. Quite the worst plan I know of was to write the numbers on pieces of paper stuck up on the wall. In Talagang wooden pegs were driven into the wall, the ends white-washed, and the numbers painted on them. But unless the pegs were extraordinarily large the numbers would be too small to be conspicuous. The plan of marking the numbers on boards hung up on the houses is probably the best for towns; for though the boards cost something (less than one anna each in Rohtak) yet they can be kept up as permanent numbers, and can be shifted as new houses are built or old ones pulled down. For sheds and gipsies' huts boards must be used; but in villages the best plan is to paint a white square on the wall, near the door-post and under the shelter of the eaves, and to paint the number on it in red (*geru*), both the white and red paints being mixed with *linseed* oil boiled with a little resin (*rál*); as if they are mixed with water or if the oil is not boiled, the paint runs and the rain washes it off. The numbers should be so conspicuous that it would be difficult to pass the door without noticing them; and where Nágri-writing enumerators are employed they should be written in both Nágri and Persian.

837. In deference to the opinion of Mr. Elliott, who thought that the preparation of the map showing the houses was unnecessary, I eventually allowed these maps to be dispensed with where they had not already been made. But in most districts the Deputy Commissioners found them so useful and so easy to make that they did not avail themselves of this option. I do not think they are absolutely essential, if sufficient care is taken that no house be omitted from the house-list. But I believe them to be *most* useful not only as rendering such omission less likely, but still more as a diagramatic guide to what would otherwise be a bare list of houses, which, in towns, often lie one within one another in a confusion which is almost inextricable without some such aid. In parts of the Province also where the houses are much scattered, they are especially valuable. Mr. Anderson of Kúlu writes:—

"Sketch-maps giving the positions of houses and homesteads in the blocks were prepared in all cases. They occupied the pat-
"wáris and gumáshtas for some time at the commencement of operations, but the time was not thrown away and I did not avail my-
"self of the permission given to dispense with these maps. I can speak of their great use from my own experience in testing the
"work. In Kúlu where the houses are very scattered, or rather where there are so many houses (Dogri or Sharani) lying away from
"the homestead which is itself compact enough, it was of great benefit to have these maps, however rough, to refer to. The supervis-
"sing officials could detect an omission at once. Without these maps I should certainly not have been so sure of the great accuracy
"of the enumeration as I am."

Major Lawrence of Kángra writes to a similar effect; Mr. Dames of Dera Ismáil "found them of great "use;" Colonels Minchin and Hall, Commissioners of Hissár and Ráwalpindi, both of whom saw much of the work while in camp and took a most active interest in it, consider them *most useful*. Mr. Frizelle of Sháhpur says they "were in no case omitted, and added greatly to the certainty of the work;" while Mr Wilson of Sirsa says that both himself and the other officers engaged in the work "found them most "useful, and that they did not take long to make." On the other hand, Mr. Hawkins of Amritsar considered them unnecessary, and did not prepare them; Mr. Steedman of Ráwalpindi made them, and found them "not very useful;" and Mr. Benton of Karnál says "I prepared them for the whole district. But I do not think them necessary and worth the trouble of making; though by means of them the house-numbering could be checked without going to the spot." They were certainly not intended to serve this last purpose. Again Mr. Stogdon of Lahore writes:—

"As a matter of fact I doubt whether the maps are of much practical utility. They are not required in villages where patwáris
"and lambardárs are well acquainted with every dwelling-house, and in the case of large towns like Lahore their preparation is a
"work which requires practical knowledge. To the best of my recollection some of the maps of the more tortuous and involved portions
"of Lahore were so badly made that they had to be rejected. Many houses were left out altogether. On the other hand, a careful su-
"pervisor who knows the locality cannot leave out many houses. If he omits a few in the first instance he is obliged to become so
"thoroughly acquainted with the circle during the course of the preliminary enumeration that he must find them out eventually.

"With regard to future operations I think it would be a good thing to prepare sketch-maps of towns showing all the
"houses and their numbers. Such maps could easily be prepared by municipal overseers, and they should be from time to time re-
"vised and brought up to date. Numbers should also be affixed to the houses, and the proprietors or tenants be held responsible for
"supplying them if necessary."

It must be remembered that what is wanted is, not a map to scale, but the merest diagram of relative position. Thus I found that some of these maps were made on enormous sheets of paper in order to show the outer boundaries of the block area. This was quite unnecessary. The limiting lines of the block external to the homestead should either be represented by lines marked with the name of the road or other feature for which they stand without regard to scale, or, perhaps a better plan, be separately shown as a small sketch-map drawn in the corner of the house-map of the homestead.

838. Definition of a house.—This was a much disputed point; and I think that the definition finally adopted caused a great deal of difficulty and confusion, and led to the inclusion of some very meaningless figures in our Census tables. We find five units in existence in the Panjáb:—

(1) The *enclosure*, including one or more buildings communicating with the public road by a common door or gateway;

(2) The *building or edifice*, a structural unit which is generally easy of identification and can be shown on a map;

(3) The *quarters*, consisting of any such portion of an edifice as has a separate means of access from without. For instance, in towns the ground floor of a building is often a shop open to the street; the first floor occupied by prostitutes with a staircase of its own, the second floor occupied by artisans with another and separate street-door and staircase, perhaps leading into an entirely different street, and so forth;

(4) The *household*, consisting of several families nearly related to each other, and occupying either common or adjacent rooms or buildings, but not being in commensality with each other;

(5) The *family*, consisting of a commensal group, denoted by a separate hearth, and occupying rooms, a room, or a portion of a room.

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Part I.—Preliminary Operations.

With respect to these five units, of which the first three are structural and the two last social, two points have to be determined :—

(*a*) Which is the unit to which a separate number can most conveniently be attached for the purposes of enumeration, and as a guide to the enumerator :

(*b*) Which units should be separately recorded for statistical purposes.

It by no means follows that the answer to these two questions is one and the same.

The Committee appointed in 1877-78 to consider the arrangements for the late Census recommended that the enclosures should be numbered, and that within the enclosures, which they called houses, families should be enumerated separately. The Government of India at first decided not to recognize enclosures, "the experience of the last Census having shown that the use of the term was either confusing or "impracticable," and adopted the house at its unit, defining it sometimes as "indicating the homestead "where members of a family live in commensality, and as comprising also the resident servants of such "family," and sometimes as "the several dwelling-rooms which go to make up the homestead, whether such "rooms are occupied by one or more joint families." But in its final instructions it abandoned commensality as a test, and decided that "house" should simply mean "a building to which a separate number has been "affixed," suggesting at the same time that each "dwelling-place of one or more families with their "servants, having a separate principal entrance from the public way," should be separately numbered. This was practically equivalent to adopting the "enclosure" as its unit, but retaining the term "house" by which to describe it ; and much confusion resulted.

In the Panjáb the staff were told to number as a separate house (translated by *áháta*, which really means enclosure) "any group of dwelling-houses, whether one only or many, which have a common "separate principal entrance from the road or street, however many families it may include ;" while the family (translated by *ghar*) was defined as "including all the people, whether one only or many, who eat "together at one common table (*chúlha*), together with such of their servants as reside with them and any "visitors who may be staying with them." At the same time it was pointed out to the superior staff that, while the outer door leading into the street would ordinarily be the sign of a house, which would include all the dwellings to which that door might lead, yet there were in many towns large caravanserais, courts, or collections of quarters entered by a common gateway, which it would be impossible to treat as single houses. The letter continued as follows[1] :—

"On the one hand the house, as now defined, no longer possesses any great statistical importance ; and is, in fact, little more "than a convenient unit to serve as a basis of enumeration. On the other hand, just because it is that unit, and because the "enumerator will be told that the house is a building separately numbered, it is extremely important that each house so separately "numbered and recorded in the house-list should be very clearly demarcated. Thus, where the definition in the rule leaves it open "to doubt what should be taken to constitute a house, or where the letter of the definition would give an unmanageably large house, "the following are the chief considerations by which you must be guided.

(1) "That the house should be clearly defined by obvious existing boundaries, so that there may be no possibility "of mistake as to what it does or does not include ;

(2) "That it should not include an unwieldy number of families, as its doing so might lead to great confusion "or error in the actual enumeration."

It will be observed that these instructions drew no distinction between the edifice and the quarters ; in fact I was not then aware how often the same building has several street-doors, leading respectively to its separate floors or quarters.

839. The result was that the house in the Panjáb Census had a signification which was by no means constant. In the villages of the greater part of the Province, where the enclosure generally coincides with the dwelling-place of a household including not more than two or three families, this was the unit most often numbered as a separate house. In parts of the Multán division and in the hills, where the rule is that each family lives in an isolated building, this latter was what was numbered. In the towns the building or edifice was in some places adopted as the unit, in others the quarters, less frequently the enclosure. In all cases, the limits of size beyond which a single enclosure was broken up and its included buildings treated as separate houses, varied immensely from place to place. But I regard this result as at once unavoidable and unobjectionable, so far as the unit to be adopted for the guidance of the enumerator is concerned. In this connection the only question is one of practical convenience ; and the variety and complexity of the conditions met with is so infinite, especially in towns, that it is impossible to lay down any uniform rule which shall work well. If the staff once understand that the one object is to number as a separate house what will be most obvious and unmistakeable to the enumerator and most convenient as a unit for his enumeration, they will have little difficulty in dealing with any conditions they may meet with.

In fact, I believe the great fault in the Panjáb instructions to have been that they did not leave sufficient discretion in the matter. In those huge courts or quarters (*katra* or *kúchabandí*) often found in large towns, where scores or even hundreds of separate houses are surrounded by a common wall and have only a single gateway as the means of access, the mere necessity of the case compelled the supervisor to number each building as a separate house. But the use of the word *áháta* in the translated instructions, and the fact that the definition given in the rule made the common entrance the index of a house, led the staff to feel far too great hesitation in splitting up enclosures into several houses ; and I often found, more especially in towns, courts containing many detached dwelling-houses numbered as single houses ; while, in order to distinguish the several dwelling-houses, and especially in order to provide for the entry of such of them as might be temporarily vacant, and which would otherwise, not being "houses" and including no family, have found no place whatever in the house-list, a series of subordinate numbers had been affixed to the buildings within the so-called house, thus causing unnecessary complication and possibility of confusion and error. So in the villages, where the supervision was necessarily less intelligent, I often found that quarters inhabited by

[1] General Letter B, section 3, Appendix D.

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the whole menial population of the village had been treated as single houses, simply because they were separated from the village homestead by and included within walls with only one gateway.

840. Suggestions for the future.—If I had to frame rules again on the point, I should adopt the "enclosure or house" as my unit to be numbered, translate it by "*áháta, makán, yá ghar*" (taking *chúlha* as my translation of family[1]), and explain that what was wanted was, to ensure that the enumerator should easily find the number and should be certain what was covered by it, and that these were the only points to be considered. This plan would of course leave the staff at liberty to select in each case that one of the three structural units mentioned in section 838 which might seem most convenient. But this is exactly what is wanted. The other two of the five units, the household and the family, are obviously unsuitable; for, depending on social relations and not on physical structure, they cannot have numbers painted on them as such, though they may of course, and often do coincide with one or other of the structural units. And of the three structural units, the enclosure is the most convenient in the case of a family homestead, the building in the case of separate houses opening upon a common courtyard or of isolated houses, and the quarters with its separate street-door in the case of town houses let in flats. At any rate I see no third alternative but, either to allow the discretion I propose, or to adopt the old double classification of enclosure and house which is said not to have worked well on former occasions. I think that something approaching to this double classification might still be usefully employed in the case of those small courtyards often found in towns, where the separate buildings included in them are let on hire to tenants who are constantly changing, and where, while it may be convenient to number the court as a single house, the entry of families in the house-list is but an unsatisfactory guide to the details. Here I would use subordinate or *zimmani* numbers; and, the court being house No. 154, would call the included buildings $\frac{154}{1}$ $\frac{154}{2}$ $\frac{154}{3}$ and so forth. But to prescribe a similar complication in all, because it happens to be the simplest alternative in a few cases, would I think be a mistake.

841. Occupied and unoccupied houses.—There is another aspect of the house question which gave rise to more difficulty than I could have supposed possible; and that is, the question of the degree of human occupation or of the probability of human occupation on the night of the Census, which should qualify a building to rank as a house. The instructions were that "buildings used for human occupation, whether by day only or by both day and night, and even though they be but huts or sheds," and temporarily vacant dwelling-houses, should be reckoned as houses; but not "ruined buildings, cattle pens, stalls, outhouses, godowns, and other buildings not occupied by man." But in many parts of the Province the buildings occupied by the family and by the cattle respectively are very similar in style; and if guests arrive the cattle are turned out of one and the guests take their place, while somebody always sleeps with the cattle in their shed as a precaution against thieves. Again, small sheds are scattered all over the fields and at the wells, which serve as a protection against the sun in the daytime, and are almost always occupied at night by the men who watch the crops to guard them against wild animals. Both these classes of buildings were very commonly entered as houses. Both were certainly used for human occupation, by night at any rate; and probably the entry was technically correct under the rules. But as the men sleeping in them at night would, under section 6 of the instructions to enumerators, be enumerated as if they were at home with their families and not in these buildings, there was no real need to enter the buildings; and it would have been better to have excluded them, not only because to include them in the statistics for houses was misleading, but also because there was a danger that the men sleeping in them might be enumerated twice over. Perhaps in a future Census the existence of a hearth (*chúlha*) would be the best test of human habitation in doubtful cases. But exceptional cases will occur for which no rules can provide. I reproached one Tahsíldár for entering a tank as an inhabited house; but he justified himself successfully by explaining that a faqír was sitting in the middle of it up to his waist in water, who avowed his intention of remaining there till after the Census.

842. Halting-places and camping grounds entered in the house-lists.—As the house-list formed the basis of the enumerator's schedules, the only method of securing that any spot likely to contain human beings on the night of the Census should be visited by him was to enter and treat it as a house. Almost every village has a well or grove close to it at which way-farers commonly encamp at night; while cartmen or men travelling with animals usually encamp in the bare space immediately surrounding the homestead. Accordingly I directed that all such spots, all encamping-grounds and recognised halting places for travellers, and such *ghâts* and mooring-places for boats as were not large enough to be treated as separate blocks, should be added at the end of the house-list and treated in every way as houses. In one tahsíl I found that almost every well in each village had been entered as a separate house, lest somebody should be sleeping at it. This misunderstanding might be guarded against in a future Census.

Thus the figures for houses include a good many buildings to which the term would hardly apply; but they are almost all included among "unoccupied houses," so that even if these figures possessed greater statistical value than in my opinion they do, the error would not be very important.

843. Entry of families.—The definition of family given in section 838 is so familiar to the native mind, that no mistake should have arisen in this respect. But the difficulty described in section 839, which was felt chiefly in towns, sometimes led the Census staff into error. They had enclosures, buildings, and families to deal with. The house-list showed two units only, houses and families. Following the rules, they entered the enclosure in the house column. Where the several buildings were permanently occupied by members of the same household the entry of families was a sufficient guide. But where they were occupied by very temporary tenants, wholly unconnected with each other, and who were not unlikely to have changed

[1] Captain Bartholomew of Jhang suggests *tabbar*, and Mr. Perkins the Commissioner agrees. The word is not used in the east of the Province. If it is better known in the west than *chúlha*, both words might be given. In the east *kúrai* is the common word; but *chúlha* would be understood, I think, by all.

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before the Census night, and especially where some of these buildings were temporarily vacant, the staff felt that a list of families would be no sufficient guide to the enumerator, and not infrequently availed themselves of the column for families in which to distinguish the several buildings included within the enclosure. And the use of the terms *ghar* for "family" and *ghar ábád* for "resident family" was unfortunate, as they do no doubt apply to the structure as well as to the family which inhabits it. I had wished to use *kúrhi*, which is certainly the word for the Eastern Panjáb; but it is doubtful whether it would be generally understood, and I think *chúlha* would be the best word to use in future.

The result of the tendency I have just described was, that the entries in columns 7 and 8 of the house-lists [1] sometimes referred to buildings rather than to families, dwelling-houses which were vacant at Census being shown in them with their owner (not occupier) as the head of the family, while a building which might contain several families had only one entry. So again the chief to whom a rest-house (*hujrah*) in a frontier village belonged was often shown as the head of a family residing in it. Again I found that people sleeping only in the building, as in the cattle sheds and field-huts mentioned in section 841, and shop-keepers who, while owning separate houses where their families lived and at which they took their meals, slept in their shops for the greater security of their goods (an apparently very common practice in towns), were often entered as families resident at their sheds or shops, whereas their real families were separately entered in their homes. This not only meant a double entry of families, but also danger of a double enumeration of persons. The use of the word *chúlha* for family would render such error unlikely.

844. Statistical units to be recorded.—But after it has been decided what units are most convenient for purposes of enumeration, the question remains whether it is worth while for statistical purposes to separate any other units in our record. Of course the house, as defined above in section 840, possesses no statistical value whatever. And I doubt much whether it is worth while collecting statistics for any *structural* unit in the Panjáb. In the villages the family homestead and in the towns the edifice coincides most nearly with the English house. But in both it is impossible to draw a line for practical purposes between the family-enclosure containing only small buildings in which the members of a household live, and the larger enclosure which includes many houses occupied by independent households. And the structural complexity which is often found in towns is most extraordinary and perplexing. It is not uncommon to find one or more houses built inside another, the last having originally had a large courtyard, but having fallen into the hands of owners too poor to be able to afford the convenience, and who have allowed others to build within it.

As for the household, the figures would be interesting if they could be easily obtained, as they would give the same statistics for the collective groups of families as the figures for the family give for the individual families comprised in those groups. But the difficulty of obtaining trustworthy figures seems insuperable. There is no outward visible sign by which the unit can be recognised; and it would be hopeless to expect that the investigation into social relations which alone could determine its limits should be profitably carried out at Census. Mr. Elliott verbally proposed to me to attempt something of the sort, by ruling that the buildings within an enclosure should be numbered as separate houses when they were occupied by independent households, but that the enclosure should be treated as a single house when the included buildings were occupied by the families of a single household. But I believe that the mixed cases, in which some of the buildings would be occupied by allied and others by independent families, would be, in towns at any rate, so common as to destroy the value of the figures; and I am convinced that the only unit that can be usefully prescribed for the purposes of numbering must be a purely structural one.

The family is, of course, a unit of the highest statistical importance. And the *chúlha* or hearth affords an easy and universally recognised means of distinguishing it. But I think that the house-list, and not the schedule, is the paper from which the most valuable statistics of families could be obtained. I had, indeed, intended to take my figures for families from this paper; but I abandoned this intention because, as already pointed out, the defect in the definition and treatment of "house" led to the family column in the Panjáb house-lists being not unfrequently filled up incorrectly. But the definition I propose would remove this difficulty; and nothing would then be easier than to obtain an absolutely correct record of resident families. If a family includes three sons, one of whom is going on a visit and is enumerated in a hostelry, another is travelling with his cart and encamped by a village on the Census night, while a third is in his boat at the time, the schedules will, unless special precautions are adopted, show four separate families. This is not what is wanted. What is wanted is a record of family *foci* or hearths; and this the house-list, and the house-list only, will give us.

845. Demarcation of urban limits.—Few expressions are more indefinite or capable of more various interpretations than "the town of so and so." There is the city, often included within walls; the houses built immediately under and outside those walls; the surrounding waste lands, which are seldom wholly free from buildings; the outlying suburbs, often very extensive; the small hamlets located for convenience of cultivation in the revenue-paying area attached to the town, and included within the boundaries of the *mauzah* or survey village; the Civil lines or cantonments, generally at some distance from the town itself; while not uncommonly the town is made up of two or more large villages or towns lying close to each other, but still sufficiently distinct the one from the other. The papers of the Census of 1868 do not enable us, except in the rarest instances, to say with certainty what the limits were which included the population returned for each town; and it is therefore difficult to compare the town population of 1868 with that of 1881. I have attempted, but not with complete success, to ensure that at the next Census a similar uncertainty shall not be felt with regard to the limits now adopted. In my first instructions (see sections 6

[1] Form B, Appendix D.

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and 7, General Letter A, Appendix D) I directed that limits were to be fixed for each town, cantonments, and Civil lines, which should include all suburbs and areas in or upon which people found on the night of the Census would properly be included in the population of the town. I pointed out that in the case of municipalities, municipal limits would ordinarily suffice; while in that of cantonments, cantonment limits should be followed. I directed that where the town had arable land attached to it and constituted a revenue-paying estate, the area included within the urban limits just described should be treated throughout as quite separate from the remaining village area, and from any hamlets that the latter might contain. Subsequently, at Mr. Elliott's desire, I stated that the urban limits of municipalities should correspond with municipal limits except under very special circumstances.

These instructions were, I am glad to say, not carried out; at least not in the manner intended by me. I had meant that limits were to be fixed for the urban area of the town of Dehli, for instance; and that everything within that area was to be labelled "Town of Delhi" without further specification. But this would have deprived us of all details of population for suburbs; and thus, while we should have been saved an immensity of trouble in compilation, we should have lost much very valuable information. It may be, even now, that the limits of certain towns and the population returned for them may include suburbs for which it would have been well to have had separate figures. But a glance at the Appendix to Table XIX for British Territory will show that such a case, if it occurs at all, is at least the exception.

846. The way in which I would treat the matter on a future occasion is this. The external urban limits in each case would be governed by the principles to be presently laid down in the section on Tabulation. Within those limits I would fix the boundaries of (1) the town proper, (2) each suburb or group of suburbs for which it was desirable to have separate figures, separating as a general rule every suburb that bore a distinct name, and which it was at all possible to separate. The town proper might be taken by wards, if such divisions existed. The limits of each section should include all encamping grounds, waste land, or other urban spaces which properly attached to it, so that the area of all the sections taken together should include all that would be included within the external urban limits.

I would then make my Census blocks follow these sections, so that no block should as a rule contain any part of more than one section. If a block contained two small sections I would have a separate book of schedules for each, and cases where boundaries crossed would be similarly provided for. For instance, part of the Ambála Civil lines lies within and part without municipal limits; and, as I have not separate figures for the two parts, I have had to treat the population of the whole as part of the population of the municipality in the present Census. In this particular case the error is absolutely insignificant; but it and similar cases would be met by having separate books of schedules for the intra and extra-municipal portions. The arrangement above described would give us separate figures for every separate group of houses which could possibly be considered as falling within the limits of the town.

The question which of these groups should be reckoned to form part of the town could then be dealt with at leisure. I shall discuss the principles on which I think it should be decided when treating of the classification of population as urban and rural in the section on the Tabulation of the figures.

847. Size of Census sub-divisions.—At the Panjáb Census of 1868 each block included 100 families (then called houses), and each circle 20 blocks in towns and 10 to 15 in scattered village areas. The Committee of 1877-78 recommended that each block should not contain more than 60 families, or 300 souls; and that a circle should not contain more than 10 blocks, supervising agency being still further strengthened on the night of the Census. I pointed out that the agency available and the distribution of the population would vary so greatly that no hard-and-fast rule could be laid down; but that the "limits "for a block prescribed at the last Census should not ordinarily be exceeded; while the supervisors should be "appointed as nearly as possible on the scale recommended by the Committee." I also recommended that not more than four circles should be placed under a single superintendent. Abstract No. 143 on the opposite page shows the average number of people included in each town and village block respectively, and the average area of the latter in each district in the Panjáb. Column 9 shows the area per block after deducting the waste lands belonging to Government, which are not indeed wholly uninhabited, but of which the nomad pastoral population is exceedingly scanty, and required special arrangements for their Census.

I had hoped that district officers would avail themselves fully of the license I gave them in my circular, and would work up to the full limits of 500 souls per block. But I find that they almost universally speak of 300 souls as "the prescribed limit," while the table shows that they did not take me at my word; and in one district I found small hamlets of only 8 or 10 houses made into separate blocks, and villages of less than 400 inhabitants divided between two enumerators. This I had altered at once; and in fact the figures of the Abstract, representing as they do the final number of blocks, do not adequately represent the degree of sub-division which was at first attempted; for blocks which had at first been separated were in many cases thrown together before the Census, owing either to my representations or to the impossibility of finding enumerators for them. Of course in hill districts like Simla, Kángra, Hazára, and Kohát, where the population is distributed over small hamlets lying at considerable distances from each other, and where travelling is exceedingly difficult, it is unavoidable that each block should include only a small number of people. And generally, distance is a very important factor. But there seems no sufficient reason why the average population of a village block should have been below 300 in Hushyárpur and Muzaffargarh, and below 350 in Dehli and Gurdáspur; or that of a town block below 300 in Ludhiánah, Hushyárpur, Gurdáspur, Multán and Dera Ismáil Khán, and below 350 in Dehli, Jálandhar, Siálkot, Gújránwála Ráwalpindi, Gújrát, Sháhpur, Montgomery, Muzaffargarh, Derah Gházi Khán, Pesháwar, and Hazára.

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Part I.—Preliminary Operations.

Abstract No. 143, *showing the* Number and Size of Census Blocks *for* Districts.

1	2	3	4	5	6	7	8	9	10
District.	Number of Blocks.			Average number of persons per block.			Average area of village blocks in acres.		Index to
	Villages.	Towns.	Total.	Villages.	Towns.	Total.	Total area.	Corrected area.	
Delhi	1,288	653	1,941	341	312	332	634	634	Delhi
Gurgaon	1,307	115	1,422	436	628	451	949	949	Gurgaon
Karnal	1,052	180	1,232	517	435	505	1,457	1,457	Karnal
Hissar	1,090	181	1,271	408	421	410	2,158	2,158	Hissar
Rohtak	1,134	250	1,384	400	398	401	1,022	1,022	Rohtak
Sirsa	713	75	788	312	413	321	2,697	2,697	Sirsa
Ambala	2,040	319	2,359	454	440	452	806	806	Ambala
Ludhiana	1,532	295	1,827	350	282	339	574	574	Ludhiana
Simla	84	118	202	250	187	2 3	137	137	Simla
Jalandhar	1,767	414	2,181	369	331	362	479	479	Jalandhar
Hushyarpur	3,268	285	3,553	253	260	254	427	427	Hoshyarpur
Kangra	3,098	109	3,207	228	225	228	1,874	1,874	Kangra
Amritsar	1,602	441	2,043	440	426	437	629	629	Amritsar
Gurdaspur	2,260	344	2,604	328	237	316	516	516	Gurdaspur
Sialkot	2,171	247	2,418	428	340	419	577	577	Sialkot
Lahore	1,331	443	1,774	541	462	521	1,754	1,547	Lahore
Gujranwala	1,655	219	1,874	321	370	329	1,000	696	Gujranwala
Firozpur	1,454	154	1,608	402	429	405	1,221	1,221	Firozpur
Rawalpindi	2,816	284	3,100	261	300	265	1,005	858	Rawalpindi
Jahlam	1,528	135	1,663	346	453	354	1,628	1,421	Jahlam
Gujrat	1,664	130	1,793	389	326	384	759	759	Gujrat
Shahpur	1,023	163	1,186	362	317	355	2,433	2,095	Shahpur
Multan	1,360	297	1,657	341	297	333	2,767	1,352	Multan
Jhang	880	84	964	407	440	410	4,047	1,352	Jhang
Montgomery	937	69	1,006	430	312	424	3,805	1,352	Montgomery
Muzaffargarh	1,430	63	1,493	221	347	227	1,405	1,188	Muzaffargarh
Derah I. Khan	1,344	209	1,553	293	229	284	4,422	3,855	Derah I. Khan
Derah G. Khan	709	135	844	452	320	431	4,057	3,057	Derah G. Khan
Bannu	589	55	644	520	469	516	4,203	4,053	Bannu
Peshawar	1,130	391	1,521	411	329	399	1,418	1,418	Peshawar
Hazara	1,301	62	1,423	285	303	286	1,429	1,315	Hazara
Kohat	452	35	487	361	519	373	4,008	3,911	Kohat
Total	46,089	6,954	52,983	357	350	356	1,487	1,256	

848. Taking the Province as a whole, I think that the blocks were too small; and more especially in towns, where the average number of souls per block is even smaller than in villages. I attribute the fact partly to a common disinclination, which I found to exist in many districts, to include two villages in one and the same block. It is of course absolutely necessary that the block should be of such a size, both as regards the number of people included within it and the distance between the points to be visited, that a single man can easily visit and correct the entries for every house in it on the night of the Census. But so long as this precaution is duly observed, the larger the blocks the better. An excessive number of blocks increases the difficulty of finding enumerators, compels recourse to uneducated agency writing strange and illegible characters, renders supervision more difficult and uniformity less easy of attainment, increases the labour and expense of compilation, and obliges the Government to call upon a larger proportion than necessary of the non-official population to perform a very irksome duty. In many districts the very greatest difficulty was experienced in finding enumerators for the outlying tracts; and in not a few cases men unacquainted with the locality and its inhabitants had to be sent out from the towns, and their expenses paid from district funds. Yet in those very districts I not unfrequently found that two blocks had been made where one enumerator would have been amply sufficient. In the towns themselves qualified agency existed in plenty; but here the houses lie so close together that it would have been well, by increasing the size of the blocks, to reduce as much as possible the demand upon private individuals for assistance in the operations. I found that, in the City of Lahore, the enumerators in the portions which I visited had generally finished their work by 10° 30′ or 11° on the night of the Census; and I believe that where the houses lie in a compact mass, a single enumerator can easily manage 500 to 600 people, for he has not to take their Census *ab initio*, but merely to check and correct a Census already taken at the preliminary enumeration. Where the houses lie far apart, as in the case of scattered hamlets and villages, the distance to be travelled, and not the number of houses to be visited, is the most important fact. Mr. Stogdon of Lahore, Major Plowden of Kohát, and Colonel Parsons of Jahlam would increase the size of the block, Mr. Stogdon remarking that "it is difficult to get enumerators, and it is better to have intelligent enumerators and large blocks than enumerators of no capacity and small blocks." Major Parker of Hissár, Mr. Hawkins of Amritsar, and Mr. Fitzelle of Sháhpur are of opinion that a block should include at least 100 to 125 families "which a single man could easily visit and correct the entries of; while where the block lies wholly in one town or village it might safely include 125 houses irrespective of families." Mr Benton of Karnál writes:—

"A large number of the enumerators, such as the patwáris lately employed in the settlement and many others, were admirably "qualified for the work, and although to meet the attainments of men generally available, it is no doubt desirable to keep the

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Part I.—Preliminary Operations.

"average size of blocks low, I do not hesitate to say that such men could easily perform the duties satisfactorily on the night of "the final enumeration for 200 or 300 houses, provided they were all together.

"I would retain the average size of block at 60 houses or 300 persons as suited to enumerators of the meanest capacity, but I "would make it optional to make over two or three of such blocks to properly qualified persons, because so long as preliminary enu- "meration is had recourse to, an active and intelligent man could easily attest 1,000 entries, when the houses are contiguous, in three "or four hours. The work of enumeration was almost completed in Pánipat before moonrise, that is to say before 8° 20' P.M."

Mr. Gardiner of Gújránwála indeed would decrease the size of the block if he could get enumerators enough; but he gives no reason for his opinion.

The statement of Census agency given on the opposite page shows that the proportion of supervisors to enumerators was as a rule largely in excess of that suggested in my instructions, in many cases being more than twice as great, and only in Lahore, Montgomery, and the villages of Derah Ismáil Khán falling materially below it. The number of blocks placed under each supervisor was on an average—

In the villages	8⅘
In the towns	7⅓
In the Province	8⅘

The average number of circles allotted to each superintendent was—

In the villages	3¾
In the towns	3
In the Province	3⅔

Here again, the agency was most numerous in the towns; and this was I think right; for the town enumerators, while or rather because more highly educated, are less intimately acquainted with the people they have to enumerate, more satisfied with their own ignorance, and less inclined to follow instructions, while the population they have to deal with is far more varied and complex. Thus supervision, while easier to exercise, is far more necessary in the towns than in the villages.

849. Statistics of Census agency.—Abstract No. 144 on the opposite page shows the number of the Census staff, and the sources whence they were drawn.

The results may be abstracted as follows :—

		Villages.	Towns.	Total.
Number of	Superintendents	1,444	314	1,758
	Supervisors	5,440	959	6,399
	Enumerators	46,029	6,954	52,983
	TOTAL	52,913	8,227	61,140
Official Staff	Superintendents and Supervisors	2,419	750	3,169
	Enumerators	3,670	1,748	5,418
	TOTAL OFFICIAL	6,089	2,498	8,587
Village accountants.	Supervisors	3,067	81	3,148
	Enumerators	4,223	100	4,323
Village headmen	Enumerators	2,759	59	2,818
	TOTAL SEMI-OFFICIAL	10,049	240	10,289
Shop-keepers	Enumerators	10,958	342	11,300
Schoolboys	Enumerators	3,084	859	3,943
Other non-officials.	Superintendents and Supervisors	1,398	442	1,840
	Enumerators	21,335	3,840	25,181
	TOTAL NON-OFFICIAL	36,775	5,489	42,264

To furnish so large an agency, every class of the community which had any pretensions to education or intelligence was called upon to contribute. Almost every official in the district was employed in Census duties, the supervising staff being chiefly drawn from this class; while private individuals formed the great mass of the enumerating agency. The superintendents were for the most part Munsifs, Náib Tahsíldárs, Qánúngos and their Assistants, Deputy Inspectors of Police, superior office staff, Chief *Muharars* and head-masters of schools, Assistant Surgeons, *Daroghas*, members of committees, Government pensioners, Honorary Magistrates, merchants, *Zaildárs*, *Jágírdárs*, and rural notables and men of influence. The supervisors were drawn from the classes mentioned above, supplemented by officials from the district revenue, judicial, and municipal establishments, sergeants of police, district school and post masters, native doctors, village accountants and headmen, *jágírdár's factors*, intelligent pupils from the higher schools, respectable traders, and land-owners and others of intelligence. The class of enumerators comprised every clerk or similar official who could be spared and whose services had not already been utilised in the higher grades; and to these were added a vast body of non-official agency, stamp-vendors, petition-writers, candidates for employment, pensioners of sorts, retired *Patwáris*, merchants' clerks, *Jágírdárs'* servants, *dharwáis*, senior school-boys and monitors, *Pandits*, *Maulvis*, village *Mullás* and *Qázis*, shop-keepers, village headmen, and land-owners who could read and write. The services of even illiterate but intelligent *zamindárs* were often utilised by attaching to them school-boys as amanuenses.

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Abstract No. 144, *showing* Census Agency *for* Districts.

Serial No.	Name of Division	No. of Supervisors.			No. of Enumerators.								TOTAL VILLAGES AND TOWNS.												Name of District.	Serial No.
		Officials.	Non-officials.	Total.	Village Accountants.	Village Headmen.	Other officials.	Village Shopkeepers.	School-boys.	Other non-officials.	Total.	Officials.	Non-officials.	Total.	Village Accountants.	Other officials.	Non-officials.	Total.	Village Accountants.	Village headmen.	Other officials.	Village Shopkeepers.	School boys.	Other non-officials.	Total.		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	1
1	Delhi	36	14	70	200	44	32	285	110	33	138	518	104	376	1,288	60	32	92	111	65	187	519	222	867	1,941	Delhi
2	Gurgaon	30	7	37	59	49	38	146	245	68	153	273	137	419	1,307	34	7	41	247	55	188	278	146	490	1,423	Gurgaon
3	Karnal	33	4	37	55	38	14	107	171	36	141	99	49	339	1,052	28	6	34	175	43	128	99	61	659	1,232	Karnal
4	Hisar	35	5	30	70	21	22	113	141	36	74	396	36	397	1,050	38	8	46	145	37	134	403	79	427	1,231	Hisar
5	Rohtak	45	8	53	148	28	8	184	275	72	118	425	84	424	1,134	34	8	42	84	33	130	454	133	505	1,384	Rohtak
6	Sirsa	18	...	18	16	22	4	112	108	11	307	153	18	84	713	23	...	23	108	46	348	154	27	105	788	Sirsa
7	Ambala	...	...	...	24	165	57	246	193	61	170	109	541	905	2,040	...	1	64	198	63	218	135	586	1,164	2,359	Ambala
8	Ludhiana	48	...	48	76	99	6	181	254	68	212	448	242	368	1,533	63	1	64	254	68	200	448	350	447	1,817	Ludhiana
9	Simla	2	...	2	4	6	2	13	1	10	10	...	10	53	84	7	...	7	1	10	40	...	19	132	202	Simla
10	Jalandhar	61	6	67	168	28	18	214	175	114	103	149	103	1,033	1,767	78	10	88	178	118	163	180	383	1,311	3,181	Jalandhar
11	Hoshyarpur	76	25	101	233	132	54	419	141	181	387	154	342	2,063	3,268	85	27	112	238	184	404	388	388	2,202	3,583	Hoshyarpur
12	Kangra	65	7	72	158	98	61	317	11	369	63	1,384	40	1,331	3,098	71	8	79	159	104	184	1,403	49	1,286	3,207	Kangra
13	Amritsar	32	51	83	153	53	27	210	271	51	109	141	79	951	1,602	44	56	100	153	51	157	141	100	1,333	2,043	Amritsar
14	Gurdaspur	33	57	80	234	26	6	298	145	182	133	90	49	765	2,266	34	66	100	238	185	238	904	75	966	2,664	Gurdaspur
15	Sialkot	33	56	88	224	33	48	278	142	137	91	958	54	789	2,171	43	62	104	239	153	153	982	64	939	3,418	Sialkot
16	Lahore	24	4	28	38	48	27	113	220	57	124	238	93	604	1,331	32	22	54	43	85	226	296	113	854	1,774	Lahore
17	Gujranwala	33	5	38	52	70	49	171	198	47	71	283	51	1,005	1,655	39	10	49	54	51	119	334	75	1,004	1,874	Gujranwala
18	Firozpur	42	12	54	54	98	33	185	227	33	51	323	55	700	1,434	45	12	57	136	51	120	334	80	812	1,668	Firozpur
19	Rawalpindi	64	28	92	160	66	77	303	134	332	99	964	56	1,240	2,816	75	30	105	161	145	333	1,600	86	1,390	3,100	Rawalpindi
20	Jhelum	49	13	62	182	13	10	205	37	110	72	115	100	1,091	1,528	54	14	68	187	20	113	196	114	1,198	1,691	Jhelum
21	Gujrat	32	11	43	117	14	14	166	75	109	99	645	101	635	1,664	37	11	48	120	21	110	196	113	1,199	1,794	Gujrat
22	Shahpur	28	3	31	98	17	9	124	38	69	81	270	53	506	1,023	31	4	38	104	25	70	299	73	586	1,186	Shahpur
23	Multan	33	8	41	71	24	47	142	171	83	95	438	58	515	1,360	38	14	52	72	45	83	447	133	643	1,657	Multan
24	Jhang	10	7	17	26	37	30	93	133	27	53	30	67	570	880	24	7	31	18	43	29	30	85	394	964	Jhang
25	Montgomery	35	7	42	135	47	32	81	138	38	55	90	80	539	937	36	11	47	5	53	28	91	83	374	1,096	Montgomery
26	Muzaffargarh	38	7	65	135	19	14	168	45	83	115	112	73	1,062	1,430	62	8	70	136	83	178	83	95	1,035	1,493	Muzaffargarh
27	Dera I. Khan	28	2	30	72	38	14	124	122	87	84	221	153	674	1,344	37	6	43	73	59	172	324	176	767	1,553	Dera I. Khan
28	Dera G. Khan	11	11	22	11	23	44	78	117	62	83	40	12	395	700	22	11	33	120	41	150	40	14	453	844	Dera G. Khan
29	Hamun	5	...	5	14	38	14	66	132	22	57	43	25	310	589	6	...	6	136	42	89	44	28	343	644	Hamun
30	Peshawar	27	4	31	74	17	38	129	217	103	61	686	20	143	1,130	33	8	41	216	37	173	596	32	387	1,521	Peshawar
31	Hazara	27	27	54	114	1	46	161	107	71	40	393	103	852	1,301	31	26	57	3	46	59	292	113	884	1,423	Hazara
32	Kohat	10	...	10	22	21	...	43	28	9	220	72	18	111	452	11	...	11	28	...	233	72	33	112	487	Kohat
	Total	1,067	387	1,444	3,067	1,362	1,017	5,440	4,223	2,759	3,670	10,958	3,084	21,335	46,020	1,875	483	1,758	3,148	1,894	1,357	6,390	4,393	2,818	5,418	11,300	3,943	25,181	54,983	

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Part I.—Preliminary Operations.

850. Aid afforded by independent Departments.—At the very commencement of operations the Panjáb Government issued a circular to all Heads of Departments[1] pointing out that "every available "officer in every department of Government who can be relied on for efficiently conducting supervision "will be expected to render assistance; and Heads of Departments are requested to place the services of "their staff unreservedly at the disposal of the District Officers for this purpose, so far as may be compatible "with the conduct of necessary business." The manner in which these orders were carried out varied much in various Departments and parts of the Province. In those districts where a settlement was in progress, *viz.*, Sirsa, Lúdhiána, Jálandhar, Hushyárpur, and Ráwalpindi, the settlement establishment afforded a ready-made supervising agency of the utmost value. The Deputy Commissioners and Settlement Officers came to an agreement about the division of the work in each case; and the most practical plan was that by which the latter undertook, in communication with the former, the whole of the purely village Census, leaving the remainder to the district staff proper. The Forest Officers rendered most valuable assistance in the difficult mountain tracts in which their duties lie; and the Customs officials in Gurgáon worked most zealously for us. The staff of the Western Jamna Canal was of great use in the final Census of some districts; while the postal, educational, and medical departments as a rule responded cordially to the call made upon them. But complaints of want of willingness to help in the work are not altogether absent. The Panjáb Northern State Railway refused to assist; in Gurdáspur the Canal staff formed a solitary exception to the willingness of other Departments; in Gújrát the educational officers were obstructive; while in Jahlam little aid could be obtained outside the district staff. In Pesháwar, the staff drawn from other Departments "did little, and never made any attempt to learn the working of the Census machinery." Major Nisbet writes from Simla:—

"One of the weakest points in the general arrangements was, the little assistance that could be obtained from other Depart-"ments for Census work, notwithstanding the orders Government promised to issue on the subject. The usual excuse was, that "office hands were generally fully employed in their own special duties. On a future occasion it might be well for Government to "ascertain beforehand the assistance obtainable from different offices, and determine the sufficiency of excuse offered for neglecting "to afford such aid."

I think it would have been well had Government issued a second circular early in December, again insisting upon the importance of help being cordially rendered. And much can be done by the Deputy Commissioner to make matters easy for other Departments. Mr. Benton of Karnál for instance used the Canal staff as enumerators only. He required them to attend at their blocks from 26th December to 2nd of January and again from 15th to 18th February, and he had the supervisors ready on the spot with the papers to explain their duties to them. In any case the brunt of the work must fall upon the Deputy Commissioner and his subordinates. I called for returns from the districts showing the amount of assistance rendered by other Departments; but unfortunately the figures are rendered meaningless by the fact that most of the Deputy Commissioners have included under that heading the district staff in the Police, Educational, Medical, Jail, Registration, and similar departments, which, being directly subordinate to the Deputy Commissioner, should have been counted as District Staff. The Government of India forbade us to employ the rank and file of the police on Census duty; and though the exception deprived us of the services of a large staff distributed all over the country, yet I think it was absolutely necessary.

851. Preparation of the preliminary record by village accountants.—It will be noticed that of the village accountants, nearly half were utilised in supervision. As a fact the great mass of the preliminary records in the villages was I believe made by these men, whether as supervisors or as enumerators. At an early stage of the operations I pointed out[2] that two classes of agency would be employed for the work of filling up the schedules, as it would sometimes be impossible to obtain for the preliminary record the services of the man who was to conduct the final enumeration on the Census night, and the staff which prepared the former would have to be supplemented for the latter operation. I had wished to put the point more strongly, and to say that in villages this would very generally be the case: but the Panjáb Government was of opinion that the objections to having the preliminary record prepared by one man and the final Census taken by another were so great that, though such a course would doubtless be unavoidable in some cases, yet it was well to treat it as a quite exceptional and deplorable necessity.

In the towns it is not only necessary but easy to arrange that the same man shall perform both operations. But in the villages this is not the case; and everything I have seen and heard in connection with this Census has convinced me that it is better to prepare the preliminary record by the best available agency, and where necessary to this end, to sacrifice the advantage of having the same man make and finally correct the entries. Of course it is most advisable that, where a really competent agent can be secured for the final enumeration, he shall also fill up his schedules in the first instance, as he thus obtains an intimate knowledge of his block and its inhabitants. But in the villages it is too often simply impossible to obtain really good men in sufficient numbers to take the final Census. The fact that of 40,000 village enumerators, 11,000 were village banyas, 3,000 village headmen, 3,000 school-boys, and 21,000 non-officials resident (for the most part, at any rate) in the villages, speaks for itself. For the preliminary record on the other hand, which is made more or less at leisure, it is comparatively easy to obtain a sufficient and competent staff: in fact the village accountants might almost prepare it unaided. Now the preliminary record is, in the villages at least, the very backbone of the Census; in fact it might almost be said to be the Census itself. I do not believe that the alterations made on the night of the Census affected 5 per cent. of the village population. But supposing the proportion to have been double this, and that in a village of 500 souls, the enumerator cut out 25 as having died or gone away, and entered 25 new men as having been born or come to the place since the preliminary record, I would prefer to have the 500 entries correctly made in the first instance, and to take the chance of the enumerator making a mess of his 25 new entries—the 25 excisions he could not well go wrong with—rather than have

[1] Section 3 of Secretary to Government's No. 7, dated 27th August 1880, Appendix D.
[2] Section 11, General Letter A, Appendix D.

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the original 500 entries made by a bungling old banya in a hand-writing which he could not read himself, much less anybody else, with all the attendant difficulty of supervision, alteration, recopying, and the probability of error in the end. He would at any rate have correct entries to guide him in entering the newcomers; and where they were members of families already entered he could hardly go wrong, even in recording them. Of course I would insist that where any other than the final enumerator made the preliminary record, he should take the enumerator over his block before the night of the final Census, visit every house with him, and read out and explain to him each entry in the schedules. This was done in the present Census; and, as I have said, from what I have heard from Deputy Commissioners and others, as well as from the final reports from the districts, I believe that the village schedules were very generally filled up in the first instance by the village accountants. In a future Census I would have this done wherever a thoroughly competent enumerator could not be found.

852. I append below opinions on the subject by several district officers. Many others make incidental remarks in support of my proposal. Mr. Wilson of Sirsa, however, thinks that it is better that the enumerator should himself make the preliminary record, as he thus understands the entries better :—

Dehli (Mr. T. W. Smyth).—"At first, following the instructions, patwáris were appointed as supervisors and superintendents, "and in this capacity they refrained from taking an active part in filling up the schedules when the preliminary enumeration "commenced. It would have been better if their services had been utilised as enumerators and they had been made to understand that "this was their principal duty. This was done to a large extent after the inspecting officers began to move about and check the work."

Kangra-Kulu (Mr. Alex. Anderson).—"Two systems were followed in this sub-division for writing the schedules. In Kúlu "the enumerators were made to fill up the schedules, while in Plách this was generally done by the patwári. The enumerators in "Kúlu were more intelligent, and sooner understood what was wanted; still the entries were all made in the presence of the supervi- "sor or patwári. In this way they required no more instruction for the final enumeration than they had got; and I found that in "Kúlu they understood what to do. At first the work was done much more slowly, but the enumerators became more proficient. In "Plách on the other hand, I found that many of the enumerators had exceedingly vague ideas of how the schedule was to be filled in, "and up to the last every effort had to be used to instruct them. The Kúlu system had one great disadvantage: the books were scarce- "ly legible even by the enumerators themselves, while in Plách they were well written and any one could read them. Translations "had to be largely made and this caused delay in sending in the returns. I think it is more satisfactory to make the enumerator write "his schedules, even though he writes to another's dictation, than to have the books written by another."

Hushyarpur (Mr. Coldstream).—"As regards the work of preliminary grouping and numbering Mr. Kensington thinks that, "had it been done gradually by a very much smaller staff, the mass of the enumerators being kept for the final enumeration only, "it would have been (in this district as under settlement particularly) much more efficiently controlled, more correct in the first in- "stance, and done in at any rate not longer time. 'As it was, the mass of enumerators rushed at it with an excess of zeal which "'defied supervision and necessitated its being in many instances done all over again, at an expenditure of labour altogether dispro- "portionate either to the amount or difficulty of what was really required.' The preliminary work would no doubt have been done "better by a better trained staff (even of smaller numbers) and the necessity for much supervision and correction and recorrection "would have been saved. I think the whole of the preliminary arrangements such as demarcation and numbering might with advantage "have been done by the settlement establishment alone. But there would have been a serious danger in using enumerators for the "final enumeration who had not been trained to do the kind of work required, and to use the forms to be used on the final night: "and perhaps the best way to teach them was to employ them in preparing the schedules. I admit, however, that the plan adopted "cost the enumerators and supervising staff much time and labour. It may be hoped that by the time another Census comes round, "there will be little difficulty in securing the services of competent enumerators, for education will then have largely spread, but they "will not be such a *cheap* agency as the present one.

"One of the most energetic and efficient of my supervising officers reports on this point that throughout the police division of "Tandah there was not a single circle of which the supervisor had not himself to write the papers of one or more of the enumerators."

Shahpur (Mr. Frizelle).—"A great many of the entries in the more rural and backward circles, or wherever the enumerators "were of inferior qualifications, were in the supervisor's handwriting; and the main use of enumerators was to check the entries and "record alterations on the night of final enumeration, but these additions or omissions were again checked by the supervisors on the "day following the final enumeration. Supervisors being in the main patwáris, and a better trained and educated (though less trust- "worthy) class in this district than is usual elsewhere, being previously well acquainted with the names and conditions of all persons "residing in their circles, and the Census *halkas* being largely identical with patwáris' circles, the burden of the work naturally fell "on patwáris; it came easy to them, and was I believe well performed by them.

"In no case were the sketch-maps of homesteads omitted, and these I think added greatly to the certainty of the work. Where "patwáris were not supervisors but only enumerators in consequence of better educated supervisors being easily available, the sketch- "maps and I believe sometimes the house-lists and portions of the index-register were prepared by patwáris for places outside their "blocks but within their ordinary official circles, as they were better practised at such work than non-patwári supervisors; and this "was the only slight departure, if it can fairly be called a departure, from the principles laid down by the Deputy Superintendent. "Perhaps a great part of the preliminary record being in some instances in the handwriting of supervisors may also be considered a "departure; but this was unavoidable once it was determined to adopt only the Persian character, and it threw no undue labour on "supervisors, especially as the blocks and circles were small and patwári supervisors already knew all about their circles. Such "entries were always made in the presence of, and were fully understood by the enumerators, and their being in the patwáris' hand- "writing in my opinion only adds to the value."

Gurdaspur (Mr. Browne).—"*Suggestions.*—I have only to suggest that in special tracts like this, on future occasions it would "be a great relief to the agency employed if they were drawn from the residents, and I would propose that each tikkali form a block: "the preliminary record be prepared during a period of about a month by the patwáris, and after check a list on separate paper for "each tikkali be given in the Thakari character to the enumerator. It would be his duty on the date fixed for final enumeration "to add to or subtract from that list in Thakari, while one man accompanying the patwári afterwards as translator, would enable "him in presence of the enumerator to correct his Persian schedule accordingly. I believe men competent for this work could be "found on the spot; they all know their own tikkali bounds, and the hardship of sending up men from the plains to work over 10 or "15 *kos* of hilly ground more than once whilst living at their own cost would be avoided; unquestionably this has caused a certain "amount of resentment."

Rawalpindi Division (Colonel Hall).—"Many of the enumerators were men of defective education and could not be "trusted, except under close supervision, to do the work, which led to the schedules being prepared by the supervisors.

"This was particularly the case in the Rawalpindi and Shahpur districts. The main use of the enumerators of inferior "qualifications in both districts was to check and alter the schedules on the night of the final enumeration.

"In this as well as in the other districts it has to be remarked that the supervisors did most of the work of enumeration. The "non-official enumerators had first to be trained to their duties.

"The final result may however be considered eminently successful, and it is therefore worthy of consideration whether in a "future Census the number of persons employed could not be considerably reduced, and the work of enumeration be entrusted from "the commencement to persons thoroughly able to do it."

Lahore (Mr. Stogdon).—"Nominally an enumerator was appointed to each block, and the probability is that almost "every one of them could read and write the Persian character; but many of them were men of little intelligence and no experience, "and it would have been quite unsafe to have left the work of making the preliminary record to them. It would have been illegible

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"and full of the most absurd mistakes and inconsistencies. As a matter of fact the patwáris made the preliminary record in very "many instances, and the enumerators accompanied them and made themselves acquainted with the condition of their blocks, so "that they were well able to make the final enumeration on the night of the Census."

Jahlam (Colonel Parsons).—"Many of the enumerators were so inefficient that their preliminary record had to be made "by the supervisors. It is probable that in rural circles the more extensively the patwáris are employed in the work of enumera- "tion, with such aid in the way of assistants as may be available, the better the work will be performed."

Karnal (Mr. Benton).—"Generally speaking I devolved the whole duty of preparing the house-lists, plans of villages, and pre- "liminary records on the revenue establishment, or on other officials subordinate to myself, as I was afraid to trust more than I could "help to other departments. The officials of other departments, and in many cases also non-officials, had the preliminary records "and other documents already complete made over to them; they had only to study them and make themselves acquainted with the "locality in the last days of December, and to make the necessary corrections on the night of the final enumeration."

Peshawar (Mr. Lawrence).—"The preliminary record was almost entirely made by patwáris who for the most part were "supervisors. The enumerators though able to read and write could not be entrusted with this work. Their writing was so bad; "and their ignorance of detail as to religion, caste, clan, and family, became evident from the trial sheets which were given them to "fill. Hence we have had to depend on the patwáris for the preliminary record and the result has been successful, and eminently so "in the Utmán Bolák tahsil."

853. Non-official agency.—It will be seen from the table on page 429 that, including village headmen, 1,850 out of 8,150 superintendents and supervisors, and 45,000 out of 53,000 enumerators were non-official. Of these last 3,950 were school-boys; excluding whom we employed on Census duty 36,500 out of a literate population of 298,800 in the villages, or one in eight, and 4,700 out of a literate population of 183,300 in the towns, or one in thirty-nine [1]. Of the non-official staff only 1 superintendent and 3 supervisors in the Teri jágír of Kohát, and 37 supervisors and 390 enumerators in the city of Amritsar were paid. In a few instances one or two rupees were given to defray actual expenses of men sent to a distance from their homes for the final enumeration; and in the city of Dehli and the Lambagráon jágír in Kángra the house-lists were prepared by paid agency. With these exceptions no payments of any sort were made on account of Census operations. The district officers bear unanimous testimony to the willing cheerfulness with which all classes assisted in the Census; and in many districts the individuals selected do not appear to have even attempted to escape from the duty. But that that duty was most irksome is beyond dispute. Every man employed on the supervising staff had to devote to the work a not inconsiderable portion of his time for from two to three months; every man employed as an enumerator had to learn his work, fill in his preliminary record, go over his block with inspecting and revising officers, and take the final Census; all had to conduct an inquisition which, though made as little distasteful as possible, exposed them in towns at any rate to the risk of insult and abuse. Practically no payment of any sort was made for the service rendered; and though the certificates issued after the work was over were much valued, yet they were no real equivalent for the duties to the effective performance of which they testified. I quote below the remarks of some district officers on the subject. Further testimony to the willing help afforded by the non-official classes will be found in abundance in the paragraphs on the attitude of the people in the section on Subsidiary Matters:—

Dehli (Mr. T. W. Smyth).—"In fact the services of every available male adult who could write legibly and "was willing to assist were utilised, according to his fitness, either as a superintendent, supervisor, or enumerator: their "services were as a rule willingly rendered, and all took an interest in the duties assigned to them. Although every "effort was made to employ Census officers in the neighbourhood of their head-quarters, yet of necessity many were "required to work at a distance from their houses and at great personal inconvenience to themselves. The operations, "including the preliminary preparations, the rough and final enumerations, extended over a period of nearly three months. The "work was to some extent onerous, as the inspection of the superior officers was frequent and conducted occasionally at unexpected "times, and every one employed was required to be constantly at his post."

Jhang (Captain Bartholomew).—"In conclusion as suggestions have been invited for future use, I would express the opinion "that in no country but India could such a thoroughly efficient Census be performed by a practically unpaid staff. Large "numbers of private individuals were pressed into the service and a great portion of their time for months was used by Government "without any remuneration."

Hushyarpur (Mr. Coldstream).—"There was a great deal of gratuitous work done, and done on the whole I believe won- "derfully cheerfully and well. The wish of Government was sufficient, there were those found who could undertake the work, and "I heard I think no repining of any sort as I went about the district.

"Of the total number of agents 1,292 were unpaid. On one occasion in the hills I called the enumerators of a circle together "and found they were some 30 Brahmins who could write Hindi and who with wonderful cheerfulness were serving Government for "nothing. The amount sent by Government for remuneration of these men is very small,[2] and would give if equally distributed but "annas 14 pie 1 for each. This sum has been supplemented from District and Municipal Funds, and it seemed but justice that some "recognition of their services should be made, and the sum which after these local contributions it was possible to give them was but "a very scant remuneration for the great pains they took.

"It was undoubtedly a difficulty even in many parts of this district (in which preliminary education is perhaps more forward "than in most) *to find competent enumerators.*

"That the work had to be done practically without remuneration greatly increased the difficulty. The exigencies of the situation "necessitated the employment of enumerators in a manner which differed in no respect from *begár* labour except that the subjects had to "labour with head and pen, and not with hand and spade. This was a real difficulty and should in fairness be mentioned here. It has "been very partially met by the apportionment by Government of a small cash allotment and by contributions towards payment of the "enumerators from District and Municipal funds. I think it would have been well if Government had from the commencement direct- "ed District Funds and Municipal Committees to set apart ₹5,000 or ₹10,000 (the sum being regulated according to population and "area) of the budget allowance for the year to defray the expenses of the Census in each district. This matter has been alluded to by "several of the supervising officers. The Tahsildár and Munsiff of Dasúyah wrote 'the impressed enumerators considered the work "as '*begár*, but they paid attention when *admonished, and came to know that an Act had been passed.*'"

Commissioner, Jalandhar (Mr. Tremlett).—"When the next Census occurs it may well be questioned whether larger sums "ought not to be raised, probably from local sources, for the remuneration of those employed. If the people are being educated to "regard it as discreditable and wrong when a private individual 'useth his neighbour's service without wages and giveth him not "for his work,' surely the State itself should not set a bad example and be thought to exact '*begári*' labour."

Simla (Major Nisbet).—"The large number of voluntary enumerators employed was recruited principally from the native "shop-keepers of Simla and other large places, many of whom undoubtedly ought to have been paid. It might not always be

[1] See Table No. XIII for British Territory, Appendix A. But a part of the agency employed in the villages was drawn from the towns.

[2] This grant was not intended as *remuneration*, but as *rewards* to those who had done well; and great stress was laid upon this distinction in my circular on the subject.

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Part I.—Preliminary Operations.

"possible for the district officer to induce persons to volunteer for the work, as it is difficult to imagine any person not actuated by "some other motives than a mere wish for employment undertaking a duty which occupied several weeks and involved considerable "physical labour and serious responsibility, while carrying with it no other remuneration than barren thanks.

"There is no question that district officers were hardly dealt with in having to do all this extra work of a complex and difficult "character without funds being placed at their disposal for necessary and reasonable expenses[1]; and on any future occasion I think, "in taking a Census of India, greater liberality in incurring necessary expense must be shown."

854. I do not for a moment agree with those officers who condemn the policy which demanded these gratuitous services from the more or less educated classes. I think it is well for the people in general to have it brought home to them that they are liable to be called upon to assist Government on occasion, and that they themselves must take a share in operations which concern the public weal. But these duties, while of a distasteful nature, carried with them no corresponding privileges or status; and I think that every effort should be made to keep within as narrow limits as possible the numbers from whom they must necessarily be demanded. If the larger limits of blocks which I recommend in section 848, and the scale of supervising staff laid down in the instructions were followed, we should have:—

	Villages.	Towns.	Total.
Population	16,410,907	2,439,530	18,850,437
Number of blocks and enumerators	32,822	4,066	36,888
Number of circles and supervisors	3,282	407	3,689
Number of charges and superintendents	820	102	922

and, taking the official and semi-official staff actually employed as still available, we should have—

	Villages.	Towns.	Total.
Official superintendents	820	102	922
,, supervisors	1,615	407	2,022
Village accountants, supervisors	1,067	...	1,067
Official enumerators	3,670	1,992	5,662
Village accountants, enumerators	5,023	181	5,814
Village headmen ,,	2,759	59	2,818
Non-officials employed	20,770	1,834	22,594

which would release nearly half the non-officials employed in the villages and two-thirds of those employed in the towns from Census duty. Of course the real effect would not be quite so great as this, as many of the officials cannot, consistently with their ordinary duties, be employed at any great distance from their head-quarters. But the figures show that a good deal could be done without, as I believe, injury to the work, to reduce the burden thrown upon the educated non-official population.

855. With regard to the composition of the non-official staff, there is not much to be said. There are 34,000 villages in the Panjáb, and probably at least 80,000 village headmen; and the fact that less than 3,000 of these were utilised for Census purposes shows how few of them must be able to read and write. The village shop-keepers probably formed as competent an agency as could be obtained in many cases, as they at least know the village and the people. The figures of Abstract No. 145 show that at least 7,450 out of the 11,300 village shop-keepers employed as enumerators wrote the Persian character; a result so surprising that I made special inquiries as to the accuracy of the figures. It appears that the new generation of shop-keepers very generally learn Persian in the Government schools. Where this is the case there can be no possible objection to their employment. But where they do not write Persian I think they should be avoided, even at the cost of sending out school-boys at some little expense. In such cases they are wholly unable to read the instructions issued; the character they are pleased to employ is often a mere system of mnemonic symbols only moderately intelligible to the writer; while their mental capacity is not seldom of the meanest. Shop-keepers ignorant of Persian were most largely employed in Rohtak and Kángra, and somewhat less largely in Dehli, Hissár, and Firozpur. In other districts they were not made much use of.

Where school-boys were utilised in the villages, an intelligent headman or other person on the spot was often made responsible for the Census, the school-boy practically acting merely as his amanuensis. But several district officers remark that a sharp school-boy made a better enumerator than almost anybody else, as he was proud of his duties, active, and intelligent, and yet willing to learn and to obey orders. In many cases the more intelligent youths from the higher schools in the towns were sent out into the villages and their expenses paid from district funds. They were most largely employed in Ambála where they constituted one-fourth, and in Lúdhiána where they formed one-sixth of the whole staff of enumerators, and in Dehli, Derah Ismáíl Khán, Gurgáon, Rohtak, Simla, Jálandhar, and Hushyárpur. In almost all these districts their use might have been, to some extent at least, avoided by raising the block to a normal size. As regards the use of non-official agency in general, in Kángra only 73 out of 3,098 enumerators and in Hazára only 43 out of 1,361 were official, while in Ráwalpindi and Jáhlam some thirteen-fourteenths of the enumerators were non-official. In Simla, Gurdáspur, Siálkot, Gújrát, Sháhpur, and Muzaffargarh, all with the exception of Simla districts where the blocks were unnecessarily small, the proportion of non-officials is large; while in Kohát it sinks to less than half, and in Gurgáon, Karnál, Lúdhiánah, and Bannu it is only about two-thirds.

856. Characters used by the enumerators.—Abstract No. 145 on page 434 shows the characters written by enumerators.

[1] Major Nisbet submitted no application for a grant. Under the peculiar circumstances of the Simla district, such an application would probably have been sanctioned if he had sent one up.

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Abstract No. 145, *showing* Characters written by Enumerators *for* Districts.

Serial No.	Name of District.	TOWNS. Number of Enumerators writing						VILLAGES. Number of Enumerators writing						TOTAL. Number of Enumerators writing						Name of District.	Serial No.
		English.	Persian.	Nágri.	Mahájani.	Gurmukhi.	Total.	English.	Persian.	Nágri.	Mahájani.	Gurmukhi.	Total.	English.	Persian.	Nágri.	Mahájani.	Gurmukhi.	Total.		
1	Dehli	3	646	4	...	...	653	...	1,006	282	...	...	1,288	3	1,652	286	...	...	1,941	Dehli	1
2	Gurgaon	...	115	...	...	...	115	...	1,077	229	1	...	1,307	...	1,192	229	1	...	1,422	Gurgaon	2
3	Karnal	...	171	9	...	...	180	...	760	292	...	...	1,052	...	931	301	...	...	1,232	Karnal	3
4	Hissar	...	177	4	...	...	181	...	469	559	22	...	1,050	...	646	563	22	...	1,231	Hissar	4
5	Rohtak	...	233	16	1	...	250	...	634	284	216	...	1,134	...	867	300	217	...	1,384	Rohtak	5
6	Sirsa	...	66	9	...	...	75	...	487	111	115	...	713	...	553	120	115	...	788	Sirsa	6
7	Ambala	...	319	...	...	...	319	...	2,004	36	...	...	2,040	...	2,323	36	...	...	2,359	Ambala	7
8	Ludhiana	1	294	...	...	...	295	...	1,405	127	...	...	1,532	1	1,699	127	...	...	1,827	Ludhiana	8
9	Simla	4	114	...	...	...	118	...	37	47	...	...	84	4	151	47	...	...	202	Simla	9
10	Jalandhar	2	412	...	...	...	414	...	1,767	...	...	...	1,767	2	2,179	...	...	...	2,181	Jalandhar	10
11	Hushyarpur	...	285	...	...	...	285	...	2,738	485	5	40	3,268	...	3,023	485	5	40	3,553	Hushyarpur	11
12	Kangra	...	66	21	...	...	109*	...	301	375	52	1	3,098*	...	427	396	52	1	3,207*	Kangra	12
13	Amritsar	3	438	...	...	...	441	...	1,602	...	...	...	1,602	3	2,040	...	...	...	2,043	Amritsar	13
14	Gurdaspur	6	338	...	...	...	344	...	2,229	28	1	2	2,260	6	2,567	28	1	2	2,604	Gurdaspur,	14
15	Sialkot	10	237	...	...	...	247	...	2,171	...	...	...	2,171	10	2,408	...	...	...	2,418	Sialkot	15
16	Lahore	7	436	...	...	...	443	...	1,331	...	...	...	1,331	7	1,767	...	...	...	1,774	Lahore	16
17	Gujranwala	1	218	...	...	...	219	...	1,655	...	...	...	1,655	1	1,873	...	...	...	1,874	Gujranwala	17
18	Firozpur	10	144	...	...	...	154	2	1,340	16	77	13	1,454	12	1,490	16	77	13	1,608	Firozpur	18
19	Rawalpindi	22	261	...	...	1	284	...	2,792	...	13	11	2,816	22	3,053	...	13	12	3,100	Rawalpindi	19
20	Jahlam	...	135	...	...	...	135	...	1,525	3	...	...	1,528	...	1,660	3	...	...	1,663	Jahlam	20
21	Gujrat	...	130	...	...	...	130	...	1,664	...	...	...	1,664	...	1,794	...	...	...	1,794	Gujrat	21
22	Shahpur	...	163	...	...	...	163	...	1,023	...	...	...	1,023	...	1,186	...	...	...	1,186	Shahpur	22
23	Multan	5	201	1	...	...	207	...	1,300	...	59	1	1,360	5	1,501	1	59	1	1,657	Multan	23
24	Jhang	...	84	...	...	...	84	...	880	...	...	..	880	...	964	...	...	...	964	Jhang	24
25	Montgomery	...	69	...	...	...	69	...	937	...	...	...	937	...	1,006	...	...	...	1,006	Montgomery	25
26	Muzaffargarh	...	63	...	...	...	63	...	1,430	...	...	...	1,430	...	1,493	...	...	...	1,493	Muzaffargarh	26
27	Derah Ismail Khan	...	200	...	...	...	200	...	1,154	...	190	...	1,344	...	1,363	...	190	...	1,553	Derah Ismail Khan	27
28	Derah Ghazi Khan	1	134	...	...	...	135	...	662	..	17	...	709	1	826	..	17	...	844	Derah Ghazi Khan	28
29	Bannu	...	55	...	...	...	55	...	589	...	...	...	589	...	644	...	...	...	644	Bannu	29
30	Peshawar	27	364	...	...	...	391	...	1,109	...	16	14	1,139	27	1,464	...	16	14	1,521	Peshawar	30
31	Hazara	4	58	...	...	...	62	...	1,335	...	20	...	1,361	4	1,393	...	20	...	1,423	Hazara	31
32	Kohat	...	35	...	...	...	35	...	452	...	...	...	452	...	487	...	...	...	487	Kohat	32
	Total	106	6,760	64	1	1	6,954	2	39,952	2,874	810	82	46,029	108	46,712	2,938	811	83	52,983	Total	

* Note.—Including 2,331 writing Pahari (2,309 in villages and 22 in towns). The schedules of Spiti were written in Tibetan and then translated.

Part I.—Preliminary Operations.

The figures are summarised in the margin. Of these characters, the only ones that are fairly constant in their character and at the same time generally intelligible are the three first. Mahájani or Landa, the character in which the shop-keepers keep their accounts, is quite impracticable: and it would have been difficult or impossible to find men to read the Gurmukhi or Pahári cursives. Consequently except in Kángra, where the proportion of Pahári writers was necessarily exceedingly large, and where a staff of patwáris who could read the character was available for employment in tabulation, no schedules were allowed to be sent to the compiling offices in any one of these three characters. Every effort should be made to avoid enumerators writing any other character than English or Persian. The Deputy Commissioners report very generally that the use of other characters caused the greatest difficulty, the enumerator being unable to read the instructions and the supervisors to read the schedules, while the last were generally exceedingly illegible even to the man who wrote them. By far the best plan, where the enumerator wrote an inadmissible character or one not understood by the supervisor, was to write the original entries in the schedules very far apart, and to fill in an interlinear transliteration. The supervisor then repeated in the Persian text all alterations or additions made by the enumerator in his original, and the former alone was used for compilation; the latter however being immediately available for comparison in case of doubt. Where this plan was not followed it was necessary to make a duplicate copy of the book of schedules in the Persian character, and this greatly increased not only labour but also risk of error.

	Villages.	Towns.	Total.
Enumerators writing English	2	106	108
„ „ Persian	37,643	6,738	44,381
„ „ Nágri	2,874	64	2,938
„ „ Mahájani	810	1	811
„ „ Gurmukhi	82	1	83
„ „ Pahári	2,309	22	2,331

The Pahári character was used in Kángra alone; the Gurmukhi in Hushyárpur, Fírozpur, Ráwalpindi and, curiously enough, Pesháwar. The Mahájani was chiefly made use of in Rohtak, Sirsa, the Bhakkar and Leiah tahsíls of Derah Ismáíl Khán, and to a less extent in Kángra, Fírozpur, and the Mailsi tahsíl of Multán. Nágri was employed most largely in Hissár; but generally to a considerable extent throughout the Dehli and Hissár divisions and in Sháhpur Kandi of Gurdáspur, and not at all in the western parts of the Province. Here again characters other than Persian were often used in districts where the average block was unnecessarily small; but in some cases no doubt, the two phænomena are both due to a scattered and illiterate population inhabiting an outlying tract difficult of access. It is not impossible that some of the men entered as having used the Nágri character may really have written in Mahájani, as the latter is sometimes, but incorrectly, called Nágri. Schedules were printed in Persian, Nágri, and Gurmukhi (the last at the Lúdhiánah Mission Press). Such as were used in Mahájani and Pahári had to be prepared by hand in the districts in which they were used, as the character varies from district to district.

The highly educated enumerator is a mistake; and though it is necessary that the supervising staff should be men of education, yet a great deal of trouble was given and inaccuracy caused by their pride of learning. An educated Hindu or Musalmán has strong theories, derived from the scriptural authorities of his religion, as to how things ought to be; and is not content to record them as they are, or to believe that each man himself knows better than any outsider, however greatly his superior in education, what are his caste, clan and religion. Mr. Stogdon of Lahore says:—

"A considerable amount of tact was needed in dealing with such a large number of voluntary enumerators, many of whom "considered that they were conferring a favour on the authorities by undertaking the work, and looked upon themselves as abler "men than their supervisors and superintendents. The most difficult kind of enumerator to deal with was the self-opinionated one "who worked on a method of his own and was above instruction or correction. In a few instances it was necessary to dispense "with the services of such men altogether. They were worse than useless."

A flagrant instance of the harm which educated men may do with the best intentions is quoted in section 37.

857. Progress of the work.—I took up my appointment as Deputy Superintendent of Census on the 4th August 1880, and the preliminary Government circular issued on the 7th of the same month. My instructions for the demarcation of Census units and the selection of agency were dated the 31st of August; while instructions for the preparation of house-lists were issued on the 15th, and for the preliminary record on the 23rd of September. The preliminary record began on the 27th of December 1880, and was completed during January 1881. The instructions for the final enumeration, which could not be issued till the Government of India had issued orders for the Railway and Military Census, were dated the 20th January 1881, and the Census was taken on the night of the 17th of the following month. The record of progress is continued at the end of Part II of the next Chapter.

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PART II.—THE ENUMERATION OF THE PEOPLE.

858. The form of schedule.—The form of schedule prescribed by the Government of India for use throughout the Empire was as follows :—

*District*_____ *Sub-Division** _____ *Village*_____ *Name of Enumerator*_____

*No. of House*_____

1	2	3	4	5	6	7	8	9	10	11	12
Serial No. of each inmate.	NAME.	Condition, *i. e.*, whether married, unmarried, widow, or widower.	Sex.	Age last birth-day.	RELIGION. Religion.	RELIGION. Caste if Hindu, Sect if other religion.‖	Mother tongue.	†‡§ Place of birth.	¶ Occupation of men, also of boys and females who may do work.	*Education.* 1. Under instruction. 2. Not under instruction but able to read and write. 3. Not under instruction and not able to read and write.	*Infirmities.* 1. Unsound mind. 2. Deaf mutes from birth. 3. Blind. 4. Lepers.

* Title to be varied according to circumstances.
† If a European British subject, add British subject.
‡ If a foreigner, state of what country.
§ If of mixed European and Asiatic parentage, add Eurasian.
‖ In this column such details of tribes, clans and septs, as the Local Government may direct, should be introduced.
¶ Boys at school, girls, small children, and women who perform no regular work should not be shown at all under col. 10.

The form finally adopted for the Panjáb is given below :—

*Tahsil*_____ *Village* *Circle No.*_____ *Block Letter*_____ *Name of Enumerator*_____

*House No.*_____ *Description.*_____

1	2	3	4	5	6	7			8	9	10	11	12
Serial No. of Inmate.	NAME.	Condition, *i. e.*, whether married, unmarried, widow or widower.	Male. Female.	Age last birth-day.	RELIGION, CASTE OR TRIBE, CLAN, GOT, &c. Religion and sect.	Original Caste or Tribe.	Clan.	Got or Family.	Mother tongue.	*†‡ Place of birth.	Occupation of men; also of boys and females who work as makes whose work is brought regularly to market.	*Education.* (a) Not under instruction but can read and write. (b) Under instruction.	*Infirmities.* (a) Unsound mind. (b) Deaf mutes from birth. (c) Blind. (d) Lepers.

* If a European British Subject, add British subject. † If a foreigner, state of what country. ‡ If of mixed European and Asiatic parentage, add Eurasian.

As the Panjáb schedules were roughly stitched up into books, the name of the district was transferred from the schedule to a label on the cover with the following headings: Name of District and Tahsíl, No. of Circle, Letter of Block, Name of Enumerator, No. of pages in the Book, Signature of Supervisor. I think that there was no need to repeat the names of the Tahsíl and the enumerator on each schedule; and that the name of the village should have been shown on the label as well as on each page.

859. The original proposal of the Census Committee was to devote a page to each family (or, as they called it, house). In the Panjáb, however, the schedules were filled in straight away, a short line being drawn below each family in the column of names to facilitate checking, and a space being left at the end of each house in which to enter any newcomer on the night of the Census. Under these circumstances I think the decision to write the No. and description of house across the middle of the page was a mistake. It led in some cases to the entry being omitted altogether for all houses save that at the top of each page, and in others to space being wasted by a whole opening being devoted to the single house entered in the top heading. I would add a column at the beginning of the schedule, headed "No. and description of House."

Column 2, Name.—In column 2, Name, the directions were to write simply the word "female" or "infant" where there was any objection to giving the name, or where a child had not yet been named. It was found in Peshawar that this caused difficulty in checking where there were several women so entered. I think it would be better to write "wife," "daughter," "daughter-in-law," &c., as the case might be, giving the woman's relation to the head of the family. There would, I conceive, be no objection to give this information. Mr. T. W. Smyth of Delhi would record the relationship of each person enumerated to the head of the family. He writes :—

"It might be an improvement to have an additional column in the schedule showing the relationship of the several members "to the head of the family, such as wife, son, daughter, mother, and so on. This would assist the enumerator when recording the "names, and would prevent omissions which are likely to occur if the names are not taken down in some recognised order. The "enumerator's first question should be 'Who is the head of this family?' This would be followed by 'Is he married, and if so "what is the name of his wife?' 'Has he any sons, if so what are their names?' 'Has he any daughters, if so what are their

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Part II.—The Enumeration of the People.

"names?' 'Are there any other relations residing in the family, mother, brother, sister, &c.; if so, what are their names?' "Are there "any visitors in the house, if so what are their names?' and so on. If the family is a large one the names of some of the members "are likely to be omitted if they are simply taken down, without any questions being put by the enumerators similar to those "noted above."

What he says is doubtless true; but I fancy there might be some difficulty in recording relationship generally, especially where concubines are kept; it is on general grounds well to keep the number of columns to be filled up as low as possible; and I would confine the entry of relationship to the cases where the name was not given.

Column 3, Civil Condition.—In the column for civil condition we had to translate "Married or unmarried" by *kwárá yá kwárí* (virgin), and frequent references were made as to how unmarried prostitutes were to be entered. I cannot suggest any better translation; but I think the point might be noticed in the instructions to enumerators. Captain Bartholomew suggest *ná-biáhí*, a mongrel word not in use. Mr. Perkins very justly remarks "here strict etymology yields to custom. The use of *kwárí* for an un- "married prostitute is not more absurd than the usage whereby a married girl of 5 years old is *not* a *kwárí*."

Column 4, Sex.—The column for sex was split up into two columns headed male and female respectively. The entry was made by simply writing the word *ek* (one) in the appropriate column. But I think it would have been better to have a single column headed "male or female," and make the enumerator write the proper word opposite each entry. The former plan saved a little trouble in compilation; but we found that in many cases the word "one" had been written in the wrong column, as was apparent from the name of the person. There would be much less risk of the wrong word being written. It would have been well to state in the instructions to enumerators that eunuchs were to be shown as males. The native does not apply the word *mard* to them, and much doubt was felt in the matter. The Patiála State added a third sub-column for them under the head of sex.

Column 5, Age.—By some extraordinary perversity of circumstances, we were instructed to record the number of years completed by each person on the birthday immediately preceding the Census. In the first place this upset the understanding of every villager of whom the question was asked; for every native calls himself 22 years old directly after he has completed his 21st year. In the second place we were directed, in the case of infants under one year of age, to enter their age in months, though no detail of age involving months was tabulated. In the Panjáb we ordered the number of months to be written in words instead of numbers, with the word "months" after them; but it is no doubt possible that the number of months was in some cases entered or read as years. The age on the *next* birthday should on another occasion be entered in the schedules; and if for instance it is wished to make the figures in the column of the age tables headed "5 to 9" include those who have finished their 5th, but not completed their 10th year, all that would be needed would be to tabulate the schedule entries from 6 to 10 instead of from 5 to 9, as would be done in England, and enter the result. This would enable children under one year of age to be entered as one year old, and *so prevent the possiblity of confusion between months and years.*

Columns 6 and 7, Religion and Caste.—These are the columns in filling up which the Panjáb enlarged most widely the Government of India scheme. The Government of India directed us to record the caste of Hindus and the sect of members of other religions. But caste, used in any wider sense than the somewhat mythical and practically obsolete fourfold division of Manu, is mainly a matter, not of religion, but of tribal descent or occupation. Under the instructions received, only 1,907,732 out of the 3,564,519 Jats and 325,215 out of the 1,436,058 Rájpúts in the British Panjáb, or just over half of the one and not one-quarter of the others, would have been shown as Jats or Rájpúts, the remainder appearing as Musalmáns of either the Sunni or Shíah sect. Accordingly we transferred sect to its proper place under religion. In another Census it should, I think, have a separate column or sub-column to itself. It is a great mistake ever to enter two distinct facts in one and the same column, for it leads to difficulty and error in tabulation, if not in enumeration.

Having thus left column 7 clear for caste or tribal divisions, we divided it into three sub-columns headed respectively "original "caste or tribe" (*asl qaum*), "clan" (*zát yá firqah*), and "*got* or sept" (*got yá shákh*). The whole question of the classification thus adopted, its merits and defects, and the best course to be adopted in future, is discussed in the Chapter on Castes at pages 188 to 190 of the Report. It was suggested that we might direct the enumerators to record only certain specified tribes and clans, selecting such as were administratively important. But I believe that such a course would be fatal to accuracy. It is true that we may hope to have before next Census classified tables of clans, from which it would be easy to select in each case those which alone appear of sufficient importance to be worth recording. But such a list would be far too bulky for practical use. Besides, I am certain it would be fatal to tell the enumerators to record clans in some instances and not in others. And a native who, if asked what his clan was, would give a perfectly correct answer, would, if asked to which of a certain number of specified clans he belonged, almost certainly name one of them, though he might really belong to none. The only practicable plan is that adopted at the present Census, namely to record everything, and to select and reject at the compilation and not at the enumeration stage.

Column 10, Occupation.—In the vernacular schedules this column was headed "Occupation of men, and of those boys and "females whose work contributes to the family income" (*Peshah mard báligh ká, wa un larkon wa larkion wa auraton ká ki jin ki mihnat se ámdani húá kartí hai.*) This was in accordance with the original instructions issued by the Government of India, but not with the amended form in which Mr. Elliott framed them (see section 718 *supra*), and which, though followed in our instructions to enumerators, was received too late to alter the heading of the schedule. I am of opinion that it would be far better to head the column with the single word "Occupation," leaving all details for the instructions. If it led to any error, it would only be the very unimportant one of recording an occupation which was not wanted, and which could be rejected at compilation. There is one strong objection to the use of the word *peshah* for occupation; namely that, where used with reference to females, it commonly means prostitution. But *rozgár*, which seems almost the only alternative, also means income; and the answer to "What is "your *rozgár?*" might be "Rs30 a month." Perhaps the phrase "*Guzára kis tarah se hotá hai?*" would have been better, especially as it was determined, after the schedules had been printed, to record source of income where the person had no occupation.

Column 11, Education.—In this column we had only two entries "Under instruction" (*likhna parhna sikhtá hai*) and "Not "under instruction but can read and write," (*likh parh jánta hai*). Where a person was neither literate nor under instruction the column was simply left blank; a better plan, I think, than entering him as "Not under instruction, and not able to read and "write," as proposed by the Government of India. The entries in this column are unfortunately few; and our plan saves much trouble in tabulation.

860. The enumerators' books.—The schedules were printed across half-sheets of royal Bally paper, the half-sheets being doubled across the middle and bound up into books measuring 12½ by 10 inches, the schedule being 12½ inches across, and the books opening like a banya's account-book, with one set of headings to each opening which ran parallel to the back of the binding. The bindings were simply two separate limp boards made of glazed calico pasted over country paper, and were made up at the Central Gaol, Lahore, at an inordinately high rate. This was a mistake. They could have been made by contract in the districts for half the money; and the heavy cost of carriage, often by post, would have been saved. But in any case sample-books should be sent to each Tahsíl, as was done in the Census, to show them how to make up their books; as incredible ingenuity is often displayed in misunderstanding written instructions. As for the form of book, I think it could hardly be improved. The loose schedules and covers were supplied to the supervisors, and they stitched up the books according to the size of their respective blocks. If it was found that there were not enough pages in the book, or that it contained spare schedules which were wanted elsewhere, the stitching was easily undone, schedules added or taken out, and the book stitched up again.

The plan adopted was to have one book for each block, the entries for different villages included in the same block being separated by a blank leaf. I think it would be better in another Census to have a separate book for each of two or more villages in the same block. It would save much trouble in tabulation

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Part II.—The Enumeration of the People.

and render error less likely to have every village with a book or books of its own. The label outside the book, described in section 858, would then show the name of the village.

Each book of schedules contained at the beginning a copy of the instructions to enumerators, a set of questions which the enumerator was to ask in each case, and three pages of sample schedules filled up by way of specimen. At the end two or three blank pages were kept in case of need on the Census night.

861. Preliminary practice of enumerators.—The blocks having been finally marked out, the houses numbered, the house-lists prepared, and the enumerators appointed, the next thing was to teach them their duties. This was effected by each supervisor collecting his enumerators at his head-quarters about a week before the date fixed for the preliminary record, and making them take a Census of a portion of the town or village, each man taking his share of the work. As each man showed that he thoroughly understood the instructions and method of work, he was sent back to his block to await the day for beginning the preliminary record. In some cases I found that the Tahsíldár had caused the enumerators to take a Census of the whole Tahsíl a month or more beforehand by way of practice. This ought to be strictly forbidden; both because it entails unnecessary labour upon the staff and unnecessary worry to the people, and because there is a great danger of the enumerators using the papers so prepared for their preliminary record instead of going over the ground again.

862. The preliminary record.—The instructions were that the preliminary record was to be begun on the 27th December and finished by the 25th January. It was to be finally tested and declared correct and complete by the 7th February, leaving 10 clear days for preparation before the Census night. I think the record might have been begun later. The curious effect upon the age-statistics of separating the mass of the entries from the date of the final Census by a considerable interval, has already been discussed in the chapter on Age. But besides this, the larger the interval which separates the preliminary record from the actual Census, the greater will be the number of alterations which have to be made on the latter date. Thus while I am convinced that for an Indian Census, it is absolutely imperative that the entries should in the first instance be made and tested at leisure, leaving for the night of the Census nothing more to do than to bring them up to date,—and this, not only because the type of intelligence enjoyed by the average enumerator is so low that most minute supervision and examination of the entries is necessary, but also because it would be impossible to obtain a sufficiently numerous staff to make the record *ab initio* in a single night,—yet it is on several grounds advisable to reduce the interval between the two operations as much as possible. Now a fairly active enumerator will complete the preliminary record of his block in three or four days. Supposing that my plan of making the patwáris prepare the record was accepted (section 851, *supra*), the outside that a patwári could possibly have to do would be ten blocks; and in that case, he being himself the supervisor, no time need be allowed for examination of the entries. I think that the best form the instructions could have taken would have been the following: the preliminary record not to be begun in any case before the 7th January, and to be completely tested and corrected in all cases by the 7th February; subject to this latter condition, the later it is begun the better. This would leave a certain discretion as to the date of beginning, and would allow sufficient time for practising the enumerators who had not themselves made the preliminary record. It is true that the preliminary record in Hissár was begun on 25th December and not finished till 10th February; but I am sure so long need not have been taken. In Sirsa it was all done within 18 days. Mr. Frizelle of Sháhpur writes as follows on this subject:—

"Indeed I may say that the great fault committed in this district was that the preliminary record and the testing of it by "supervisors was over too soon, so that before the final day came the ardour of the great body of the staff and the interest they took "in their duties had somewhat cooled down, they had probably forgotten a good many facts in the mean time, and more alterations "were necessary in the entries in the schedules than would otherwise have been the case."

And Mr. Lawrence of Peshawar says:—

"I would suggest that the preliminary record be completed in 10 or 15 days. In this district during Census operations the "drought continued. Had rain fallen in January the greater part of the preliminary record of the Hashtnagar, Mardán, and Utmán "Bolák Tahsíls would have had to be corrected, for the Gújar population would have left Hashtnagar for Yúsufzai. As it was the "preliminary record had to be corrected, for in the month allotted to this work there were many changes."

When the enumerator began the preliminary record he went round his block with the house-list in his hand, and entered all residents for the time being, whether permanent or visitors only (save in hotels, hostelries, encamping grounds, and the like, where only permanent residents were entered) for each house, and, within the house, for each family. I think it was a mistake to enter mere temporary visitors. The preliminary record must necessarily be separated from the final Census by an interval of some weeks; and the rule followed caused a good deal of needless excision on the Census night, while it greatly increased the danger of double enumeration. I would not enter visitors whose stay was not intended to exceed two or three days. In fact I suspect such persons were very often not entered at all, the rule to the contrary notwithstanding. Mr. Stogdon of Lahore notes that wedding processions present at the village at the time of the preliminary record were not entered.

Every house, whether empty or not, was entered in the book, a note being recorded where necessary that the house was unoccupied or only occupied by day, and a small space being left for possible inhabitants on the night of the Census. So where the residents were to be allowed to give in their own schedules, the house was entered and a note made to the effect that a private schedule had been sanctioned for it. The large number of shops, and the cattle-sheds and field-huts mentioned in section 841 wasted a good deal of room; but unless every house had been entered the house-list would have been no check upon the enumerator, and I think the plan adopted was the right one. The method of entry of houses and families is described in section 859.

While the preliminary record was in progress the supervisors were constantly moving about their circles, inspecting and testing the entries. When the supervisor had satisfied himself of the correctness of the entries on any page, he signed it at the top or foot. In some cases the enumerators were so incompetent that their schedules had to be wholly recopied after correction by the supervisor; and this was always

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done with any schedule which was much cut up by corrections, the stitching being opened and the schedule replaced by a new one. When the preliminary record was complete and correct for any block, the supervisor numbered each page above his signature, entered the total number of pages in the label on the cover, and signed the label.

863. Date of the final Census.—The precise date of the final Census was fixed by the Government of India with reference to the various conditions of the several provinces of the Empire. They determined that it should be some time in February; the necessity for a moonlight night confined the choice within still narrower limits; and the consideration which dictated the choice of the 17th as the night of the Census was the advisability of avoiding any days on which large masses of people would be assembled at fairs, where their enumeration would have been most difficult. In the Panjáb the only fairs of any importance that took place between the 10th and 18th of February are shown in the margin.

11th—Dehli	. 50,000;	Siálkot.	. 4,000
12th—Dehli	. 25,000;	Siálkot.	. 13,000
13th—Dehli	. 3,000;		
14th—Dehli	. 3,000;	Amritsar	. 15,000

But it may be noted that, supposing it to be impossible to avoid a date on or about which large fairs were held, it would probably be better to take the actual date of the fair itself than the day immediately preceding or following it. The difficulties of enumerating a large number of people collected at a fair would doubtless be great. But they would not be insuperable, and they would at least be known exactly, and could be provided for beforehand. But on the night before or after a large fair there would be a great number of travellers scattered about the villages and roads in the neighbourhood; it would be quite impossible to foretell where they would be found; and local agency might be insufficient to meet the unexpected demand upon it, while there would be a danger of such travellers escaping enumeration altogether. Mr. Hawkins of Amritsar writes:—"A large local fair was to take place in the village of Khatra in the Ajnála "Tahsíl on the night of the 17th; but it was put off to the 28th by order of the Deputy Commissioner, and "no difficulty whatever was experienced in the matter, as timely notice had been sent out to the "neighbouring Districts of Lahore, Síálkot, and Gurdáspur, from which the people come to the fair." This procedure might usefully be adopted in other cases if necessary.

But there is another consideration which apparently escaped notice altogether. Certain days are *lagan*, or auspicious dates for the celebration of Hindu weddings. They vary much in the degree of pre-eminence attached to each. But it is evident that, were a first-class *lagan* chosen for the night of the Census, a considerable proportion of the Hindu and Sikh population of the Province might be absent from their houses at wedding processions; and it would be necessary to cut their names out of the preliminary record, and add them in the schedules of the place where the wedding was taking place. This consideration is not so important in the Panjáb as in the North-West Provinces, first because a much larger proportion of the population is sufficiently Islamized not to attach any importance to *lagans*, and secondly, because weddings are forbidden for three only out of the twelve months in the greater part of the Panjáb, instead of seven as in the North-West Provinces. But it is a point which should be borne in mind on future occasions. So again, Thursday is a usual day for Mahomedan marriages, and some Thursdays especially; and the day before or after a *lagan* is almost as objectionable as the *lagan* itself, as wedding processions often have to make more than a day's journey. In the Dehli territory a good deal of trouble was caused by wedding processions on the Census night.

Mr. Benton of Karnál remarks that in Pánipat much of the work was finished before the moon rose (8° 39′ P.M. local time); and that a day on which the moon rose earlier would have suited better.

It may be noted that, in the vernacular instructions, the time for the Census should be described as "the 17th of February, after the lamps are lit," and not as "the night of the 17th of February," which many natives would understand to mean what we call the night of the 16th.

864. The final Census.—During the ten days which intervened between the 7th and the 17th February, the Census staff were busy in making preparations for the final event. Enumerators who had not themselves made the preliminary record had to be taken over their blocks in detail; supervisors had to satisfy themselves that the enumerators understood exactly what was required of them; red ink had to be distributed through the supervisors to the enumerators; proclamations had to be issued; and arrangements (to be described hereafter) had to be made for the Census of nomad population, and of travellers by road and river. The supervisors had to satisfy themselves, during the 15th and 16th and 17th, that every enumerator was at his post, and had his preliminary record complete and everything in readiness for the final Census. Immediately before the Census a proclamation was issued requesting the people to go home early and stay there as far as possible on the night of the 17th; and directing the head of each family to remain awake and keep a lamp burning till the enumerator had visited him. Village headmen and watchmen were also warned that they must accompany the enumerator on his rounds.

On the 17th of February, as soon as the cattle had gone home from the fields and the lamps were lit, the enumerator went out on his rounds to visit every house in his block. After he had finished them, he examined the space round the village, and all spots where travellers were likely to be encamped. During the day he had ascertained whether there were any gipsies or squatters camped in the area of his block; and if so he had to visit them also. The cardinal rule for his guidance was, that he was not to alter any entry made in the preliminary record save by scratching out the whole of the entries appertaining to persons who had died or gone away. This was absolutely necessary, as all the entries had been carefully tested by the supervising staff, and an ignorant and conceited enumerator might have played great havoc with them. He was also, of course, to add anybody who had been born or come to the place since the preliminary record was made. It would be interesting to know how many excisions and additions were made on the night of the Census; but this information has not been collected.

Newcomers were added at the end of the house, when they were staying in a house already entered, without reference to the family to which they belonged. To have left a blank space for their entry at the

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Part II.—The Enumeration of the People.

end of each family would have wasted much room and served no useful purpose. On this subject Mr. Wilson of Sirsa writes:—

"I think that too much importance is given to the 'house,' which is often an accidental conglomeration of families, and too "little to the 'family' which is bound together by natural ties of great force. For instance if a baby was born after the preliminary Census and before the 17th February, it was then entered at the end of the house, perhaps a long way from the entries "relating to its father and mother and below entries relating to a totally distinct family. I would class individuals into families, "and for this district at least would not attempt to class them into houses."

I agree with him as to the relative statistical importance of houses and families. But the record of families is taken from the entries of the preliminary record; and the addition of the baby at the end of the house did not affect the figures. Moreover, I propose (in section 844) to take these figures in future from the house-lists, and not from the schedules. If however it is desired to take statistics for families from the schedules, it would be well to direct the enumerator, in cases where whole families had come to the house since the preliminary record, and especially in hostelries and the like, to separate off such families from isolated individuals in the new entries made on the Census night by two lines across the name column, just as families are divided off at the preliminary record. This would make the record of families more accurate; but the primary object of distinguishing families in the schedules is, not to obtain statistics, but simply to facilitate checking.

All excisions and entries made on the night of the Census were made in red ink. Mr. Elliott feared that some difficulty would be experienced in supplying this to the enumerators; but this was not found to be the case. The common magenta aniline dye (*shíshah ká rang*) is cheap, sufficiently permanent for the purpose, goes a long way, and is procurable in the bazaar of any considerable town. One Deputy Commissioner estimated for some hundreds of rupees worth, enough to dye all his enumerators scarlet; but this was unnecessary. The colour was distributed through Tahsíldárs to the supervisors and by them to the enumerators. And the use of red ink on the Census night was exceedingly valuable, as it enabled the supervisors to distinguish at once the entries of the preliminary record, which he had already tested, from those of the final Census which he had still to test, and prevented enumerators from playing tricks with the entries.

865. It was proposed, at a late stage in the operations, to instruct the enumerators to distinguish residents from visitors in the schedules. The late date of the proposal would of itself have prevented its adoption in the Panjáb. But in any case I do not think it a good one. The permanent population of any town or village=its population on any night+absentees on that night—visitors on that night. Now, though it would be easy enough to distinguish the visitors, yet it would be practically impossible to obtain any record of the number of absentees under any system compatible with a synchronous Census, in which each person is enumerated wherever he may happen to be on a given night. In large towns and even in villages, save under special circumstances such as the occurrence of a fair, the visitors and absentees would be about equal in number at any given time; and the numbers actually present on the Census night would probably give a far more accurate idea of the normal population of the town than would those numbers after the exclusion of visitors. If it were really desired to obtain an enumeration of the permanently resident population of each place, the proper plan would I think be, to accept my proposal that temporary visitors should not be entered at the preliminary record (see section 862), and further to direct that all persons only temporarily absent should be entered as if at home in the preliminary record. These last would of course be cut out again at the final Census if they had not returned meanwhile; but if the number of excisions and additions made in red ink on the Census night were tabulated, they would give by a very simple sum the permanent population as it stood at the time of the preliminary record, births and deaths occurring between that time and the final Census not being taken into account. The only danger would be that the names of persons temporarily absent at the preliminary enumeration might be left standing at the final Census, although they had not returned meanwhile.

On the night of the Census the supervising staff were strengthened wherever possible (chiefly, I suspect, in the towns) and each enumerator was visited while at work wherever distance would allow of its being done. Where this was impossible, the supervisors were instructed to direct their attention to the larger villages with miscellaneous populations, and to those on frequented roads where many travellers might be expected to stop. Next morning they started off to go round their circles and test on the spot every excision or entry that had been made on the Census night. They were allowed to cut out or add any person who appeared from enquiry to have been wrongly left in or omitted; but they were not to alter the details recorded concerning any person rightly entered, and who was not before them, unless they could make perfectly sure of what should have been recorded.

This completed the operations, so far as they concerned the enumeration of the people.

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PART III.—SPECIAL MEASURES.

866. Municipalities.—The Census of the Municipalities, which embrace almost all the towns with more than 5,000 souls, and many with smaller populations, was taken through the Municipal Committees. In the large Municipalities the arrangements were most thorough and complete; and in several cases advantage was taken of the opportunity to prepare permanent maps and house-registers. The cost of the Census up to the time that the books were sent to the tabulating offices was borne by the municipal funds in all Municipalities of the first and second classes, and in such of those of the third class as had at the Municipal Census of 1875 a population of not less than 10,000 souls; forms and instructions, however, being supplied at Government expense. But even here, where funds were available, it was not found necessary to employ paid agency save for the preparation of house-lists in the city of Dehli, and for the whole operations in the city of Amritsar. Agency was, of course, easy to procure; and the only respect in which the Census of the large cities was not specially accurate was the entry of religion, caste, and clan. Here the educated city enumerators knew far too much to record accurately; and from what I saw and heard, I suspect that the poorer classes in cities do often lose in a far greater degree than do those in the villages, their regard for and memory of caste distinctions. A not inconsiderable proportion of them often consists of a miscellaneous collection of low-caste men from the North-West Provinces who are indifferently classed as Púrbias by the people of the Panjáb; and they themselves seem in many cases to have grown to consider the classification as sufficient.

867. Cantonments.—The Census of cantonments was taken through the Cantonment Magistrate; and the cost was borne, as in Municipalities, by cantonment funds. The degree of success with which the work was done varied immensely, and depended wholly upon the character of the Cantonment Magistrate. Some of the very best work I saw was in cantonments. In other cases the Magistrate was lazy and indifferent, the greatest pressure had to be put upon him to induce him to move at all, and in one cantonment at least the Census would hardly have been taken at all if the district officers had not stepped in at the last moment and done the work themselves. In others again no care was taken in selecting the agency, no supervision of any sort was exercised, and the staff had to be threatened with criminal proceedings before anything could be done. The position is a difficult one. The Cantonment Magistrate is subordinate to the district officer only in his judicial and not in his executive capacity, while the Cantonment Committee which control his action in the latter capacity is of course wholly independent. But, as the appointment of the Magistrate rests with the Panjáb Government, I think on a future occasion a lazy and indifferent man might be given clearly to understand that the work had got to be done, and that he must do it, and do it properly. In Jahlam, Talágang, and Naushahra the whole work, excluding of course the Census of military limits, was done by the district staff with the consent of the cantonment authorities, who placed the cantonment staff at their disposal. Where this arrangement is agreed to nothing could be better. Messrs. Beckett and Lawrence of Peshâwar both recommend that either the work should be made over absolutely to the civil authorities, or that they should be relieved of all responsibility for it. Probably the best way would be to empower the Deputy Commissioner, in case proper arrangements are not made by the cantonment authorities, to do the work himself by paid agency and charge the cost to cantonment funds. The population of cantonment bazaars is more mixed than any other population in the Province, and its Census presents very peculiar difficulties as to the entries in some of the columns. It is, however, insignificant in extent.

868. Military Census.—It was, of course, absolutely necessary to take the Census of troops through their own officers. At the same time it was advisable to throw as small a burden as possible upon the military authorities. It was therefore decided that the Census of only "regimental lines, quarters, barracks, or other premises in which troops or others subject to military discipline are quartered" should be taken by the military authorities; "that of regimental and sadr bazaars, and of all parts of cantonments except "the quarters just mentioned, being taken by the civil authorities acting through the Cantonment Magis-"trate." Civil officers were directed to fix, in consultation with the military authorities, the limits within which each was respectively responsible for the Census. A proposal was made to define the several responsibilities of the two departments by making the military authorities answerable for the Census of certain specified *classes of persons*, and of them alone. And in the Railway Census I found a similar standard proposed in one case, whereby to distribute the responsibility for the Census among the several departments into which the railway staff was divided. I am convinced that any such rule must result in confusion and error. Where two persons have to take a Census side by side, the only safe plan is to fix *topographical boundaries*, and to make each responsible for the enumeration of all persons of whatever description found within his limits. At the same time, to reduce the risk of error, the military authorities were requested to clear all military limits, between sunset and sunrise of the Census night, of all persons not temporarily or permanently living within them; and this, I think, was useful.

869. The general plan of preliminary operations was somewhat simplified for the military Census. The rules on the subject will be found in Appendix D. Military quarters had been treated as houses in the general Census of cantonments, and entered as such in the house-lists [1]. This, as Mr. Elliott pointed out, was a mistake; and it would have been better to have treated them as blocks. It had not been decided, when I issued the instructions, within what limits the military Census should be confined; but in future it might be explained to the civil authorities that the military Census would be confined within the narrow limits already described, and that such limits should be treated as blocks for the Census of which the military authorities would be responsible. No separate index-register was prepared by the Military Department;

[1] Section 9, General Letter A, and Rule 6, Part I, Instructions to Supervisors, Appendix D.

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Part III.—Special Measures.

no maps were needed; and a simplified form of house-list was used. But of course the schedule and the instructions for filling it up were the same as those used elsewhere, and a preliminary record was made. As soon as the entries of the Census night had been finally examined and corrected, the schedules were made over to the district office, and the military authorities had no further concern with them. Troops on the march were censused by their officers, the schedules being sent, in the case of a detachment to the regimental headquarters, and in the case of a regiment to the station at which they were last cantoned. The schedules of a regiment on its return from service beyond the frontier, which was censused at Karnál, were included in the population in that place. This was done by order. I should have preferred to include them in the population of the nearest cantonment.

A European civil officer was appointed in each case to advise the military authorities, and to exercise such general supervision over their operations as was necessary to secure uniformity in the results. It was found exceedingly difficult to get the authorities to move in time. It was not that they were indifferent or neglectful; in fact as far as I can judge (and I have examined all the English schedules page by page) the Census of military lines was carefully taken; far more so than that of many cantonments. But the military Census was such a simple matter that the whole of the work might have been done within the week immediately preceding the Census night. Thus the preliminary arrangements were put off to the last. But printed forms and instructions had to be provided; and in some cases the indents for these had not been received by the beginning of February, and I had to estimate the probable requirements as well as I could, and send off such a supply as I thought would certainly be sufficient. I at least, if no one else, should have been saved much anxiety if the operations had not been so delayed, that not till the day before the Census did I receive news of the completion of the preliminary record for the Province.

870. The Khaibar troops.—At the last moment Mr. Elliott informed me that he had, in consultation with the Government of India, decided to include the troops and camp-followers then lying in the Khaibar Pass in the Census of the Indian Empire; as, though situated beyond British Territory, they formed an integral part of its population. I accordingly sent them schedules and instructions. When the schedules were returned to the tabulating office it was found that for the most part only four columns, namely those showing name, sex, religion, and occupation, had been filled in. The military authorities informed the Deputy Commissioner that they had authority for thus curtailing the entries; but, though I reported the fact, I am unaware what orders had been issued on the subject. In my tables the Khaibar has been treated as an eleventh division, and its people included in the population of the British Panjáb whence they were for the most part drawn. Where the entries in the schedules were left blank, the numbers are shown as "unspecified."

871. Public institutions and gangs of workmen.—The Census of all public institutions, such as gaols, police-lines, dispensaries, hospitals, asylums, and the like, and of gangs of workmen employed on railways, canals, &c., was taken through the officers in charge. They were treated as single houses in the house-lists [1]. Mr. Elliott was of opinion that they would better have been treated as blocks; but I think that except where very large, they were best treated as houses of which the occupier filled up his own private schedule. Schools and other similar institutions, and police, canal, forest, customs, and other outposts were treated in a similar manner. Gaols [2], outposts, and similar buildings were entered as houses in the general house-list of the blocks in which they were situated, and their resident families recorded therein. Separate house-lists were prepared for gangs of workmen on large works. In both cases the preliminary record was deferred to within a few days of the Census on account of the unstable nature of the population.

872. Private schedules.—The Government of India directed that private schedules, to be filled up by the householder himself and not by the enumerator, were to be given only to "highly-educated people and members of the European community," as it had been found at previous Censuses that they were often filled in most correctly. They were also distributed to all schools, outposts, and similar houses mentioned in the preceding paragraph. They had instructions for filling them up printed on the back, and a specimen will be found in Appendix D. The number of houses to which they were given in each district is shown in Abstract No. 146 on the opposite page; but the figures are somewhat misleading, as in some districts the schedules given to police posts and the like have, contrary to my express instructions, been included in the return.

I found that Deputy Commissioners had not in all cases observed the direction to prepare lists of the houses for which private schedules were sanctioned at an early stage of the operations. This was a mistake; as till the enumerator has an extract from this list for his block, he cannot tell what houses he is to exclude from his enumeration. The schedules were distributed two or three days before the Census night, the headings having been first filled up in the district office. On the morning after the Census the enumerator collected all the private schedules for his block, carefully examining and correcting and completing the entries in each. He then stitched them into the end of his book, and the supervisor, after testing and signing them, added the number of new pages on the outside label. It was originally proposed that the householder should enter in his private schedule all persons residing in his compound. But at a late stage of the proceedings I succeeded in procuring sanction to the rule that he should enter only the residents of his bungalow, the native servants being enumerated in their outhouses by the ordinary enumerator of the block. This was, I am convinced, the right rule to work upon. It would be hopeless to expect an officer of a Queen's regiment or a European tradesman to enter correctly the particulars regarding his native servants. But of course the distinction should in future be drawn in the first instructions issued to enumerators. Several Deputy Commissioners bear testimony to the good effect which the rule had upon the correctness with which the schedules were filled up, and point out that the chief difficulty experienced was where officers in camp, for instance, had to fill up the schedules for Natives as well as Europeans. In many cases these had to be returned for correction or completion.

[1] Section 9, General Letter A, and Rule 6, Part I, Instructions to Supervisors, Appendix D.

[2] Section 11, General Letter B, No. 257 of 16th November 1880, Section 11, Part I, Instructions to Supervisors, Appendix D.

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Part III.—Special Measures.

Abstract No. 146, *showing* issue of Private Schedules *for* Districts.

1	2	3	4	5	1	2	3	4	5
Serial No.	Name of District.	No. of Houses to which private schedules were given.			Serial No.	Name of District.	No. of Houses to which private schedules were given.		
		English.	Persian.	Total.			English.	Persian.	Total.
1	Dehli	59	1	60	19	Rawalpindi	291	106	397
2	Gurgaon	14	13	27	20	Jahlam	40	9	49
3	Karnal	15	7	22	21	Gujrat	6	142	148
					22	Shahpur	7	2	9
4	Hissar	14	48	62					
5	Rohtak	10	30	40					
6	Sirsa	10	5	15	23	Multan	112	21	133
					24	Jhang	7	12	19
7	Ambala	196	...	196	25	Montgomery	7	6	13
8	Ludhianá	17	43	60	26	Muzaffargarh	7	19	26
9	Simla	206	...	206					
					27	Derah Ismail Khan	33	1	34
10	Jalandhar	60	12	72	28	Derah GhaziKhan	37	138	175
11	Hushyarpur	9	27	36	29	Bannu	29	33	62
12	Kangra	70	30	100					
					30	Peshawar	155	4	159
13	Amritsar	64	224	288	31	Hazara	46	85	131
14	Gurdaspur	71	58	129	32	Kohat	23	44	67
15	Sialkot	71	31	102					
						Total	1,915	1,263	3,178
16	Lahore	122	70	192					
17	Gujranwala	16	2	18					
18	Firozpur	91	40	131					

873. The private schedules issued to natives appear to have been as a rule correctly filled up. And it is curious how little desire seems to have been felt to claim the privilege of self-enumeration. In the whole of Dehli only one private schedule was asked for or granted to a native. In Lúdhiánah private schedules were offered to several natives and refused by them, as they preferred the enumeration to be made in the ordinary way; and in Gurdáspur these schedules were almost wholly confined to the district head-quarters, "where the native officials encouraged applicants." On the frontier, where the *pardah* feeling is exceptionally strong, one would have thought that the privilege would have been eagerly sought after. But in Hazára the Deputy Commissioner notes that the schedules arrived somewhat late, and meanwhile many of the houses had been censused in the ordinary way by enumerators without any sort of difficulty or objection. Mr. Dames of Derah Ismáíl Khán writes :—

"Private schedules were issued to European residents and to officers in camp. But few were issued to Natives as they expressed no desire for them. Most of the leading families reside in Derah Ismáíl Khán, and as they are well represented on the committee they preferred to have their Census taken through it, which was done without interfering with their privacy. One was issued to the household of the Nawáb of Tánk and one to Nawáb Alla-Dád Khán, Extra Assistant Commissioner at Bhakkar."

and Mr. Lawrence of Pesháwar says :—

"It was determined not to issue private schedules to any one in the district. We had taken measures to appoint men of family to act as enumerators for villages in which the Arbábs and Kháns reside, and the enumeration took place without any obstruction. In this district it would have been very hard to draw the line, and much discontent and ill-feeling would have resulted from using private schedules."

It almost seems that there would be no difficulty in refusing private schedules altogether to natives if thought advisable. At the same time the Deputy Commissioner of Síálkot is of opinion that their employment "lightened the work;" and Mr. Hawkins of Amritsar writes "the grant of private schedules saved "the establishment, and the entries made were as a rule correct. They might have been more largely "used with advantage."

But the English schedules were as a rule very badly filled up, especially by military and non-official Europeans. The Pesháwar authorities report most unfavourably of the inaccuracies which characterised these returns. Colonel Jenkins of Kángra writes :—

"The same cannot be said regarding the English schedules, the majority of which were wrongly filled in. At least this was the case regarding the schedules distributed in the Dharmsála Civil Lines and Cantonments, which were collected by Mr. Tucker. "It is perhaps to be regretted that those distributed to the planters scattered over the valley were not subjected to some sort of "scrutiny. It would have been easy for them to have been sent to the Sadr, and no great delay would have been occasioned thereby. Indeed though the greater part of the arrangements for the Census have been so elaborate and excellent, there appears to be considerable room for improvement with respect to Europeans.

"It must be remarked that the distribution of private schedules in no way lightened the amount of labour, but that the contrary was the case."

and Major Nisbet of Simla says :—

"In taking the Census of a considerable town like Simla, and that of the non-official European and Eurasian population of the hill cantonments of Kasauli, Dagshai, and Sabáthu, the number of private schedules issued no doubt appears very small. It will be remembered, however, that a Census of the town of Simla was taken so recently as 20th September 1878; and profiting by the experience then gained, I felt it was useless to trust many Europeans and Eurasians to furnish correct Census returns, if private schedules were made over to them to fill up. In the case therefore of a majority of families, the Census returns were filled up and prepared for them on the night of the Census by English-speaking enumerators specially engaged for the purpose."

It is, however, absolutely necessary that private schedules should be used for a large portion of the European population, as native enumerators would be treated by them with neglect, if with nothing worse. I shall presently make some suggestions on this subject.

874. Officers in camp.—Officials who were in camp on the night of the Census enumerated the whole of their camps in private schedules obtained from the nearest Deputy Commissioner. Their camps were held for Census purposes to be included in the block in which their head-quarters lay; and where two officers had a joint camp, the head-quarters of the senior officer were taken for this purpose.

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Part III.—Special Measures.

875. European Census in general.—I believe it is the experience of all Indian Censuses that the Census of the European community is much worse done than that of any other portion of the population, and certainly the Panjáb Census of 1881 was no exception to the rule. The mere enumeration was no doubt accurate enough. But it appears hopeless to expect the average European house-holder to read the instructions for filling up the schedules; and the proportion of Native to European population in any given block is so overwhelming that the enumerators and supervisors are necessarily chosen for their knowledge of the natives.

I examined every entry in every European schedule during the course of tabulation. I found few instances of silly or wilfully incorrect entries. But I found—

(1) that the sect, and the fact of being a European British subject were very generally omitted:
(2) that the nationality was often not rightly entered:
(3) that the name of some obscure English village was often given in the birthplace column:
(4) that the occupation was very generally insufficiently described.

The schedules returned by women were generally far more accurately filled up than those prepared by men. The ages of women were of course stated incorrectly. One lady returned her age as "over 30." I have no reason whatever to doubt the correctness of the entry; but it was not so precise as I could have wished.

I do not think it would be impossible to have, as suggested by Colonel Jenkins (section 873), all the European schedules examined immediately after the Census by a properly qualified European officer, and returned for correction where necessary. The largest number of Christians other than Native Christians in any single district is 4,018, and the greater part of these are soldiers, whose schedules can be run through very rapidly.

6		7	
Religion	Sect	Nationality	European British Subject.

I would also slightly modify the schedule for Europeans and Eurasians, substituting for columns 6 and 7 the columns shown in margin. As the Government of India schedule stood, the nationality and the fact of being a Eurasian or a European British Subject were to be entered in the birthplace column, which not unnaturally confused such people as read the foot-notes directing the entry.

876. Railway Census.—All Railway premises were from the first excluded from the block-system. It would I think have been better to have treated and entered the Railway premises in each Tahsíl as a separate circle, for the enumeration of which the Railway authorities were responsible. The Railway Census was divided into two distinct portions (1) that of the resident Railway population; (2) that of passengers by rail on the night of the Census. The rules on the subject will be found in Appendix D.

The Railway premises, for the Census of all persons residing within which the Railway authorities were responsible, included the roadway, all stations and signallers' and gatesmen's huts, all large quarters occupied solely by Railway employés such as exist at Lahore and Philaur, and all colonies which owe their existence entirely to the railway, like those at Adamwahán and Sher Sháh Bandar. In some cases it was found more convenient to enumerate isolated signallers and gatemen with the village near which their huts were situated; but in many parts, and especially where the Multán line passes through the *bár*, the distances were too great for this to be possible, and these men and their families were enumerated by an official who trollied down the line on the day of the Census; and they were assumed to be present at night.

The scheme followed was identical with that in general use, save that no house-lists were prepared. They were omitted under orders. But I think it was a mistake, as the Railway quarters at Lahore for instance include a very considerable number of houses; and in such places at any rate they should have been drawn up. Each station was treated as a block for the enumeration of which the station-master was made responsible, and at small stations a house-list was undoubtedly unnecessary. At large stations the premises were divided off into blocks, the responsibility for which was distributed among the various Departments. The schedules of each block, when completely tested after the Census, were sent to the Deputy Commissioner of the district in which the block lay.

The passenger Census was conducted on this wise. Between 9° P.M. on the 17th and 6° A.M. on the 18th, each traveller by rail was enumerated at the station at which he alighted. On the morning of the 18th all passengers found in the train, and who did not appear to have been already enumerated, were counted at the first considerable station at which the train stopped at or about 6° A.M. Private schedules were used for first and second class passengers, and for Europeans travelling third class. The few passengers who alighted at wayside stations were easily dealt with by the local staff. Where trains arrived at a terminus during the night, the clerks employed at the terminus during the day were kept on the spot to take the Census. Where the six o'clock Census had to be taken at a station which was not a terminus, sufficient clerks were sent there for the purpose. I saw the Census of the 11° mail taken at Lahore. Pairs of clerks, accompanied by a porter bearing a lamp, were standing on the platform as the train drew up. Each pair took a third class compartment, one taking one side and one the other, and beginning at opposite ends. As soon as they had finished their carriage its occupants were allowed to get out; and the clerks moved down the train to the nearest carriage which had no enumerators at it. The door of each carriage was kept locked till its enumerators were ready to begin work. In this way the Census of the passengers was taken quickly and well. The passenger schedules were kept entirely separate from those of the resident railway population, and were treated separately at tabulation.

877. The instructions for the Railway Census were not issued by the Government of India till the end of December 1880, which left us very scant time to make the necessary arrangements; and I did not hear of the completion of the preliminary record for the whole Panjáb railway population till the day of the Census itself. The arrangements on the Scinde, Panjáb and Dehli and Indus Valley Railways were admirable, and the Census of the small part of the Rájpútána State Railway lying within Panjáb limits was

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Part III.—Special Measures.

taken punctually and well. I cannot say from personal knowledge what arrangements were made on the Panjáb Northern State Railway, as the Chief Engineer neglected to report them notwithstanding constant applications on my part, and was reported to the Government of India for his neglect; but Mr. Gardiner, the Deputy Commissioner of Gújránwála, writes as follows:—

"The Railway Census was not satisfactorily performed, the necessary particulars having been omitted wholesale in very many "cases; and I was told that the special office which had been created to superintend the Census was abolished immediately after the "night of the Census itself, so that no arrangements were made to ensure that the schedules were promptly submitted nor was any "clerk kept to ensure their being submitted at all.

"The arrangements for the Railway Census should certainly be improved, and if no other more efficient supervision can be "found than that in force on the present occasion, District Officers should be authorised to control all Railway arrangements. In "many cases inefficient enumerators were appointed, and travellers detained for hours on arrival at their destination before they were "enumerated and allowed to leave the Railway premises."

The authorities of the Panjáb Northern State Railway seem to have been negligent and obstructive throughout the Census operations.

878. Boat Census.—At the original demarcation of blocks, every ferry, *ghát*, and mooring-place which was not so insignificant and so near a village that the enumerator could easily include it in his rounds, was marked off as a separate block. Where there was a bridge across the river or a mere ferry without a mooring place, the Census of both ends was taken by a single enumerator posted at the end where the bridge or ferry establishment was stationed; and this even where the ends lay in different districts or provinces, the population concerned being quite insignificant. In Dehli tickets were, at the request of the North-West Provinces authorities, given to all persons enumerated as they crossed the Jamna; but the precaution seems to me unnecessary, as the number of travellers must have been inappreciable. But where there were mooring places on both sides of the river, and no bridge, the enumerators were posted on each side. Each such enumerator went to his post at 9° P.M. on the 17th and enumerated all persons spending the night at the *ghát*, whether in boats or encamped. Where there was a bridge or a ferry, he stayed at his post all night and enumerated all travellers who did not appear from their statements to have been already counted at their homes. In any case he remained at the *ghát* during the whole of the next day, and enumerated all persons arriving in boats (but not any travellers by land who might cross over the ferry) unless they had a boat ticket or appeared to have been counted at their homes on the previous night. To each boat which he enumerated he gave a boat-ticket on red paper signed by him, and specifying the No. of the boat, the name of its master, the number of people enumerated, and the date and place of enumeration.

879. River turnpikes for boat Census.—The object of keeping the enumerator at his post on the 18th was to pick up any boats that might have escaped enumeration during the night by being moored or stranded at a distance from any *ghát*. This precaution was sufficient for the local traffic. But on the lower course of the Indus, Chanáb, and Satluj, a very considerable through traffic is carried on, the boats engaged in which carry a store of provisions and often do not put in at any mooring-place for several days. In fact on the Baháwalpur and Derah Gházi Khán portion of the Indus there is often no such place for many miles together. This traffic is smaller in the cold weather than at other times; but still sufficient to necessitate special arrangements being made, otherwise the boats engaged in it might have altogether escaped enumeration by being out in the main stream on the night of the 17th and during the whole of the 18th. The boat bridges at Kálábágh on the Indus, at Fírozpur on the Satluj, and near Jhang at the junction of the Chanáb and Jahlam, fixed the limits beyond which such traffic was impossible. Below these points the following stations were selected to serve as river turnpikes:—

On the Indus.—Mithankot, Koreshi, and Derah Ismáil Khán:

On the Chanáb.—Sher Sháh:

On the Satluj.—Bindrewála and Bhílá Ghát:

each station being arranged for by the Deputy Commissioner who managed the ferry at the spot. The Deputy Commissioner of Muzaffargarh proposed Khángarh Doma, below the junction of the five rivers, the Deputy Commissioner of Multán proposed Bandar ghát on the Chanáb, and the Deputy Commissioner of Montgomery proposed Chichawatni on the Rávi as additional stations; but I informed them that, while there was no objection to their extending the special arrangements to these places if they thought it advisable, it did not seem necessary to do so; and as a fact Bandar ghát was the only one of them at which a turnpike was established. At the stations thus selected, enumerators were posted and instructed to question all boats which passed their station up to the evening of the 28th February, and to enumerate all persons in them who did not appear to have been already enumerated elsewhere.

880. Travellers.—All travellers moving along main roads during the night of the 17th were enumerated at fixed points on those roads, turnpikes and police posts being utilised for the purpose. In this instance special permission was given to utilise the rank and file of the police for purposes of enumeration. People merely coming home from the fields along the road were of course left to be enumerated at their houses. The enumerators were especially instructed to visit all likely resting-places for travellers within their blocks after going the round of their houses, and to revisit all hotels, rest-houses, hostelries, encamping grounds and the like early on the morning of the 18th, to find out whether any unenumerated traveller had come in after they had gone their rounds. In the case of large city hostelries and encamping grounds it was ascertained early on the Census night whether an unusual number of travellers had arrived; and an additional enumerator was put on where necessary.

881. The numbers of people travelling by road in a February night is ordinarily so small that it was not thought worth while to organise a system of check-tickets for travellers; but the word of each man was to be taken, where he was not obviously answering falsely, as to whether he had already been enumerated elsewhere or not. The Deputy Commissioner of Jahlam, indeed, reports that above 3,000 travellers were enumerated on the Grand Trunk Road in the Jahlam Tahsíl alone; but it seems almost impossible that these figures should be accurate, as they would give one traveller to about every 20 yards of

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Part III.—Special Measures.

road. In the Dehli district check-tickets were used; but I doubt the necessity, though I do not know that there is any objection to the system beyond the additional complication it introduces. In the city of Amritsar every person coming out of the city after dark had to satisfy clerks stationed at the gates that he had been already enumerated; while persons entering the city were enumerated in the gate-schedule and given a ticket to prevent their being counted again. I think this last was a mistake, as persons absent from their houses for a few hours only were to be enumerated as if present there, and the plan adopted probably led to a good deal of double enumeration. In Simla the Census of people travelling by tonga or bullock train was taken before they started, and a special mark placed on the vehicle; and this was again checked at an intermediate point in the journey within the district. In Pesháwar one *jamadár* travelled up the road from Attak and another down from Pesháwar, taking the Census of the roadside posts, and at the same time enumerating travellers by *dák*.

882. Nomad population.—The great grazing grounds of the Western Plains, known as the *bár* tracts, are inhabited by nomad pastoral tribes who settle on a spot till the grazing within reach is exhausted, and then move their encampments to new grounds. These camp-villages (*jhoks* or *rahnas*) were allotted to enumerators, generally drawn from the establishment in charge of the grazing preserves, who made the preliminary record. A day or two before the Census the exact position of the camps was ascertained from the headmen or forest officials (who usually acted as supervisors), and the same enumerators, mounted on horses or camels, took the Census of their blocks on the night of the 17th without difficulty. The Deputy Commissioner of Montgomery proposed that mounted enumerators should sweep the *bár* during the daytime of the 17th, but this seemed unnecessary. The exact position of the camps can easily be ascertained, and in February the whole population is to be found within them at night. The similar camps of the *Pawindahs* in Derah Ismáil Khán called *Kirris* appear to be fixed and not moveable, and to have needed no special measures.

The enumerators were instructed to make particular inquiries from the village headmen as to the existence and locality of any temporary squatters, such as gipsies, hunters, travelling gangs of artisans, and the like within the limits of their block. Such encampments are always known to the villagers, as the squatters come into the village to buy food and make money.

883. The Census of Teri and Tanawal.—The Census of feudal Tanáwal in Hazára, held by the Nawáb of Amb and the Khán of Phúlra, and of the Teri jágír of the Khattak Nawáb of Teri in Kohát, presented some difficulties. In both cases the Chiefs were semi-independent, exercising all functions but those of police and criminal justice within their territories; in neither had any village survey been made, while men able to read and write were few and far between. In Teri the Nawáb was ready enough to do all that was possible, and only needed assistance, no revenue staff existing in his country. Accordingly a supervising agency at the cost of Rs. 900 was sanctioned, the cost of which in view of the unfavourable seasons which had preceded the Census it was thought advisable to throw upon Government. The actual expenditure was only Rs. 561, and was really a political rather than a Census charge.

In Tanáwal there was more difficulty. The Nawáb had raised objections in 1868, stating that Census operations would involve him in difficulties with his border villages. His objections had, however, been overruled. On this occasion he confined himself to inactivity, which was almost as formidable a difficulty as active resistance, as much of his Territory had never been visited by Europeans. The matter was neglected more than it should have been in the first instance. But eventually pressure was brought to bear by the Commissioner, and the Deputy Commissioner summoned the Nawáb's Tahsíldár for instructions to Abbotábád and sent his own Superintendent to supervise and inspect the work, while he himself marched through the territory. The Deputy Commissioner had the preliminary record sent him for examination before the final Census, and he reports that the work was in the end fairly well done.

884. Montane and sub-montane tracts.—I have already noted that the rule by which the division into blocks was made one of *area*, the collective blocks comprising the whole area of the district, was laid aside in the more mountainous portions of the Province. It would have been absurd to divide the snowclad and inaccessible sides of uninhabited mountains into blocks; but all the villages, hamlets, and outlying farms or huts were carefully allotted among the several supervisors and enumerators, so that there might be no doubt as to who was responsible for the Census of each. It was here that, as noted by Mr. Alex. Anderson, the Assistant Commissioner in charge of Kúlu, the block maps showing the position of homesteads and homes were specially valuable; indeed without them it would have been almost impossible to exercise any effective check, or even to ensure that the enumerator knew what he had to enumerate. I would insist upon these maps being prepared in hill-tracts at least, even if nowhere else.

The scattered nature of the hamlets made it in some cases impossible to confine the blocks within the limits of village boundaries. If, for instance, three outlying hamlets belonging to three distinct villages lay close together and at a distance from the parent villages, it was necessary to give them to one and the same enumerator. But in this case the books of schedules for the various villages or parts of villages were kept distinct, so that no confusion resulted. This plan was only had recourse to in a very few villages of Kúlu, Murree, and Sháhpur Kandi.

885. A still more serious difficulty was the snow, which in many parts of the Panjáb lies thick on the ground in the middle of February; even where the snow was not an obstacle, the scattered nature of the habitations in the hills, coupled with the special difficulty of obtaining competent agency in these backward tracts, made the distances which each enumerator had to travel necessarily great; while the spots which he had to visit were often difficult of access even in the daytime. Owing to these causes it was decided that in certain mountainous tracts it was impossible to take a Census at night, and the final enumeration was conducted by daylight on the 17th, and, when that did not suffice, continued to the 18th. The tracts to which this was extended were, Kágán in Hazára, including the valley of the Kúnhar itself, Bhogarmang, Upper Konsh, and the snow-clad parts of Agror; the Tanáwal hills, also in Hazára; the whole Tahsíl of Murree in Ráwalpindi and 50 villages of the Kahúta Tahsíl in the same district,

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including all north of the river Ling and some villages in the east of the Tahsíl in the mountainous country adjoining the river Jahlam ; the Sháhpur Kandi tract in the Gurdáspur district ; some few villages in Kúlu; that part of Seoráj included in the Mandi State ; and nearly the whole of the Chamba State. In these cases the question asked by the enumerator was, on the 17th, "Who will sleep at your house "to-night?" and on the 18th "Who slept at your house last night?" In these parts, and at that time of year, the population is very stationary ; and the departure from an absolutely synchronous Census which circumstances compelled us to make was one in name rather than as regards the facts recorded. In Chamba some few remote farms on different mountains were not visited at all by the enumerator, the heads of the families being summoned and made to give an account of their households.

In Kágán indeed, in Kothis Swar and Kohr in Chhota or Bír Bangáhal of Kúlu, in Mandi-Seoráj, and in Pángi of Chamba, it had been feared that the country would be wholly inaccessible during February ; and orders had been issued to take the Census at the latest safe date before the passes were closed by the snow, and after visitors who did not intend to winter there might be presumed to have left, and to revise the schedules immediately they opened again in the spring, and before people could have had time to enter ; recording in the first instance only those who declared their intention of wintering in the villages, and adding in the second instance only those whom local inquiry showed to have been present in them on the 17th February. But the season was exceptionally mild, and the Census was taken as in the other hill-tracts specified above, the only exception being that in Kágán and Tanáwal the houses were not numbered. The Deputy Commissioner reports that there was no time for this operation—why I know not. There was as much time here as in similar tracts in other districts ; though no doubt the difficulties of procuring agency, and in other respects also, were very great—perhaps greater than anywhere else in the Panjáb.

In Kúlu the Census was particularly well taken under the superintendence of Mr. Alex. Anderson. He found it impossible to prepare maps of circles and charges, as indeed no village survey was in existence. But, as he remarks, the village boundaries being strictly adhered to, it was easier to describe the boundaries of large areas in this mountainous country in writing than on a map. The block-maps however showing the position of homesteads and houses were prepared, and proved most invaluable. One village, Mulána, was isolated by snow before the Census ; so a good enumerator was sent to it before the pass closed, and remained there till it opened again. The Census of the great mass of the tract was taken at night on the 17th without difficulty, as was that of the Simla Hill States, including even the wild villages of Bashahr ; and this fact makes me doubt whether the license to take the enumeration by day was really necessary in all the tracts in which it was granted. However, as already remarked, the tracts were precisely those in which the point was most immaterial owing to the fixed nature of their population in February.

886. Census of Lahul and Spiti.—The Tibetan valleys of Láhul and Spiti, lying north of the great chain of the Mid-Himalayas, and approached by passes the lowest of which is more than 13,000 feet high, are wholly inaccessible during the winter months ; the valleys themselves contained no sufficient number of educated men to take the Census ; while, even if our establishment could have been induced to winter there, travelling would have been impossible. The Census was therefore, as in 1868, postponed till the passes opened in the spring. As soon as access was possible a portion of the Kúlu establishment was moved into the valleys, and the Census was taken in the ordinary manner, only those being enumerated whom local inquiry showed to have been in the valley on the 17th of February, and the work being pushed on vigorously so as to anticipate as far as possible the influx of summer visitors. The Láhul enumeration was finished by the 9th of June, having been supervised by seven men from Kúlu, but all the 34 enumerators having been Láhulis ; on the 16th of the same month the supervisors and 14 of the enumerators moved on into Spiti, where no local agency whatever was available ; and they completed their work by the 4th July. The season appears to have been an unfavourable one. One great difficulty was the prevalence of the Tibetan language, many of the schedules being of necessity written in that language and character, and having to be translated afterwards.

887. Other special cases.—Political considerations required that the Census of the village of Garhi Mián Zaid Gul in the Pesháwar district should be taken by two different agencies, the Census of the Mohmand population of the village being taken by the Deputy Commissioner of the district, while the Kohát authorities enumerated the Kalla Khel Afrídis. The schedules of the latter were however transferred to the Pesháwar office and incorporated with the other schedules of the village. The Census of the Biloch tribes of Derah Gházi Khán was taken through their tumandárs ; while the trans-border Wazíri nomads who were present in Kohát at the time of the Census were enumerated, more or less roughly I fancy, through their headmen. More elaborate arrangements would, it was feared, have alarmed them. As it was the Durkáni section of the Gurcháni Biloches in Derah Gházi Khán threatened to leave the district and return to their homes before the Census night, though they eventually thought better of it. A small outlying hamlet called Dhok Pathán lies within the confines of the Ráwalpindi district, while the main village of the same name, with the greater portion of its lands, is situated in the Talágang Tahsíl of the Jahlam district. Both in 1868 and now, the outlying hamlet was included in the Census of the main village. Strict adherence to territorial limits would have caused inconvenience ; and the population in question was inappreciably small. On the Pathán frontier it was found necessary to obtain from the Commissioner a ruling as to whether certain villages were or were not to be considered as within British Territory for the purposes of the Census, as "the border round this district (Peshawar) has never been "defined." I have been unable to obtain particulars of the line eventually adopted ; but any possible variation at future Censuses would be too small to materially affect the comparison with the present results.

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Part I.—Abstracting the Figures.

CHAPTER XII.
THE PREPARATION OF THE FIGURES.

PART I.—ABSTRACTING THE FIGURES.

888. Village Abstracts.—The first step after the complete testing and correction of the schedules was the preparation of the preliminary abstracts. These were generally prepared for villages by the patwáris, and in towns by the enumerators. Railway schedules were dealt with by the district staff, and military blocks by the cantonment office. A separate abstract was prepared for each block, and for each village included in the block. The utility of these abstracts is discussed in section 900.

Their form is given in general letter E. of Appendix D. It consists essentially of three parts. In the first are shown the number of families and of occupied and unoccupied houses, of males, females, married, unmarried, and widowed, of persons of each sex whose occupation is recorded, of persons who can read and write, persons under instruction, and persons suffering from infirmities. The numbers of houses and families thus arrived at furnished the statistics under these heads, as the figures were not again abstracted. The remaining numbers furnished a check upon the abstracting carried on in the divisional offices, the totals of which should correspond with the figures of the preliminary abstract. The figures for males and females also enabled us to publish the preliminary results of the Census far sooner than could have been done in any other way. The second part of the abstract showed no figures, but gave the *names* of the religions, sects, languages, birthplaces, and occupations entered in the schedules. These names were used to fill into the registers used by the abstracting staff of the divisional offices, so that they should find headings ready written for them when they came to abstract the figures into the registers.

The third part consisted of a classified abstract of all tribes and castes and their sub-divisions entered in the schedules, together with the number of persons in each. This was the most important part of the abstract. The other two parts could have been dispensed with, as all the figures with the exception of those for houses and families were got out again by an independent operation. But we accepted the figures of the third part as they stood, and thus saved an immense deal of labour. These figures, too, were prepared by men who, in the villages at least, knew every person entered in the schedules, and were intimately acquainted with the clans they belonged to. The importance of this has already been pointed out. In my instructions I left it to the supervisor to arrange whether he or the enumerator should prepare the preliminary abstracts, making him however sign each in attestation of its accuracy, and pointing out to district officers that in the villages the patwáris would be the proper agency. This was in accordance with the general tenour of the Panjáb scheme (see section 851). I believe that, like the preliminary record, these abstracts were as a rule prepared by the patwáris; but I think that on another occasion this should be made an invariable rule in the case of villages, each man doing his own circle; partly because the patwári could be held responsible for inaccuracies detected later on, which is impossible in the case of the non-official enumerator; but especially because intimate knowledge of the people enumerated is so important in the preparation of at least the third part of the abstract.

889. Transliteration and collection of schedules.—All schedules and abstracts that were written in any other character than English, Persian or legible Nágri were, save in the case of Kangra, where as already noted an exception was made, transliterated into the second of those characters unless an interlinear or other transliteration already existed, in which case it was carefully compared and checked. The enumerator himself in all cases read out his schedules to the transliterator. The schedules and their copies, if any, were then tied together; and as soon as the schedules and abstracts for his circle were complete the supervisor brought them, together with the house-lists, maps, and other papers, to the tahsíl, where they were compared with the index-register to see that they represented all the blocks and villages of the circle, and were examined to ensure that the series of pages were complete, that all the private schedules were stitched in, and that the total number of pages corresponded with the entry on the label. This work was intended to be completed by the end of February, and I believe that, except in the case of towns where there were difficulties, it was generally finished by then.

890. Tahsil abstracts.—Meanwhile the Tahsíldár had (or should have) prepared a tahsíl abstract in the form given in General Letter E. of Appendix D. This consisted of a list of all the blocks in the tahsíl, but arranged not by circles but by villages, the villages again being arranged in the fixed order of the revenue office according to which all village records are kept and village statements prepared. The names of the villages were entered, and opposite each the letters of the block in which it was included or the blocks into which it was divided, with their circle numbers, were filled in from the index-register. After the detail of villages followed similar details for all towns which, not being estates annexed to land-revenue, had no revenue number, all cantonments and civil lines, and all Railway stations. The former population of each village and town was entered in a separate column from the Census papers of 1868. As the schedules and abstracts for each circle were received in the tahsíl, the present population for each block was entered up from them and totals given for each town and village.

In Karnál all the schedules were examined as they were received "by the Tahsíldár and a few of his ablest assistants," and any imperfections were at once inquired into and removed where possible. This

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was an excellent plan and ought to be generally adopted on a future occasion; it being distinctly laid down however, that only omissions or obvious departures from the instructions are to be corrected, and that entries of caste, religion, or occupation are not to be meddled with because they do not fit in with the Tahsíldàr's theories on the subject.

891. Despatch of schedules to the Divisional office.—As soon as the figures for a circle had been entered up in the tahsíl abstract, the village abstracts and house-lists were put inside each book and the books were tied up together with the circle copy of the index-register. They were then packed up by complete circles and sent off to the divisional office, which all schedules should have reached by the 15th of March. The tahsíl abstract also was totalled and sent to the divisional office, together with the index register of the tahsíl.

This work, or at least so much of it as was done in the tahsíls, was worse done than any other portion of the Census work. The *figures* of the preliminary abstracts were fairly correct; quite sufficiently so for practical purposes. In only eight districts was the difference between the preliminary and final results as large as one-half per cent., and even this error was more often due to mistakes in the tahsíl than to errors in the abstracts. But the tahsíl part of the work was often carelessly and sometimes exceedingly badly done. The lists of blocks and villages were incomplete, the entries were wrongly made and totalled, the books of schedules were sent off anyhow, and paper after paper had to be returned for correction and completion. This carelessness did not affect the ultimate results; but it immensely increased the labour of arriving at them; and I allude to it only in order that it may be guarded against in future. I do not think it was unnatural. I have in a former chapter remarked that the Census operations were, to speak plainly, an unmitigated nuisance to everybody concerned. They threw a very great deal of additional labour of a monotonous and uninteresting nature on an already overworked staff, without any hope of additional remuneration; they took men away from their ordinary work, which naturally fell into arrears, to be cleared off afterwards; and they disturbed the whole working of district routine. But they were exceptional in their nature, Government attached great importance to them, and the whole Panjáb staff laid aside all other duties that could be temporarily neglected with safety, threw their whole heart into the Census work, and willingly and zealously strove to ensure a successful result. I do not think that the manner in which the work was done can be praised too highly. But, the Census once taken, the tension suddenly relaxed; the Tahsíldárs thanked Heaven it was over, and turned their attention to working off the arrears which had accumulated. They knew that they would be in no way responsible for the tabulation of the results; and it is hardly to be wondered at that they neglected subsequent operations of a purely routine nature, the only object of which was to save trouble to another Department.

The allowing the Tahsíldárs to send up the schedules and returns direct was a great mistake. It was done to save time; for Mr. Elliott allowed no margin in his estimate of the time necessary for the completion of the work. But in the end it would have saved time to have had all the papers sent through and checked in the district office; and I strongly recommend that this course be adopted in future.

892. Machinery for abstracting and tabulating the figures.—At the Census of 1868 the figures for each district were tabulated by patwáris in the district itself, and a single clerk compiled the tables for the Province from the returns sent up in English by the Deputy Commissioners. Sir Robert Egerton had intended to follow the same general plan, so far at least as tabulation was concerned, on the present occasion. But Mr. Elliott was strongly opposed to it, as he was of opinion that it would be almost impossible to secure uniformity of method, that a really effective supervision could only be ensured by the deputation of an Extra Assistant Commissioner for each district, which would be exceedingly expensive, that there would be great danger of the work being badly and slowly done, and that a single dilatory Deputy Commissioner might delay almost indefinitely the results for the whole Province. He was of opinion that the work should be done at a central office at Lahore under my immediate supervision. This, I objected, would be next to impossible. We could not bring local officials, such as patwáris, so far from their homes; and, without local knowledge, no central staff could either read or understand the terms for castes, clans, and occupations used, for instance, in Dehli and Pesháwar respectively. The employment would be very temporary; it would be difficult to collect a sufficient staff; we should have but little hold over the men when collected; and much time would be lost in taking on incompetent men, trying and rejecting them, and finding better to take their place. Even the mere matter of office accommodation and floor space would be a difficulty. Above all, the constant references to districts which would be necessary for the explanation of words and discrepancies would cause the greatest trouble, confusion, and delay. I verbally consulted two-thirds of the district officers of the Province, when collected at the Imperial *darbár* of 1881; and though it was to their interest to have the labour shifted off their shoulders to those of another, all but one were of opinion that the work would be done better, more quickly, and at less trouble to all parties in the districts than at a Central office.

Eventually a sort of compromise between the two schemes was agreed on, and accepted by the Panjáb Government. A separate office for tabulation was constituted for each division, at which the figures of the whole of that division were dealt with. Each office was put under the charge of a picked Extra Assistant Commissioner, Superintendent of Settlement, or Tahsíldár, the qualities taken into consideration in selecting them being familiarity with figures, experience in the management of large establishments, and driving power. The offices were manned by patwáris and qánúngos selected rateably from the staff of each district in the division, a smaller number however being drawn from districts under settlement, or having large areas under fluctuating assessment or subject to river action. The whole staff thus selected numbered 92 out of the 389 qánúngos and 935 out of the 7,668 patwáris of the Panjáb. A Sadr qánúngo was also selected in each division as assistant to the divisional officer. The divisional officers were seconded in the lowest grade of their respective ranks; while arrangements were made, where necessary, to provide substitutes for patwáris and qánúngos at as low a rate as possible. District officers were directed to select the very pick of the revenue staff for this work; and most of them honestly complied with the demand, though here and there the election was left to *Tahsíldár ji*, who shot his rubbish into my hands. The

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officers were located at the head-quarters of divisions except those of Pesháwar, Deraját, and Hissar, which were stationed respectively at Ráwalpindi, Multán, and Dehli so as to be easily accessible by rail[1].

893. I am inclined to think that, on the whole, the plan adopted was the best. The single central office is, I am convinced, quite out of the question for the Panjáb. The only staff that could hope to deal successfully with the infinite variety of local terms is a local staff such as we employed. Take a single instance—that of occupation. *Faqír* means a tenant in Pesháwar, a beggar elsewhere. *Daftari* means an office menial save in Pesháwar, where it is a landed proprietor. *Páli* and *Banjárá* mean respectively a herdsman and a cattle merchant in the east, an oil-presser and a pedlar in the west, and so on *ad infinitum ;* while the local terms which are absolutely unintelligible to any outsider are simply innumerable. So, too, we found local knowledge invaluable in the grouping of blocks and hamlets under the proper villages and towns, and of villages in their proper places ; though if the tahsíl abstracts had been properly prepared this should have presented no difficulty. But the necessity for local knowledge is felt at every stage of this portion of the work.

The choice then lies between district and divisional tabulation. I do not attach much weight to the objection that uniformity would be impossible if each district tabulated its own figures. So long as the operations are confined to *tabulating*, not *classifying*, the various entries in the schedules, the process is purely mechanical, and uniformity is only convenient because it is well that the easiest and best method should be followed everywhere. A scheme would be drawn out beforehand ; and a good Tahsíldár would do the work more quickly, better, with a smaller expenditure of power, and with infinitely less hindrance to current work than would be possible were it done in a divisional office. But there are bad Tahsíldárs, and those not a few. It would be difficult for the district staff, already burdened with arrears of work accumulated during Census operations, to exercise the continual supervision which alone would be effective ; it would be impossible to put a superior officer on special duty for each district ; and though in a large portion of the Province the work would be admirably done, yet bad work would be an absolute certainty from some tahsíls.

The two dangers are delay and fudging. The first should be easily avoidable. We took three and a half months abstracting and tabulating with one-eighth of the Provincial revenue establishment ; and if we had tabulated for tahsíls only instead of for districts and divisions also, we should have saved three weeks out of that. The work could be done in the tahsíls in a month or six weeks with the greatest ease. But the danger of fudging would be infinitely greater. As I have noticed in section 891, the strain of Census operations once relaxed, there is a tendency to scamp further work done in the districts ; while in the divisional offices the staff take a fresh departure. It is true that the inconvenience and discomfort entailed upon the patwáris by living away from their district for three months is so great that there is a strong temptation to make the figures agree by any means, and so have done with them. But native clerks will fudge under all circumstances if they can ; and everything depends upon the supervision. Now it is possible to select ten first-class supervisors for the divisional offices ; but it is not possible to procure thirty-two such men for the districts. Thus, on the ground that there are some Tahsíldárs under whom the work would certainly be fudged, and on that ground alone, I am inclined to think that the divisional offices offer the best practicable solution of the problem. The consideration that weighs most strongly with me on the other side of the question is, that if the tabulation were done in the districts, errors detected in the tables would be sent to the districts to be put right ; and the small compiling staff would not be, as it now was, overburdened with the labour of correction. But I think that the burden might in future be largely reduced by measures to be described later on, and in any case the difficulty could be met by increasing the compiling establishment.

894. Preliminary examination of schedules.—When the schedules reached the divisional offices, they were examined in order to make sure—

(1) that we had a book of schedules for every block in the Province ;
(2) that the series of pages in each book was complete, and that their number was not less than the number of pages entered on the outside label ;
(3) that the series of house-numbers was complete, and that each house had an entry of some sort against it, even if only to state that it was empty ;
(4) that the private schedule for every house which was noted as being allowed such a schedule was stitched into the book.

Where the schedules were found defective in any of these respects the Tahsíldár was referred to for an explanation or requested to send the missing papers. This examination assured us that we had a Census entry for every house entered in any house-list throughout the Province. Having this assurance, we assumed that those entries were correct save in the very few cases noted in section 899. Some of the offices began making a more minute examination ; for instance, to ascertain whether all the families and heads of families shown in the house-list were entered in the schedules. But I put a stop to this. The house-list was prepared some months before the final Census ; it was quite possible that the apparent omission, where it occurred, was correct ; and if we had examined every entry to discover whether it afforded room for suspicion of inaccuracy, our labour would have been endless while the errors corrected by it would have been inappreciable in number and importance. I was astonished to find in how extraordinarily few instances did this preliminary examination afford any grounds for inquiry, and how hardly ever did that inquiry disclose any real omission. In tahsíl after tahsíl not a single apparent flaw could be detected ; and the schedule record of houses, at any rate, was practically perfect.

895. The process of abstracting.—The abstracting was done by abstractors who took out the figures

[1] Probably by next Census the facilities for communication and even the district and divisional boundaries will have altered considerably. But I am not at all sure that it would not be better to tabulate the figures for the Bannu district at Pesháwar and not at Multan not only because the distance is smaller, but also because it is a Pathán and Pashto-speaking district.

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Part I.—Abstracting the Figures.

from the schedules and entered them in registers devised for the purpose. These registers were eight in number, and showed the following details in addition to the sex, which was distinguished in each :—

1. Age. Civil Condition. Religion.
2. Musálman Sects.
3. Birth-place. Language.
4. Caste. Certain Religions.
5. Occupation, under or over 15 years of age.
6. Education (including uneducated). Age. Religion.
7. Infirmities (four kinds). Age. Religion.
8. Sects and Races of Christians.

The original plan of working was this. Four men took a book of schedules: blank registers were distributed among them: one man read out the entries opposite each name, and as each entry was read the clerk in charge of the register in which that entry had to be abstracted made an upright stroke in the appropriate column. Each fifth stroke was drawn diagonally across the preceding four, so that they were grouped into fives and could easily be counted.

I do not propose to enter into minute details of the method of working. But there are one or two matters of principle which it will be necessary to discuss here. Some of the registers were very complicated; for instance, the one which showed civil condition and age had the age-periods printed down the side and the form ruled off into spaces for the various religions, each of which was again divided into three spaces, *single*, *married*, *widowed*, under each of which were two columns for *male* and *female*. Thus the clerk, on hearing the entry "married, male, 9 years, Hindu," had first to find the horizontal line on which entries for the age-period "5 to 9" were to be made, then to follow that line till he came to the space for Hindus, under that to select the column for married, and in that the sub-column for males. There he made a tick. In filling up registers of this complicated nature it was found indispensable that the entries should be read out; as if a single clerk attempted to abstract straight from the book, he lost his place in the register every time he looked at the book, and delay and error resulted. But in simple registers such as that for Musalmán sects, where the clerk had only to distinguish sex and five sects, it was far quicker and easier for him to abstract straight from the book. Now in the Hissár division, out of a total Musalmán population of 286,316, no less than 285,668 are Sunnis: that is to say that, probably in 99 out of every 100 blocks, every single soul was a Sunni. But a block containing 499 Sunnis and 1 Shíah could be abstracted in two ways; the clerk could either make a tick for each entry, that is to say 499 ticks in the Sunni and 1 in the Shíah column; or, he could run his eye down the column for sect, make a tick for the one Shíah, and find the number of Sunnis by subtracting him from the 500 Musalmáns which he would find entered in the religion register. Exactly the same two methods were possible with birth-place, language, education, and infirmities, in all of which the enormous majority of entries in each block are one and the same, and the exceptions can easily be picked out.

Mr. Elliott permitted either of these two plans to be followed, and, availing myself of this license, I largely followed the latter. I have since had reason to believe that Mr. Plowden objected to its adoption on the ground that it afforded no complete check. This is true. If there were two Shíahs in the block and one of them escaped the eye of the clerk, the result would not show that there was an error; whereas if he made a tick for each entry, it is less probable that he would make a tick under Sunni for the second Shíah; and unless he did this the total would be wrong and the figures would be re-examined. For similar reasons Mr. Plowden insisted upon our abstracting figures both for the educated and for the uneducated separately for each age-period used in the age-register; not because the details of age were needed in combination with education, for they were not even tabulated; or because the details of uneducated were of any use; but because, by abstracting the details of age twice over for the whole population independently, we obtained a check upon the abstracting, as the two results had to agree or the work was done over again.

896. The system of check.—And this leads to the whole question of check on the abstracting. The checking staff was, of course, kept quite separate from the abstracting staff, the work done one day by the latter being handed over next morning to the former to be checked. Here the totals and details of the different registers were compared with one another and with the figures of the village abstracts (which last were never allowed to pass into the hands of the abstracting department at all), and if any differences were detected the figures which were doubtful were re-abstracted; while even if there were no discrepancies, a certain proportion of the figures were tested by being got out again by the checkers, the proportion varying with the degree of accuracy with which the men whose work was under examination were found to abstract. Now my first rule was that under no circumstances whatever was any preliminary or intermediate check to be allowed; that the registers were to be totalled in the checking, and not in the abstracting department; and that till they passed into the hands of the checker, no examination of any sort was to be made. This practically amounted to saying to four clerks who sat down together to abstract the figures of a single book: "You must get out your figures independently of each other, and must not examine "them to see if they agree with one another. I will do that for you; and if they do not agree you will "have to do them again, and will be punished besides." The plan was admirable in theory; but I ought to have known that, with natives at any rate, it could not be worked in practice. No supervision could prevent the men comparing their figures as they went along, and I was compelled to recognise and regulate the practice. If the men found, at the end of abstracting a block, that one register gave 253 and 219 and the other 252 and 220 for Hindus and Musalmáns respectively, the discrepancy could be removed either by re-counting the Musalmáns, or by assuming that one register was right and altering the other to agree with it. The former method gave the correct results; the latter was fudging. What I had to aim at was to make first the honest correction so easy, and secondly detection of the dishonest correction so probable, that it was not worth while to fudge.

Part I.—Abstracting the Figures.

The first end I attempted to attain by making them compare the figures for each opening of the book—say 25 to 50 souls—as soon as it was abstracted. If any detail did not agree, it was almost as easy to run the eye up the column and get the right figures for the detail in question, as to fudge the corrections so as not in the act of doing so to produce a second discrepancy. For the detection of fudging (or of honest error) there were three means. First the comparison of the figures of the registers one with another. This only availed when the fudging was clumsily done, and was really nothing more than a very necessary check upon and repetition of the comparison already made by the abstractors. Secondly the comparison of the abstracted figures with the figures of the village abstracts. This was certainly a check in that it often led to re-examination of the figures; but it extended to only a few totals and really gave us much unnecessary labour, for in the enormous majority of the cases of difference, the village abstracts were wrong and the register right. Thirdly the testing the work by taking a certain number of items from each register and getting out the figures for them from the schedules independently. This was the real safeguard; and upon the efficiency with which this test was applied and the extent to which it was carried depended the measure of confidence which could be placed in the accuracy of the results.

But it is evident that the allowing the abstractors to compare the figures as they went along very greatly reduced the value of any inter-check between the registers, for they were practically no longer independent. If the clerk who left out a Shíah in section 895 found his Musalmáns short by one, he might look at the book again and detect his omitted Shíah, or he might save himself the trouble by adding on a Sunni; but in no case would the register show that there had been a discrepancy. Thus it was almost as safe to allow him to pick out his Shíah at once, and take the rest as Sunnis without counting them. So, too, it made the combination of age details with education of very small value as a check; for though the existence of the two registers showing age would perhaps disclose an error if it had occurred, yet it disclosed it to the abstractor, and the only way of finding out whether he had corrected it or fudged it was to re-abstract the figures. And it made the retention of the working party of four independent abstractors unnecessary, if any speedier method could be devised.

897. Proposed system of abstracting.—We tried many ways of combining our clerks and registers, for I allowed considerable license in this matter within the limits which I considered necessary to ensure accuracy, and which I insisted upon being strictly observed; as for instance, when I directed that the age registers were in all cases to be filled up from dictation and not by a single clerk. I found that men got accustomed some to one method and some to another, each preferring his own; and above all that certain men worked well together in parties of two, three, or four as the case might be, and gave the largest and best outturn when working in their own fashion. A slow or stupid man was a terrible drag upon sharp companions, who could really turn out more work without him. But on the whole the following is the plan which I believe to be the best, and which I would recommend for future adoption. Two clerks take two books of schedules; one reads out the entries while the other fills up the register for age, religion, and civil condition. Then each takes a book and fills up the entries for education, infirmities, sect, language, and birthplace, only counting those who are educated or infirm, or who belong to a strange sect, speak a strange language, or were born out of the district. In the same way each man fills up the registers for caste and occupation. It is true that in these two last registers *every* caste and occupation has to be entered. But they are nevertheless better filled up by one man than from dictation, as the caste and occupation is very generally constant for each family, and often for a house or even a group of houses. The clerk notes how far down the page the entry is the same, counts up the items he wants within those limits, and enters them at once. The education register we had to prepare from dictation because we were compelled to take out age statistics in it. But this I think was unnecessary.

The system I have sketched above lessens the value of the check afforded by comparison of registers one with another, because it does away with the working party of four independent abstractors. But as I have already pointed out, the value of that check is almost destroyed by intermediate comparison, which as I have explained above it is quite impossible to prevent. I am convinced that the only really effective check is the examination of a percentage of all work done by doing it over again; and during the earlier stages of the work, before the men become expert, the percentage so examined should be large. After a time it is marvellous how accurate the work becomes, and how rarely an error occurs. There is, of course, one way in which absolute accuracy can be secured. That is, by having each register prepared perfectly independently and by a separate staff of abstractors for all the blocks, and not allowing the staff who are abstracting one set of figures to have any knowledge of the results obtained by those who have abstracted another set. The registers would then be compared; and if two sets of figures did not agree they would again be abstracted by two new sets of men who had no knowledge of the figures obtained in the first abstraction. There are many practical objections to this system. There would, I think, be danger of confusion, delay, loss of papers, and omission of portions of the process. And I doubt whether, with native agency, it would be possible to prevent the results being communicated for the purpose of fudging. But an absolute fatal objection, to my mind, is the enormous expense attending it. I understand that it has been followed at Madras, and has been so costly that the Imperial Government have declined to pay for it.

898. Cost versus accuracy.—And here I would fain deliver my soul as to the ethics of fudging. The practical question that has to be considered in all operations of this sort is, what is the attainment of any particular degree of accuracy worth; given that a certain amount of extra labour and extra check will secure a certain enhanced measure of accuracy, is the increase in accuracy worth the cost of attaining it. And I think one very obvious limit is afforded by the consideration, that under no circumstances can it be worth while to tabulate material with a degree of accuracy greater than that which the material itself may reasonably be supposed to possess in the first instance. I have heard that in a certain Census the whole staff was engaged for six weeks in examining and removing a discrepancy of *four* in the total population of a Province. There were special circumstances which justified the delay in this case; but under ordinary circumstances I should look upon it as waste of time, and should consider conscientious fudging

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Part I.—Abstracting the Figures.

to be a duty. Now I fully admit that it is quite out of the question to allow such agency as we employed in our offices any license whatever in this direction. The rule must be stringent that every error, however small, is to be removed by retabulation; though however strict it may be, it is perhaps hopeless to expect that it will be observed in all cases. Fudging is like cribs, which are invaluable in the hands of a man who knows how and when to use them, but must not be given to school-boys. But it is possible to carry refinements of check so far that they cost more than the extra accuracy which they secure is worth; and this is certainly the case when the errors which they prevent are smaller than the original error in the material under treatment, and when the system of check can be relaxed without allowing any discretion or license to the establishment, either directly or by implication.

Now he must be a very sanguine man who imagines that the mean probable error in the enumeration of souls at the Census of 1881 was less than, say a quarter per cent., and the error in the entry of details will have been larger; in some respects much larger. I do not however wish to be understood to think that all errors of not more than one quarter per cent. may be safely neglected in the tabulation of our Census statistics. If this is the logical conclusion from my argument, I am quite content to be illogical on the subject. But I do think that time and labour are both wasted in removing minute discrepancies which bear no appreciable proportion to the figures involved, or to the much larger errors inherent in those figures. During the months of April, May, and June I spent some 60 days in inspecting my divisional offices, the greater part of which were devoted to checking abstracted figures by personal reference to the schedules themselves. Knowing the want of accuracy which often characterises native work, I had expected to find errors in great numbers. But I was utterly astonished at the rarity with which I detected even a single error in the items I examined. And when I did find errors they were, as far as I remember almost without a single exception, errors of one single unit in excess or defect. So far as I saw, the errors in tabulation by sex alone certainly did not reach anything approaching 1 per 1,000. The errors in other details were proportionally larger, because the figures in which they occurred were necessarily smaller. But I do not believe that the errors in the entries of age even, the most difficult item to abstract, ever amounted to one quarter per cent., even if their sign be neglected. And the errors in the totals for tahsíls and districts must be of course many times smaller than the errors in individual blocks, for errors of excess and defect correct each other to a very large extent.

It is a question of opinion and judgment, and not capable of calculation or proof: for we know exactly neither the amount of error in the original material nor the degree of inaccuracy which each relaxation of the check will introduce. But it is my deliberate opinion, founded upon a very intimate knowledge of the working of our divisional offices, that the system described in section 897, if supplemented by an examination of a proper proportion of the figures, will give us figures corresponding more nearly with the Census entries than those entries correspond with the actual facts, and therefore possessing all the accuracy which it is worth while to aim at. What proportion of the figures should be examined it is impossible to say exactly: but it should be large, probably something like 25 per cent., at the beginning of the work, may be very much reduced as the men gain skill and certainty, should be smaller for the sharper and larger for the inferior workers, and should never fall lower than two or three items in each register. The smaller items are as a rule those most liable to error, while they can be examined most easily; and figures which have been altered and corrected are especially open to suspicion. I have not hesitated to dwell at some length on this point, for it is important as largely determining the duration and cost of the work of abstracting; and I understand that Mr. Plowden is of opinion that at a future Census far more elaborate registers should be used with the sole purpose of rendering the check more stringent. I have therefore thought it right to record my opinion, based upon my personal experience, to be taken for what it is worth.

899. Departure from the entries in the schedules.—A question which had to be decided at an early stage in the operations, and one with regard to which it was most difficult to lay down any satisfactory rule, was, how far it was allowable to depart from the facts entered in the schedules when they appeared to be contradicted by other entries. The easiest, and perhaps the commonest case was that of sex. When we found "Musammát Farídan, married woman, midwife," with a "one" opposite her name in the male column, we had no hesitation in entering her as a female. But when we had "Mahamdu, caste "Mughal, tribe Máchhi, occupation fisherman," or "Bhagwána, caste Rájpút, tribe Máli, occupation gar- "dener," it was more difficult to decide whether their caste should be taken as Mughal and Rájpút or as Máchhi and Máli. In the Eastern Panjáb, where these cases were chiefly reported, it was practically certain that the latter was the true entry, and that the men had entered themselves as of the higher caste (as several Deputy Commissioners have reported to have often been the case) either as a joke or in accordance with the traditional origin of their present inferior castes. In the Western Panjáb such cases were not as a rule reported; and the reason doubtless was that, though such false entries were probably at least as common there as in the east, yet it was by no means safe to assume that in the Mahomedan portion of the Province a man was not a Mughal or a Rájpút merely because he worked as a fisherman or a gardener. In some of these cases the village patwári, who knew the individual well, had in preparing the village abstract of castes and clans entered him as a Máchhi or Máli; and here I had no hesitation in following the village abstract. But where this guide was wanting I decided that, however probably wrong, the entry must be assumed to be right unless the patwári of the village was present in the office, or other similar evidence was available. I do not think any other rule would have been safe.

900. Utility of the preliminary and village abstracts.—In section 888 I explained that the second part of the village abstracts contained the names of the religions, sects, languages, birthplaces, and occupations occurring in the schedules. They were utilised thus. A clerk took a book of schedules with its abstract, and filled in the headings into each register. For instance, in the occupation register under the heading "Name of occupation," he wrote down all the occupations mentioned in the abstract one under the other, leaving a large space in which to enter ticks opposite those occupations which a glance at the schedules showed him to be most prevalent in the block. Thus the abstractor found his headings ready

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entered to his hand when he came to fill in the register. Where the working-party plan was adopted this was a convenience, as otherwise the other three men would have to wait while the fourth wrote in his register each new heading as it occurred for the first time. But when the registers were filled up by a single clerk and not from dictation, this inconvenience was not felt; and in some cases I found that the men did not think it worth while to utilise this part of the village abstracts. There is undoubtedly one strong objection to their use. The list of occupations and birthplaces given in the village abstract was seldom absolutely complete. Now when a clerk took a blank register and wrote down his own headings as they occurred, he naturally expected to have to fill in each new one as he came across it: moreover, he knew exactly where each one was to be found in his lists, no unimportant point so far as speed is concerned with a list of 30 or 40 occupations. But when he had his headings ready filled in for him, he not only had to search for each the first few times it occurred, but when he came to one which had been omitted from the list and his search failed to discover it, he was more likely to make a tick opposite some other occupation already entered, instead of adding the new name to the list as he should have done. On the whole I should be inclined to omit this portion of the village abstracts as useless; and if the plan of abstracting described in section 897 is substituted for the working-party system, I should most certainly do so.

With regard to the first part of the village abstracts, it was of but little use as an independent check. The figures for sex and civil condition were very generally accurate, probably because they checked each other; but the details of education and infirmities were often slightly wrong, while the number of persons whose occupation was recorded was often very incorrectly stated. So far as the figures are needed for the purpose of publishing preliminary results as soon after the Census as possible, it is absolutely necessary to prepare them; but the remaining figures must be abstracted with much greater accuracy than at the late Census if they are to be more help than hindrance.

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Part II.—Tabulation of the Figures.

PART II.—TABULATION OF THE FIGURES.

901. The village-tables.—The abstracting finished, the next step was to collect the figures into village-tables, the top headings of which were so arranged as to exhibit the details needed for the final tables, while on the left-hand side of each table was printed a list of all the villages or towns of the tahsíl with their component blocks, arranged in the same order as in the tahsíl abstract (see section 890), opposite each of which the figures found in the register were entered in the appropriate column. Two separate tables were prepared, one for villages and one for towns, in each tahsíl; and each of the larger tables consisted of many parts or volumes, the number of columns being far too great to be shown in a single opening. These tables were 17 in number, and exhibited the following details, sex being distinguished in all:—

A.—Houses, families, and population—
I.—Civil condition by religion and age.
II.—Age by religion.
III.—Religion.
IV.—Mahomedan sects.
V.—Christian sects.
VI.—Miscellaneous religions.
VII.—Caste by religion.
VIII.—Language.
IX.—Birthplace.
X.—Occupation, over and under 15 years of age.
XI.—Education by religion.
XII.—Insanity by religion and age.
XIII.—Blindness, do. do.
XIV.—Deaf-dumbness, do. do.
XV.—Leprosy, do. do.
XVI.—Totals of specified occupations and of infirmities.

No. XVI was merely a check table, while No. III was added for convenience only, and contained no fresh information.

902. Mode of filling up the tables.—The work of filling up the tables for age, civil condition, Musalmán sects, education, and infirmities, was simple enough. The columns, if numerous, were at any rate fixed in number; and all that was needed was care that the figures were not entered in the wrong column. This was not the case with the tables for language and Christian sects; but even here the number of different entries in the register was so comparatively small that there was no difficulty in finding the appropriate column for each. But in the case of birthplace, of castes, and of occupations, the number of entries in the register was so enormous that the greatest difficulty was experienced in tabulating them.

The first step taken towards meeting the difficulty was to sub-divide the birthplace table into separate tables for Panjáb districts (putting those of the division under tabulation first), Panjáb Native States, other parts of India, and foreign countries. Similarly the occupation tables were separate for agriculturists, traders, service, artisans, and others. And I think it would have been well to have split up the castes in a similar manner into some such classes as priestly and ascetic, land-owning, mercantile, artisan, menial, and vagrant and outcast. Of course this classification was adopted purely to render the tabulation easier; it was exceedingly rough, by no means uniform in the various offices, and was wholly disregarded in the final compilation. The plan then adopted was as follows. In each tahsíl, and especially in the rural portions, there were a certain number of castes and occupations which recurred in almost every village. These were known to the local staff, and were entered at the head of the columns of the first part of each table. The entries which occurred less frequently were similarly entered in a second part, and so on till all but the very rare entries were disposed of, the entries in the later parts of the table being often alphabetical[1]. A tabulating clerk was in charge of each part of the table, and the registers were passed from one to the other; each one posting up the entries belonging to his headings, a numbered list of which lay by him, and ticking them off on the register. When the register had passed through the hands of these clerks, the remaining entries if any, which consisted wholly of those of rare occurrence, were entered on rough slips framed on the pattern of the village-table; and when the tahsíl was completed, these were arranged and entered into a final part of the table. After the entries for a whole circle or group of circles had been tabulated, the tables were taken and the entries in them read out to clerks who sat with the schedules in their hand and ticked off each as it was read. Every entry was thus gone over twice in opposite directions, first from the schedule to the table, and afterwards from the table to the schedule.

903. System of check.—Every table or part of a table had a last column headed total males and females, and as each page was entered up the figures were totalled horizontally and vertically, and the results compared. When the table was finished, the page totals were collected at the end and added together to get the final totals. These last were then arranged side by side in a check-abstract devised for the purpose, such that all figures which, derived from different tables, ought to correspond, were exhibited one under the other, and any discrepancy could at once be detected. The points in which the figures disagreed were then examined page by page and, within the pages which did not agree, block by block, till the error was run to ground. In the larger tables errors were in some offices so frequent, that it was found simpler to compare all entries in the tables page by page as they were filled in, and while the papers were in hand and the entries fresh in the memory. But I cannot help thinking that this was really waste of time; and that if sufficient care had been taken in posting, it would have been enough to examine only such details as a comparison of the final totals showed not to agree. The other method involved examination of figures which need not have been examined in detail at all, since their totals agreed already (see further section 904).

Here the absence of a precaution of which I ought to have thought caused a good deal of trouble. I had ordered the lists of villages and blocks which formed the side-headings of the tables to be printed and cut up into slips, which slips were pasted on to the side of each table. But I did not direct that the pages

[1] These "parts" were, as it were, separate volumes of the same table.

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Part II.—Tabulation of the Figures.

of all the tables should be made of one uniform depth, so that, if the third page of any one table contained blocks numbering 41 to 60, the third page of every other table throughout should contain the figures for precisely those same blocks. This plan should be adopted in future, as it immensely facilitates the page by page comparison. Exceptions, however, will be necessary, as I proceed to show. I had at first directed that every single table or part of a table should contain all the blocks (rural or urban, as the case might be) of the tahsíl. But I found that this increased so enormously the bulk of the tables, and consequently the labour of filling them up and the risk of error, that I had to modify my orders. For instance, in tahsíl Bhiwáni there are 3 Sikhs in a population of 103,556. Every one of the ten tables which distinguished religion had therefore to include a part headed "Sikh;" but the entries in each were confined to a single block, and it would have been absurd to exhibit in each all the remaining blocks of the tahsíl. So on the Pathán frontier there are hundreds of villages without a single Hindu; while the entries for Christian sect and race and for infirmities are everywhere few and far between. Thus it became necessary to lay down a rule for each division as to which tables and parts of tables should exhibit all the blocks, and which only those blocks in which there were figures to be entered.

904. The tabulation is a most important portion of the work, and one which it is exceedingly difficult to supervise or examine effectively. It is easy enough to take a book of schedules with its abstracting registers and to compare the one with the other; but to find out whether the items in any given one of those registers have been honestly tabulated, it is necessary to pick them out one by one in some 30 or 40 tables, some of which contain columns by the hundred. At the same time if the figures are honestly totalled, the check is almost automatic. The comparison of totals is an absolute check against omission; while a comparison of total males and females is an absolute check against what the baboo calls "miscolumning" where, as is almost always the case, the entry is misplaced by one column only; for an entry shown one column to the right or left of its proper place will fall under the wrong sex[1]. Thus the only errors which a mere comparison of totals does not at once bring to light are those in which an entry, while falling under the proper religion, age, or other head occurring in two or more tables, is misplaced by an even number of columns, two or more. And with the most ordinary care such errors should seldom or never occur.

The real danger to be apprehended is that the clerks, when they find totals disagree, should arbitrarily alter them to correspond with each other; or, in the larger tables, should save themselves the trouble of searching for the proper column or of adding a new one, by entering the item in a column to which it does not properly belong. The sub-division of tables into parts described in section 902 is a valuable precaution against this last practice. But I see no way of detecting or preventing dishonest work at this stage beyond careful selection of the tabulating clerks, and incessant and minute supervision. Re-examination of the work is, as I have said, infinitely difficult in the larger tables.

And this is one reason why I object to a page by page comparison of totals. If the final totals only are compared, the comparison is made by a separate staff; and the correcting department has to bring the error when found to the supervisor, and to satisfy him that it really has been detected and accounts for the discrepancy in the totals. But if the tabulators compare the totals page by page as they go along, this is impossible. On the other hand, the comparison of final totals does not reveal errors which are neutralised by other errors of equal amount but opposite sign; while the page by page comparison will bring to light all such errors as do not balance one another within the limits of a single page, and thus affords greater security against internal mistakes. The latter is the better plan if the agency can be trusted; the former if it cannot.

As a rule I believe that this work was honestly and thoroughly done. In some cases a grievous amount of labour and time was expended in removing errors absolutely trivial and insignificant in amount, but which it would not have been safe to allow the staff to disregard.

905. Classification of entries.—It is in this stage of the work that the question of classification has first to be considered; a question by far the most difficult and important of all that present themselves during the operations subsequent to enumeration. In the abstracting stage it hardly arises; for the strict rule was that the abstractors should in all cases show the entries exactly as they were entered in the schedules, and the only case in which that rule was waived was where the difference between two entries of one and the same occupation consisted merely in the omission in one of them of an insignificant word. Thus, "spinning," "spinning raw cotton" and "spinning (cotton) thread" (*kátna*, *rúi kátna*, and *sút kátna*) were allowed to be taken together. But even this license was seldom needed, for the entries in each book of schedules being made by a single enumerator, the term used for any given occupation was generally constant.

But when the entries came to be arranged in the village tables, the number of birth-places, castes, and occupations described in different words but bearing the same meaning was found to be so enormous, that some degree of classification was absolutely necessary if the tables were to be filled up at all. It would have been absurd to make the Kángra patwáris enter Jawálamukhi and Kángra, both within the district, as separate birthplaces; to have separate caste columns for Punwár and Túnwar, both well-known Rájpút clans; or to show *nái, hajjám, hajámat karna,* and *bál kátna* as distinct occupations. The rules adopted in each of these three cases were as follows:

For birthplaces, lists of the districts in the North-West Provinces, Bengal, and Sindh, and of the Native States of Rájpútána were sent to the Divisional offices; and any entry of birthplace corresponding with any one of these names was assumed to refer to that district or state, and was shown in the corresponding column. So again, any entry corresponding with the name of a Panjáb district or state was assumed to mean that district or state. This was probably wrong in some cases. Many people showed the *village* of birth, instead of the *district*, as they should have done under the rules; and probably some "Sháhpurs" for instance entered in the schedules, really referred to some one or other of the countless villages of

[1] For this reason the occupation tables should not be prepared separately for males and females, as was done on this occasion; because in relinquishing your double sex-columns you lose an invaluable check.

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that name which are scattered over the Province, and not to the Sháhpur district. But the only safe plan was to assume that the instructions had been carried out. I myself went through the European schedules, in which, too, unknown British villages were constantly entered in the birthplace column; referred such as I could to England, Ireland, or Scotland, and classed the rest as British Isles. Where a birthplace not falling under any of the above heads, and not known as the name of a considerable town, corresponded with the name of some village in the district in which the man was enumerated, it was assumed to refer to that village. All birthplaces not disposed of under the above rules were tabulated separately.

906. In the matter of castes, difficulty was caused by the innumerable instances in which, especially in the Western Panjáb, tribes or clans had been returned instead of castes. In the Multán and Deraját divisions the so-called castes entered in the schedules were literally numbered by thousands. Here I was compelled to leave much to the discretion and local knowledge of my Divisional officers. They had been selected with special reference to their acquaintance with the divisions the figures of which they were respectively tabulating, and the number of cases was so enormous that reference to the people themselves through the district authorities would have practically stopped our work for months. Accordingly I permitted the Divisional officers, when they were satisfied beyond all reasonable possibility of doubt, either from their own knowledge or from the local knowledge of their staff or by examination of the entries in other schedules of the neighbourhood, that a certain tribe or clan belonged to a given caste, to include that clan under that caste. But where there was reasonable room for doubt they were to show the clan separately; and it was strongly impressed upon them that it was always better to show too much than too little detail. Further, they were to furnish me with lists showing all the clans or other entries which they had included in each caste-column.

With regard to occupation, the rule laid down for the Divisional officers was that, where the occupations were identical, the difference being purely one of language, they were to be taken together; but that different occupations, however nearly related, were to be separated. Certain rules were issued for the classification of Government servants, the number of whose occupations would otherwise have been unmanageable. Here, too, lists were ordered to be kept showing all the different terms which had been taken as one and the same occupation and included under each heading of the table. The instructions to enumerators directed them to enter one occupation only for each person, save where agriculture was combined with another calling. As a fact, however, double occupations were recorded in very many instances, and one of them had to be selected for tabulation. The rules by which the tabulators were to be guided in these cases were—

(1) where one occupation was permanent and the other temporary, the former was to be taken; thus a man who was entered as reaper and water-carrier was tabulated as water-carrier:

(2) where the more important occupation could be identified, it was to be selected; thus a man entered as oilman and cotton scutcher was counted as an oilman:

(3) in all other cases the occupation first mentioned was to be taken, in the absence of any guide to selection.

907. Errors in classification.—As I have frequently pointed out in the course of my examination of the results, mistakes were made in the classification which took place during tabulation. But I do not see how they could have been guarded against. The number of separate entries with which we had to deal was far larger than I or anybody else had conceived possible. Unfortunately the detailed figures have been destroyed; and I dare not speak from memory, for I should be accused of absurd exaggeration. But some idea of their number may be gained from the fact that, *after* the Divisional officers had done all they could or might do to reduce the number by classifying them under the rules given above, the number of *separate headings still remaining*, and returned by them in their village tables, was as follows:—

Divisions.	Religion and Sect.	Caste.	Language.	Birthplace.	Occupation.	Total.	Divisions.
Dehli	25	314	24	148	940	1,451	Dehli
Hissár	16	145	9	69	327	566	Hissar
Ambála	27	398	40	113	576	1,100	Ambala
Jálandhar	28	235	18	70	369	720	Jalandhar
Amritsar	24	210	30	92	393	749	Amritsar
Lahore	33	354	30	84	523	1,024	Lahore
Ráwalpindi	23	989	35	123	382	1,552	Rawalpindi
Multán	27	2,551	51	178	709	3,516	Multan
Deraját	21	517	31	219	784	1,572	Derajat
Peshawar	34	871	30	274	618	1,827	Peshawar
Native States	108	1,808	136	742	3,426	6,240	Native States
Total	366	8,392	460	2,121	9,038	20,377	Total
Number in final tables	31	333	32	151	334	881	Number in final tables

These figures show, not only that some such license as was allowed to the Divisional officers was absolutely necessary, but also that it was not as a rule abused by them. Many hundreds of these headings covered a few individuals only, often fewer than 50, and very often fewer than 100. Setting aside the treatment of castes in the Derájat office, which was quite justified by the local custom of that division, the one unfortunate exception was the Amritsar office, where the liberty of judgment allowed was most certainly abused in order to simplify or shorten the work. This office had done its abstracting in quite first-rate style, the officer in charge was a picked man possessed of the highest qualifications for the work, and I not unnaturally gave it somewhat less supervision than other offices where the previous work had not been so good. But the Divisional officer had apparently set his heart upon finishing before any other office. In this he succeeded.

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but at the expense of accuracy, and his tables were sent in in a state of disgraceful incompleteness. All errors in tabulating and totalling were detected and removed in the central compilation office. But cases in which separate occupations or castes had been wrongly included under one heading unfortunately could not be detected till the results in their final form came to be discussed; and it was then as a rule too late to alter them (see further, section 918), though we did in the worst instances go back to the original schedules, at the cost of great time and labour.

908. But it must be remembered that this Census was, as far as the Panjáb is concerned, practically a first experiment. Nothing on the same scale or of the same degree of minute and accurate classification had been attempted before, and it was of the necessity of the case that errors should occur and mistakes be made. Those errors and mistakes I have endeavoured to set forth faithfully, not only in order to show how far the figures really are what they appear to be, but also in order that at a future Census their repetition may be guarded against. It was proposed by the Government of India that a list of castes and occupations should be prepared before the Census for the use of the enumerating staff. Anything more fatal to accuracy and fulness of detail I can hardly conceive. It would have been wholly impossible to prepare even approximately complete lists, or to use them if prepared, for they would have been of perfectly impracticable bulk. Moreover, to put a list of the sort into the hands of the enumerators would, where it did not simply bewilder them, cause them to enter, not what the people told them, but what a very cursory examination of the list might suggest as being somewhere about the truth. So, too, a native who, if asked what his clan was, would give a perfectly correct answer, would, if asked to which of a certain number of specified clans he belonged, almost certainly name one of them, though he might really belong to none. And it would be fatal to tell the enumerator to record clans in some instances and not in others. The only safe method is that adopted at the present Census, *viz.*, to *record* everything, leaving the process of selection and rejection for the after-stages of the work where the agency is of superior quality, supervision is closer, and uniformity can be secured. Moreover, while fairly complete classified lists would have been of immense value in the compiling offices, it would have been, as the Panjáb Government pointed out, quite beyond the bounds of possibility to prepare such lists; for not even the merest rudiments of the necessary material were in existence, while the Census itself afforded the only possible means of obtaining it. The result has shown how correct this opinion was; far more so probably than any of us conceived at the time.

At a future Census the case will be very different. The lists of castes and occupations included under each heading of the tables, to which I have already referred as having been prepared in the divisional offices, will, taken together with the table-headings themselves and the MS. tables of tribal sub-divisions, form a very fairly complete detail of the entries in the schedules: and this material renders possible the preparation of classified lists for use in future compilation. The Panjáb Government has approved of my proposal to devote my leisure for the next few years to the preparation of such lists; and I hope that I shall be able to complete them, and that they will be found useful. But I am convinced that the only stage in which they can legitimately be used is in the process of tabulation and compilation, and that any attempt to use them for guidance in enumeration will be fatal to accuracy. We cannot record facts at enumeration in too minute detail; we can classify and compress afterwards. To attempt to classify in the first instance through a widely-spread agency of inferior average qualifications and subject to no central supervision might perhaps tend to uniformity, but it would be a uniformity of error.

909. Units of tabulation—Towns and Villages.—Perhaps no single point throughout the Census has given me so much trouble in proportion to its importance, as the distinction between urban and rural population. There were two questions to be decided—

(1) What places should be classed as towns:

(2) What limits should be taken for the urban area in each place.

The correspondence and instructions issued on the subject will be found in Appendix D. The conclusions finally arrived at were—

(1) that all municipalities, civil lines, and cantonments were to be treated as towns, whatever their population:

(2) that all places including a population of as many as 5,000 souls dwelling together within a practically conterminous collection of houses and suburbs should be classed as towns:

(3) that places not falling within either of the above classes, but the population of which was urban rather than rural, not essentially consisting merely of a village community with its attendant menials and traders, but also including a considerable extraneous society drawn to the spot by the presence of a shrine of peculiar sanctity, by the existence of a centre of local commerce, or the like, should be reported for orders.

A good many cases were reported under clause (3) above, in pursuance both of my General Letter A and of my letter of 1881. But in all such cases the Panjáb Government declined to depart from the numerical standard of 5,000 souls. I myself should have been inclined to consider the circumstances in some of these cases sufficiently special to justify an exception, if any exception was to be made at all. But I would, on the whole, make no exception. It may be said generally that, for reasons explained in detail in sections 62, 63, it is often difficult to decide in respect of any given place whether its population is rather urban or rural in its composition; and an arbitrary numerical standard seems to be both necessary and unobjectionable.

With respect to the limits of urban areas, the cantonment boundaries were followed in the case of cantonments; while the Deputy Commissioner fixed limits for civil lines where not already included within the boundaries of a town. In the case of municipalities the municipal boundaries were followed in all cases except that of Lahore, where those boundaries included several villages purely rural in their nature and lying at a very considerable distance from the town, and which were excluded from the urban population. A detail of them will be found in the Appendix to Table XIX. There is no doubt that Lahore is not the only case in which the municipal limits include small villages which would, *per se*, be better included

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in the rural population. But it is so important to know exactly what has and what has not been included within each town, that it is I think right, save under very special circumstances, to follow the only boundaries available which are capable of exact ascertainment. The point was specially referred to the Deputy Commissioners for opinion with reference to each town in the Province; and the only case in which exception was thought advisable was Lahore. As already pointed out in section 845 the details of suburbs were, at all events as a rule, distinguished in our tables.

920. **Units of tabulation—village boundaries.**—In the case of all places not being cantonments, civil lines, or municipalities, the boundary of the village area was taken as the limit of the unit of tabulation, whether classed as town or village. There is no doubt that, as explained in sections 65, 65, these limits are in some degree and in some parts of the Province misleading as regards the size of the village. But no other limits would have been practically possible. In the first place it was absolutely essential that the tables should be based upon the *mauzahwár* lists of village areas which are kept at the revenue record-rooms; and this, not only to ensure that no portion of the Province was omitted, but also to prevent the recurrence of an error such as occurred in the Census of 1868, when a wholly imaginary town of the name of Májra, with a supposed population of 7,908, was created in the Rohtak district by adding together the small populations of the numerous outlying hamlets which are commonly known to the people as *májras* in that part of the Panjáb. In the second place the villages of the Province present every gradation of compactness or the reverse, from the single homestead of the Dehli territory, with its outlying temple, monastery, perhaps some gypsies' or cattle-traders' huts, and occasionally a hamlet or even two containing 10 to 20 houses each, to the *tappa* of Kángra and the *kothi* of Kúlu where each family inhabits a separate hamlet, or the *mauzah* of parts of Multán and the Derájat where the population is scattered all over the area at isolated wells or on lowlying plots of cultivation. Now in the former class the outlying population of say 4 to 80 souls are necessarily included with the parent homestead. In the latter class there is no alternative between either treating the village as a whole, or tabulating separately countless little hamlets, each with a population of from 20 to 100. In all cases it would be wholly impossible to draw any line which could be worked upon in practice. In the hill districts, indeed, there are well-recognised and clearly-defined sub-divisions of the *tappa* and *kothi*, known respectively as *tikas* and *phátis*: and these might perhaps prove more convenient units of tabulation than the larger areas. But Mr. Lyall tells me that in most cases they would be impracticably small. It has been suggested that an arbitrary limit of say 500 souls should be adopted, and that all hamlets reaching this size should be tabulated separately. But, besides the other objections just stated, there is the, as it seems to me, fatal obstacle that it would be necessary to tabulate all hamlets separately before it could be discovered which of them reached the limit in question. At the same time the information if it could be obtained would be most interesting, as it would gauge the progress of the process described in section 170. At present our figures for villages are almost meaningless in some parts of the Province.

The village area being accepted as the unit, it became important to exclude from the list of towns all villages, such as those of Derah Ismáíl Khán mentioned in paragraph 66, in which the population, though exceeding 5,000 in number, was widely scattered over a considerable area. Here again every degree of diversity existed, the principal homestead containing any number from 400 to 4,000 souls, and the remaining population lying at various degrees of proximity to it. The case was one which only local knowledge could deal with; and lists of doubtful cases were sent together with a copy of the correspondence on the subject to district officers, who decided in each case whether the population was or was not sufficiently compact for the place to be treated as a town. Table XIX shows 132 towns of over 5,000 inhabitants: Table XVIII shows 164 towns and villages of that size in British Territory. The difference consists of 32 villages excluded from the list of towns on the grounds just stated, in the following districts:—Jálandhar 1, Kángra 13, Derah Ismáíl 6, Derah Gházi 4, Bannu 6, Hazára 2.

921. **Towns and villages—precautions adopted and suggested.**—Such were the principles on which the classification was made. But they were very difficult to apply in practice. As already noticed in section 845, my instructions regarding urban areas had not been understood, and the schedules of suburbs which lay within municipal limits had nothing to show that they properly belonged to any town. The first thing therefore was to prepare, with the help of the tahsíl abstracts and the local knowledge of the staff, lists of all places apparently falling within the definition of town, showing all the component parts of each and their populations. These were sent to district officers with a copy of the correspondence on the subject, and they were asked to report—

(*a*) whether any places had been included which should not be classed as towns:

(*b*) whether any places had been omitted which should be classed as towns:

(*c*) whether all suburbs or outlying homesteads situated within municipal limits or the boundaries of cantonments or civil lines had been included:

(*d*) whether any of the subsidiary parts shown lay outside those limits or boundaries:

(*e*) whether any of the subsidiary parts lying within municipal limits lay so far from the town that it would be well to exclude them on special grounds.

If the answers to these inquiries had been absolutely accurate, my table of towns would have been perfect. But I had occasion later on, when the final figures were past alteration, to circulate a rough copy of the Appendix to Table XIX in print in order to have the spelling of names of suburbs corrected; and it then became apparent that in two or three cases, which an examination of column 4 of the Appendix will disclose, the answers had been inaccurate, and that I had consequently included certain small suburbs which lay beyond and excluded others which lay within municipal limits. And I cannot feel sure that we may not in other cases also have omitted suburbs which ought to have been included; for in the index-register and tahsíl abstracts, which were all we had to guide us, suburbs were sometimes shown as independent villages; and the fact that only in two or three cases did Deputy Commissioners report that we had omitted suburbs proves either that our local knowledge was singularly sufficient, or that enough care was not taken in examining our lists. In any case, however, the errors thus caused will have been comparatively insignificant in amount; for no suburb of any practical importance can have escaped inclusion with the town.

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At a future Census the plan of operations I would recommend would be something of this sort. The urban limits would be governed in all cases by the considerations just discussed. They would then be split up into sections and blocks in the manner described in section 846. The Deputy Commissioner and one of his Assistants would naturally visit every town in the district during Census operations, and would then examine the manner in which this work had been done on the spot. If he found reason to think that municipal limits should be departed from, he would report his proposals; but I think very strong grounds should be shown before this was allowed. When it was finally settled what was to be included in each town, special labels of a different colour from those used for rural books of schedules would be pasted on the urban books, in which, instead of the name of the village, the names of the (1) town (2) suburb or other sub-division should be shown separately. And it would help the compiling offices if the bindings of urban books were also covered with cloth of a different colour from that used for the rural blocks. The index-register of towns would be kept separate from that of villages; and in it care would be taken to show the name of the town in column 3 and that of the suburb in column 6 only; while in the tahsíl abstract for towns, which should be separate from that of villages, there would be two columns, one for the name of the town and another for that of subsidiary parts. These arrangements, however, would apply in general to municipalities, civil lines, and cantonments only. Village areas including a population of 5,000 or more cannot in all cases be picked out till at any rate the preliminary abstracts have been prepared and the population thus ascertained; they seldom possess any suburbs worth separating; the village boundary is followed as a matter of course; and all that is necessary is to exclude cases in which the population is scattered.

912. Floating population.—There was a certain floating population to be dealt with, generally but not always inconsiderable in amount, consisting of passengers by rail and boat, and of large gangs of workmen collected upon canals, railroads under construction, and the like. These were not included in the population of any village or town, but were entered in the tables at the end of the village population of the tahsíl in which they were enumerated. The resident Railway population was of course treated exactly like other residents.

913. European schedules.—The mass of the work being done in the vernacular, I was in considerable doubt as to how to deal with schedules filled up in English. I had at first thought of collecting them all at Lahore and abstracting and tabulating them by a separate English staff. But the divisional officers objected that this would involve two difficulties. In the first place we did not know where all the English schedules were, a considerable number being scattered about through the books of the Province; they would have been constantly turning up in small numbers which would have been very difficult to deal with, while there would have been some danger of loss. And it would have been very difficult to arrange for the incorporation in the figures of the village tables prepared by the Divisional offices, of the details of the English schedules tabulated by a central staff. In almost all the Divisional offices there were one or two men who knew English; and where this was not the case I sanctioned the entertainment of a couple of clerks for the purpose. These men made translations of the English schedules into Hindústáni, and read them out to the abstractors. On my visits of inspection I paid special attention to the subject, and especially to the translation of names of occupations, of which I caused lists to be shown me together with the translations used. But the result was not satisfactory. The age and sex were of course tabulated with ease; but there were difficulties in respect of religion, sect, language, nationality, and birthplace; while the tabulation of occupations was exceedingly unsatisfactory. At a future Census I should prefer to keep the population enumerated in English wholly separate. The great mass of it is European; and so long as the numbers of each sex were communicated to the Divisional offices to be incorporated in the village tables, the exclusion of the other details from those tables would not impair the utility of the figures for general district purposes; while supplementary tables could be supplied to the district office where large European populations existed. I would then have all the English schedules for each district collected together by the Deputy Commissioner in the first instance, each bearing plainly the number of its circle and block and the name of the village or town to which it belonged, and send them all to the central office to be tabulated by English clerks under the direction of the Deputy Superintendent. The figures would be added to those furnished by the Divisional office for each district; while only the number of each sex in each village or town would be communicated to the district offices for incorporation by separate entry in their tables.

914. Tabulation of tribal sub-divisions.—I have already explained (section 888) that we did not re-abstract the figures for tribal and caste sub-divisions, but accepted the figures of the third part of the village abstracts prepared by the enumerators. Nor did we tabulate them for any smaller unit than the tahsíl. The number of entries was so enormous that getting out even this detail involved very great labour. The number of sub-divisions shown in the tables for Jats and Rájpúts in British Territory are given in the margin. No other caste approached these numbers. The method adopted was as follows. A clerk took a bundle of village abstracts, and a number of blank slips having the name of the caste at the top and two columns to the left hand for the names of the two sub-divisions. Taking up Bráhmans for instance, he wrote all sub-divisions beginning with A on one set of slips, those beginning with B on the next, and so on, heading all these slips with the word Bráhman. If in the village abstract he found 5 males entered opposite a certain sub-division,

Number of Tribal Sub-divisions for Jats and Rajputs.

No.	District.	Jats.	Rajputs.	No.	District.	Jats.	Rajputs.
1	Dehli	2,878	722	19	Ráwalpindi	679	1,631
2	Gurgaon	732	433	20	Jahlam	969	813
3	Karnal	1,579	700	21	Gujrat	2,946	506
				22	Shahpur	1,130	2,253
4	Hissar	1,936	847				
5	Rohtak	2,986	590				
6	Sirsa	2,159	798	23	Multan	560	1,437
				24	Jhang	1,813	1,978
7	Ambala	1,863	560	25	Montgomery	1,170	1,403
8	Ludhiána	1,526	513	26	Muzaffargarh	3,734	306
9	Simla	56	186				
10	Jalandhar	398	245	27	D. I. Khan	3,739	149
11	Hushyárpur	337	587	28	D. G. Khan	6,400	329
12	Kángra	274	985	29	Bannu	2,862	486
13	Amritsar	258	225				
14	Gurdáspur	365	417	30	Peshawar	740	300
15	Sialkot	639	381	31	Hazára	109	249
				32	Kohát	380	495
16	Lahore	2,572	1,232				
17	Gujránwála	2,264	621				
18	Firozpur	3,167	1,548		Total	52,380	23,925

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Part II.—Tabulation of the Figures.

he wrote the figure 5 opposite the name of that sub-division: if 7 more males occurred in the second abstract, he cut out the 5 and wrote 12, and so on, not entering the individual numbers, but always adding on the new number to the total of all previous entries. When the space opposite any entry was filled up, the entry was cut out and re-entered at the bottom of the slip, the last total of the first entry being made the starting point of the new one. The castes with very numerous sub-divisions were worked through first one by one for the whole tahsíl, the sub-divisional slips for each being kept in alphabetical bundles. The minor castes were then taken out from each village abstract, the slips for each caste being tied up together. For each of the smaller castes the figures for a whole tahsíl were worked out on only two or three sets of slips; but in some parts, and especially in the west, the number of entries in Jats and Rájpúts was so large that by the time a circle or two had been finished the slips had become unmanageable, and a new set was begun, the whole being totalled up for the tahsíl at the end.

No exact check upon these figures was possible. Even the total population would not agree exactly, as there were small errors in the figures of the village abstracts. And, as before noted, the enumerator had occasionally departed from the caste entries in the schedules, where he knew them to be wrong. But the totals for each sex and caste were compared with the tables, and where the discrepancy was considerable the figures were re-examined. Thus the results can only be taken as approximately accurate. But they are not needed for the Imperial tables; and the errors will be so small in comparison with the numbers involved, as only very slightly to impair the practical value of the information. As it was, the Divisional officers estimate that, on an average, a month of their time was taken up by this work; while the compiling office spent at least six weeks more over the figures. To have abstracted them from the schedules and brought them into perfect accordance with the other figures would have taken the offices six months at a moderate computation.

915. Preparation of divisional tables and tahsil summaries.—When the village tables were complete, the Divisional officers prepared from them tables in duplicate for each district and division in the form of the final Census tables, showing all details contained in those tables separately for towns and villages in each tahsíl, together with district and divisional totals. These were filled up from the village-tables by direct transfer of the totals for each tahsíl, save where details were given by religion in the village and not in the final tables, in which cases the figures for the various religions were added together in separate forms devised for the purpose in order to obtain the figures for total population. Check sheets exhibiting side by side all figures taken from separate tables that should correspond with each other, were prepared for these tables and sent up with them. Further, as the village tables were not only very numerous but also, encumbered as they were with details for each separate block, exceedingly cumbrous, abstracts were prepared for each tahsíl exhibiting for each town and village in the tahsíl the following figures, with details of sex in each case:—

(1) Total population.
(2) Under 10 years of age.
(3) Unmarried.
(4) Hindu.
(5) Sikh.
(6) Musalmán.
(7) Male agriculturalists over 15 years of age.
(8) Under instruction.
(9) Able to read and write.
(10) Numbers of each of the twelve most numerous castes in the tahsíl.

No. 7 showed the number of adult males who had returned themselves of any occupation which was classed in the Divisional office as agricultural; while generally the classification of entries effected during tabulation will cause discrepancies between the details of these abstracts and of the final tables. But the differences will be unimportant, and the abstracts will be convenient for ordinary purposes of reference. They are, however, only extracts, and have not been subjected to any check beyond the ordinary comparison with the original; and the figures of the village tables alone can be considered as authoritative. Both the village-tables and the abstracts have been sent to the district offices for record.

916. Progress of the work.—The schedules and staff ought to have been collected at the Divisional offices by the 15th March at the latest. But they were really not all there till the end of the month. Meanwhile the examination of the schedules had been pushed on, and abstracting was in full swing by the beginning of April. By the end of the month we had abstracted the figures for ten million souls. During May we dealt with 8 million more, leaving less than a million for June; while all but two of the offices had finished their abstracting before the end of that month. The Jálandhar office was short-handed owing to the fact that two of the three districts were under settlement, while the patwáris of the third (Kángra) were often ignorant of the Persian character. Accordingly part of the abstracting for this division was done by other offices which had finished their own work early. We found the average number of souls for which complete sets of figures were abstracted by each abstractor in a working day to be as follows:—

From starting work to end of April	369
Last week of April	518
Month of May	680

These were, I believe, the highest averages obtained in any province in India. The averages in the several offices differed considerably, being smaller in those districts which included large towns with their numerous occupations, and in which several religions were commonly found in a single village.

During the month of June the entering up the entries in the village-tables and the tabulation of tribal sub-divisions was completed save in Jálandhar and Pesháwar, which were about a week behindhand; and the work left to be done in July was the final check, the preparation of Divisional tables and tahsíl summaries, and of certain tables needed for the information of the Registrar-General of Great Britain. The patwáris were gradually drafted off to their homes from the beginning of July, and the offices finally closed at various dates between the 3rd and 24th, except that of Jálandhar which only completed its work on the 6th August, having had to wait for the schedules of Láhul and Spiti which were not received till the end of July. The Divisional tables were translated in the central compiling office and issued together with one set of the vernacular tables to Deputy Commissioners by the first week of August, and to Commissioners some ten days later. The record of progress is continued at the end of the next part of this chapter.

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Part II.—Tabulation of the Figures.

917. Changes in the instructions issued.—Great inconvenience was caused throughout the process of abstracting and tabulating, by the changes which were made from time to time in the forms to be employed and the nature of the information to be exhibited. In February 1881 Mr. Elliott, who had previously taken the opinions of Provincial Deputy Superintendents in the matter, issued final instructions regarding the whole process of collecting and displaying the results of the Census, together with a complete set of registers, village-tables, and final tables, these last varying in a good many details from those proposed by the Census Committee of 1877-78. His consent was obtained to certain modifications which were necessary in order to display additional information desired by the Panjáb Government; and work was begun on the lines so fixed and was pushed on vigorously. Meanwhile Mr. Plowden had taken charge of the office of Census Commissioner; and he issued a new set of instructions, with new registers, village-tables, and final forms, which reached this office on the 10th April. These necessitated our doing over again the whole of the abstracting for religion, Christian sect and race, age, civil condition, education, and infirmities. When the orders reached us we had abstracted the figures for four millions of people. I was away at the time inspecting my offices. I had to obtain the orders of Government on the alterations made, to revise my whole instructions which had been issued in a complete form for all the work to be done in the Divisional offices, to devise new forms based on Mr. Plowden's but embodying the local modifications for the Panjáb, and to translate and print them all,—the forms by thousands. All this took time. The result was that about half our abstracting and a good deal of our tabulation had to be done over again; and the same was the case in Native States, a result which was particularly to be deplored. Orders regarding the tabulation of floating populations were only received on 6th June, and on the same day I received instructions to the effect that the tables we had prepared for the Registrar-General in accordance with instructions received in the end of April, were useless and must be compiled afresh on a new basis. Many minor orders and suggestions were received from time to time during the progress of the work. Orders on the classification of occupations were not received till 13th April 1882, when all our tables were already complete.

918. Evil effects of these changes.—Of course it was quite unavoidable, from the very nature of things, that something of this sort should have happened. A general Census of the Empire had been taken for the first time, and as difficulties arose and experience accumulated during the progress of the work, new orders were necessarily to be expected. I only notice the point because it had a very serious effect upon the progress of our work, and because I wish to show how important it is that a complete set of instructions should be issued in the first instance on a future occasion. The way in which it affected the work of the Tabulating offices was this. I had at first issued the most stringent orders that, as soon as the figures for a tahsíl were abstracted they were to be tabulated at once, and that as soon as the figures for a district were tabulated the final forms for the district were to be filled in and sent to me. In this manner we should have slowly gained experience in tabulation as the work progressed, we should have found out what the difficulties were and how best to meet them, doubtful points would have been referred for orders, only picked men would have been employed as tabulators, and, most important of all, the final tables would have been examined district by district as they came into my office, errors of method would have been pointed out, to be avoided in other tables, and discrepancies would have been detected and the tables returned to the Divisional offices for correction. But directly I received the new set of forms and instructions from Mr. Plowden I had to stop all tabulation, as the old tables would have been useless. Instructions and forms for abstracting on the new lines had to be got out, first for the figures which yet remained to be abstracted, and secondly for the revision of the work which had been already done; and the result was that, by the time instructions and forms were issued for the new tabulation, the abstracting was in such splendid swing and so nearly finished, that my offices protested with reason against taking any of the staff off the work to begin the tabulation. Consequently the tabulation was done quite at the end of the work for all the districts at once, the whole staff was employed on it, the final tables for the whole Province were passed into my office within a fortnight, and immediate examination was impossible and would have been useless, for the Divisional offices had been closed and the staff dismissed to their homes. One result was that the manner in which the Amritsar final tables had been prepared was only gradually brought to light as the Provincial tables were compiled one by one. If it had only been detected earlier, I should have had no hesitation in recalling the whole establishment and making them do much of the tabulation work over again. Indeed, if I had been able to follow my original system the difficulty would never have arisen, for an examination of the first tables would have disclosed the errors in system, and they would have been corrected in them and avoided in others.

919. The peculiar circumstances of the Divisional offices rendered it especially unfortunate that the preparation of the district tables was thus crowded into the last two or three weeks. The staff consisted of patwáris, accustomed to live in their villages where their work was light, their credit with the village grocer unlimited, food cheap, and lodging and fuel free, while pickings and perquisites of sorts were not wholly unknown. These men were brought in some cases more than 200 miles by road, to live separated from their families in towns where rates were high and everything had to be paid for and that in cash, while for 3½ months they were kept at arduous and incessant labour. In Pindi something like famine rates prevailed. I did what I could for them; as many as possible were lodged free in the buildings we took as offices, small grants in aid were made to those who had to come long distances, their regular pay was advanced them, and a monthly allowance of R2 per month was given them. This sum was, I am convinced, not enough to cover the additional expense they were put to; and many months after the work was all over a bonus of R2 per month was given them in addition. But this did not affect their feelings at the time. The men were generally picked men, the work was new, the supervision close, the competition between the different offices keen, and the way in which they worked was beyond all praise. But as the work drew to a close and the men saw hopes of speedy release, their one idea was to have done with it and get away home. And it was exactly at this stage that the final checking was done and all the

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Part II.—Tabulation of the Figures.

Divisional tables were prepared. In the Lahore office, indeed, it was found that the totals of certain tables did not agree, false check statements having been prepared by the supervising staff, while the men in question had been dismissed to their districts. They were at once recalled, and the tables corrected. I reported the men in fault to the Commissioner; but apparently very light punishment was awarded to them. But this instance, which was brought to light by the office itself, was the only one which occurred; and I believe that, save in the Amritsar office, even this last work was on the whole well and conscientiously done; though it would have been done better had it been done gradually.

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Part III.—Compilation of the Figures.

PART III.—COMPILATION OF THE FIGURES.

920. The Central compilation office.—The vernacular staff of the central office consisted of 30 patwáris, 3 qánúngos, and a sadr qánúngo, selected from the best men in the Divisional office staff, and as far as possible representative of every part of the Province. These patwáris received an allowance of ₹ 7 each in addition to their ordinary pay. The translating staff consisted of 12 clerks obtained in Lahore, and was temporarily supplemented when necessary. I had roughly estimated that their services might be needed for three months at the outside; but it will presently be seen how widely I had miscalculated. After four months the Deputy Commissioner of Karnál, who had lent his sadr qánúngo at great inconvenience to himself and hindrance to his work, was compelled to recall him; and the loss to me was most serious. The office was superintended by an Extra Assistant Commissioner who was selected on account of his general ability and knowledge of English.

921. Translation of the Divisional tables.—The first thing to be done was to translate and circulate to Deputy Commissioners and Commissioners the tables for their districts and divisions. The tables were translated exactly as they were received from the Divisional offices, without examination or classification of entries. It was important to furnish as quickly as possible the material for the district Census reports: and the errors afterwards detected and corrected in the central office were so small as in no way to affect the general significance of the figures.

922. Compilation of the figures.—The final tables may be divided into two classes. In those displaying total population, age, civil condition, Musalmán sects, education, and infirmities, the number of columns and their headings had been fixed from the first, classification was unnecessary, and all that had to be done was to transfer the figures of the tables furnished by the Divisional offices to the Provincial tables, to add areas, statistics of former population, and the like, to calculate such averages as were needed, and to translate the results. In the other tables we had, before this process could be begun, to classify the headings of the Divisional tables, which were in all cases far more numerous than those which it was proposed to exhibit separately in the final tables. Now if the Divisional tables had been prepared with absolute accuracy the compilation of the tables of the first class would have been simple and easy. But unfortunately this was not always the case, and on comparing them numerous small discrepancies were detected. To remove these we had first to go back to the village tables, which gave us figures for each separate religion only and not for total population, and gave them in 252 sets of items, one for towns and one for villages in each tahsíl. In many cases the totals of the village tables did not agree, the check having been made page by page, and the final totals not having been sufficiently carefully given; and in these cases we had to examine the entries for individual blocks, of which there were 53,000 in the Province. But only in the Amritsar division had we to go back to the schedules themselves, or rather to the registers abstracted from them, before we could remove the errors. The same remarks apply to the second class of tables, with the addition that in them an error in any one of our items had to be searched for among all the headings of the village-tables which had been included in that item—in some cases a very large number.

923. Adjustment of errors.—Check sheets were kept up showing the plus or minus differences between the various entries in each pair or more of tables that ought to agree; and as each error was run to ground, it was entered in a memorandum of *corrigenda* and the difference originally shown in the check sheet reduced by its amount. The *totals* of the village tables were examined in all cases of discrepancy, however small. When this had been done the check sheet showing the residual differences was brought to me. I then decided which of these differences were sufficiently important to necessitate our going behind the totals in order to remove them, and examining the figures for the individual blocks, of which there were an average of 420 in each tahsíl, and many more in some. Where the difference was minute I adjusted (*vulgice* fudged) it in the largest item concerned, or, if it was capable of distribution, distributed it rateably over the few largest items. This course I have already justified in section 898. No adjustment of the sort was made except by me with my own hand; and the danger that the knowledge that I did adjust minute differences might lead the clerks to take a similar license, was guarded against by the invariable rule that every error detected, corrected, and the correction taken credit for in the check sheet, was shown to the superintendent, who was personally responsible that the error and correction were genuine. In making these adjustments I did precisely what is daily done by every surveyor. No survey closes with absolute exactness, and the closing error, when inconsiderable, is distributed over the whole survey. It is indeed possible, by repeating observations over and over again, to reduce the probable error to insignificance; and this is the course adopted in great trigonometrical surveys which are to form the basis of further work. But these are exceptional; and in all ordinary field work small errors occur and are adjusted without wasting further time upon them.

To give a concrete instance of my procedure. The details of age in Tables VI and VII gave us more trouble in this way, and displayed a larger number of discrepancies than any other pair of tables. Neglecting all totals, and taking districts and separate religions only, Table VI contains 14,256 and Table VII 8,976 separate entries which had to correspond with each other, the total single, married, and widowed of any given age, sex, and religion in Table VI having to agree with the number of that age, sex, and religion in Table VII. The corresponding entries in the *totals* of the village-tables from which these figures had to be got numbered 109,728 and 69,080 respectively, while the corresponding number of entries for individual blocks in the village-tables were 22,888,656 and 14,411,376. When the totals of the village tables had been examined, the details of the differences that remained were brought to me. Of these I directed a certain number to be further reduced by examination of the block entries. Eventually there remained 67 differences, all of which I considered too minute to be worth further expenditure of time and

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Part III.—Compilation of the Figures.

money, and adjusted them arbitrarily. Of these 67 differences, in 47 cases the discrepancy was of 1 only in 15 cases of 2, in 3 cases of 3, and in 2 cases of 5. In adjusting them I assumed that Table VII, as giving greater detail of age, was right and Table VI wrong; and I made all adjustments up to 19 years of age in males and up to 14 in females in the table of single, above 49 years in females in the table of widowed, and in other cases in the table of married; that is to say, in the table which contained in each case the largest numbers. The two largest errors, of 5 each, were in the Hindu females of the Amritsar district. That is to say, my tables gave me the figures shown in the margin. In this case, instead of comparing block by block the entries of the two tables, I wrote 10,313 instead of 10,318 and 4,204 instead of 4,199 in Table VI. Perhaps, so far as the entries in the schedules go, I was wrong, and the corrections should have been made in Table VII. But this much is mathematically certain, that the corrections being far within the limits of the mean error of our figures, the probable variation from the actual truth was identical for both pairs of entries, and that the one was therefore, within the possible limits of our knowledge, exactly as accurate as the other.

		Age 25-29.	Age 50-59.
Table VI	Single	85	20
	Married	10,318	3,339
	Widowed	845	4,199
	Total	11,248	7,558
Table VII		11,243	7,563
Difference		+5	−5

924. With the single exception of the caste table (VIII A), the above is a fair sample of the differences thus adjusted. They were always minute, seldom being larger than 3. But in the caste table I took a somewhat wider license. The figures were needed for Provincial and not for Imperial purposes; while the number of entries which we had to re-examine and re-compile in order to correct errors which remained after examination of the village table totals was enormous. The caste entries in the several Divisional tables amounted to 8,392: each included two sexes, and almost all from two to four religions; and the separate items for each block had to be examined. Where the error was considerable this labour was of course undertaken. But eventually I adjusted a certain number of residual errors, of which all such as exceeded 10 are shown in the table below. In the greater number our totals were in defect, and the difference was credited to "Unspecified." Where they were in excess I distributed the difference rateably among the two or three most numerous castes of the religion in error; usually between Jats and Rájpúts, the two largest castes, and the two which included the greatest number of small items, in adding which together there was most danger of mistakes. The effect of this adjustment in the case of the largest difference, that in the Musalmán females of Montgomery, which happens to be one of excess, was as follows. In Musalmán Jat females I changed 18,388 to 18,346; and in Musalmán Rájpút females, 24,366 to 24,311:—

ADJUSTMENTS EXCEEDING 10 ARBITRARILY MADE IN TABLE VIII A.

1	2	3	4	5	6	7	8	9	10	11	12	13	14
	Hindu.				Sikh.				Musalman.				
	Males.		Females.		Males.		Females.		Males.		Females.		
Districts.	Total No.	Difference.	Total No.	Difference.	Total No.	Difference.	Total No.	Difference.	Total No.	Difference.	Total No.	Difference.	Districts.
Dehli	260,046	22	...	...	...	...	...	...	...	...	...	...	Dehli
Karnál	...	...	...	...	...	...	...	...	82,485	76	73,698	68	Karnál
Hissar	...	...	176,276	62	...	...	...	...	60,820	15	...	...	Hissar
Rohtak	253,113	11	...	...	...	...	...	...	...	...	...	...	Rohtak
Sirsa	72,045	19	...	...	...	...	12,624	11	...	...	...	...	Sirsa
Ambala	382,006	23	307,606	14	...	...	...	...	163,631	11	140,492	68	Ambala
Ludhiana	153,572	11	121,668	48	...	...	...	...	113,878	33	100,076	61	Ludhiana
Jalandhar	186,736	14	151,556	43	...	...	...	...	191,425	13	...	...	Jalandhar
Hushyárpur	293,470	14	256,715	61	...	...	25,930	12	...	...	...	...	Hushyárpur
Kángra	357,610	23	330,025	89	...	...	...	...	21,231	18	...	...	Kángra
Amritsar	...	...	...	...	...	...	92,946	29	222,927	24	190,280	19	Amritsar
Gurdáspur	...	...	...	...	42,293	26	...	...	209,429	86	...	...	Gurdáspur
Siálkot	...	...	...	...	...	...	...	...	353,581	87	...	...	Siálkot
Lahore	...	...	83,021	39	...	...	...	...	...	...	274,805	39	Lahore
Gujránwála	...	...	57,804	28	...	...	...	...	242,466	81	...	...	Gujránwála
Firozpur	93,910	37	...	...	...	...	...	...	168,073	77	142,479	63	Firozpur
Ráwalpindi	52,464	11	...	...	...	...	...	...	382,856	82	...	...	Ráwalpindi
Jahlam	33,609	14	...	...	...	...	...	...	272,977	16	243,768	25	Jahlam
Gujrát	38,677	23	...	...	...	...	...	...	318,239	19	289,286	26	Gujrát
Shahpur	30,330	26	28,696	18	...	...	...	...	188,714	13	169,028	78	Shahpur
Multan	...	...	...	...	...	...	...	...	238,591	88	197,310	12	Multan
Jhang	34,696	12	30,196	22	...	...	...	...	177,680	63	149,230	77	Jhang
Montgomery	45,486	16	...	...	...	...	...	...	180,274	94	150,221	97	Montgomery
Muzaffargarh	...	...	...	...	...	...	...	...	158,815	94	...	...	Muzaffargarh
Derah Ismail Khan	29,711	11	...	...	...	...	...	...	207,282	27	177,962	12	Derah Ismail Khan
Derah Ghazi Khan	...	...	...	...	...	...	...	...	173,671	69	...	...	Derah Ghazi Khan
Pesháwar	26,544	11	...	...	...	...	...	...	296,717	48	249,400	48	Pesháwar
Hazara	...	...	...	...	...	...	...	...	206,004	19	179,755	78	Hazara
Kohat	...	...	...	...	...	...	...	...	92,221	21	...	...	Kohat

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Part III.—Compilation of the Figures.

925. Classification of entries.—In the case of religion, birthplace, language, caste, and occupation, before we could begin to compile the Provincial from the Divisional tables, we had to classify the headings in the latter. The table on page 457 shows the number of separate headings included under each main group in the several Divisional tables. Of course, to take a single instance, some of the 2,551 caste headings of Multán and of the 989 similar headings of Ráwalpindi were identical. But the great majority were not. Thus, the list of separate and distinct occupations for British Territory alone numbered as many as 2,975. The bottom line of the table gives the number of headings in the final Provincial table; and shows that, for instance, the 8,392 caste entries had to be grouped together and classed under 333 final heads.

The method adopted was as follows. Take the caste table as an example. The caste headings in each Divisional table were numbered consecutively. An alphabetical list was then drawn up in vernacular containing in the margin every caste shown in those tables. The body of the list contained twenty-two columns, two for each division; and opposite each caste were entered, under each division in which it occurred (1) the serial No. of the caste in the Divisional table, (2) the number of souls entered in that table under the heading. The castes in this list were also severally numbered. I then classified the headings, and sent into the office a memorandum showing the castes which I had fixed for exhibition in the Provincial tables, and opposite each of them the several numbers borne by those caste headings of the above list which I proposed to include in it. Thus suppose Chauhán was number 357 in the list of caste headings. In the classification memorandum I prepared, 357 would be found under Rájpút, together with the numbers of all the other caste headings I had classed as Rájpúts. The list would be referred to; and opposite No. 357 would be found for each Divisional table the serial number of the heading Chauhán. The compiler had vernacular slips for each division, showing on top the name and number of the caste to be shown in the final tables, to the left hand the serial numbers and names of the caste-headings of the list which were to be included in that caste, and in front of these, columns corresponding with the columns of the final table for each district in the division. He would enter Rájpúts at the top, 357 Chauhán to the left hand, and opposite it the figures for Chauhán in that division; and so for each of the items classed by me under the heading Rájpút. The totals of these items would be the entries for the division in the final table under the head Rájpút. These totals were totalled for the whole table to check the figures, and when the slips were correct they were made over to the translating branch, who filled in from them the sheets for the press. The object of giving, in the first list, the number of souls under each heading, was to enable me to decide whether it was worth while to show the heading separately, or, if not fully understood, to make inquiries about it from the district.

It will be noted that almost throughout this process the entries were identified by numbers merely. The compilers were told to class No. 357, which comprised No. 27 in Delhi, No. 98 in Hissár, and so on, as Rájpúts. I made my grouping in English from the lists read out to me in vernacular, and my grouping had to be retranslated into vernacular. Thus although the most careful comparison and check was exercised throughout, there were many opportunities for error. For this reason the compiler's slips showed the *names* as well as the numbers of the headings. When they were complete the slips for each division were carefully examined by picked men from the division, who could hardly fail to detect any error in classification. Besides this the numbers grouped were also compared to see if any number had been omitted altogether, or grouped twice over in different places. Thus, if Chauhán was No. 357 and Agarwál No. 257, and under Rájpút I entered 257 instead of 357 by mistake, the result would be that (1) No. 357 would not be classed at all, (2) No. 257 would be classed twice over, once under Banya, and once under Rájpút, (3) the slip for Rájpúts would contain the entry Agarwál which would be obviously wrong. It is exceedingly unlikely that any material errors escaped this triple check. All cases of this sort were sent back to me for further orders, and thus I personally decided upon the classification of every single entry in the tables.

926. Principles of classification.—This is equivalent to saying that the whole classification depended for its value upon my personal knowledge of the subject-matter, added to such information as I could obtain by enquiry within a very limited time. With respect to religion and sects there was no difficulty, if I except the very extraordinary forms which the names of Christian beliefs and sects sometimes assumed in the hands of vernacular writers, which often compelled us to refer to the original schedules before we could identify them.

The number of languages was small, but my knowledge of the subject was still smaller; and I found that very little was known about the hill languages by any body to whom I could refer. I had to class them as best I could; and even the little knowledge I have since been able to acquire has shown me that I made mistakes.

The birth-places I dealt with with the assistance of Mr. Baness's "Index Geographicus," which I found most useful. Every name which coincided with the name of an Indian district or state I assumed to refer to that district or state. Where the name agreed with that of a considerable city I assumed it to refer to that city, and where there were several towns of the same name I took the most important of them. But there were still a large number of entries, especially in Derah Gházi Khán, for the most part including only two or three persons each, which I could not identify. Where the form of the name was distinctly Indian, as Xnagr or Ypur, I classed it as "India, part unspecified." So where the language was obviously English, I classed the place under "British Isles." But there were still many wholly illegible or unintelligible entries which had to be consigned to the limbo of "Birthplace unspecified."

With occupations the chief difficulty was the immense number, my list showing not less than 2,975 entries for British Territory alone. This large number was due more to the great diversity of local terms than to any essential difference in meaning. The instructions of the Census Commissioner were that our classification was to be based upon that adopted for the English Census so far as it was applicable to the circumstances of the Panjáb. I first translated or, where that was impossible, transliterated the whole 9,038 entries of the list, with the serial number of each. I then affixed to each the number of the order and sub-

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order in which I classed it. These entries were then collected into separate slips for each sub-order; and those I again classified into the groups which I proposed to show as separate occupations in my final tables. A very considerable proportion of the terms used were wholly unintelligible to me. Where they related to village life, my staff, drawn as it was from all parts of the Province, could generally supply the requisite information. But the technical terms used in the industries of cities were often either unknown to my staff, or my own knowledge was insufficient to enable me to understand their explanations. Dictionaries gave but little assistance; and my Munshi used to come to office every morning with his pockets stuffed with samples of fabrics, sweetmeats, drugs, and merchandize of sorts; while I spent many hours in the bazaar consulting with friendly traders and artificers. The number of terms I had eventually to refer to the district for explanation was not large. In the case of double occupations, of which a large number were shown in the tables, especially for mercantile men, I followed the same rules which I had laid down for the guidance of the tabulators, and which are given in section 906.

927. Classification of caste entries.—But the classification of castes was the one, not only which presented the greatest difficulty, but the results of which must be regarded with the greatest uncertainty. As already stated, the name of the tribe or clan had been in numberless instances entered in the schedules, instead of the name of the caste. These had been to a certain extent classified under the castes to which they belonged during the process of tabulation in the Divisional offices. But, as explained fully in the Chapter on Castes, the tribe has almost superseded the caste in the Western Panjáb; and many tribes are held to belong to one caste in one district, and to another in a second and adjoining district. In still more instances the tribal name is one common to many castes, and therefore tells us little or nothing as to the place which the people should occupy in the caste tables. The plan followed was this. I first went right through the caste lists and classed all the names with which my own knowledge, limited unfortunately to the people of the Dehli Territory, and a careful annotation of every Panjáb Settlement Report in print and of such other authorities as I had access to, enabled me to treat with any confidence. I then collected my whole staff and went through the remaining names in committee; not acting, however, upon the statement of any *patwári* or *qánúngo* save in the case of general consensus as to a well-known and numerous tribe, unless he came from the part of the country to which the entry belonged and was personally acquainted with members of the tribe.

The entries still remaining unclassed numbered many hundreds; but a great part of them covered only a few persons each, very often fewer than 50, not unfrequently even less than 10. I next circulated to all Deputy Commissioners in the Province a table containing 84 of the largest of these entries, and supplementary tables of 293 smaller entries in the Multán, Ráwalpindi, Pesháwar, and Ambála divisions, especially the three first, asking for their opinion as to their classification. At the same time I circulated to all districts a list of 201 castes which I proposed to distinguish in my tables, together with a detail of the principal headings which I had included under each. With respect to these I asked (1) whether I had joined together what ought to be separated; (2) whether I had separated what ought to be joined together; (3) whether any caste was omitted from the list. I further gave a list of cases about which I felt doubtful, and of castes about which I especially needed information. I distributed these circulars widely throughout the Province to both officials and non-officials who I thought might help me; and the replies and criticisms I received were both numerous and bulky, and in many cases of the greatest possible value. In this manner I referred for local opinion the classification of every entry numbering not less than 50 souls, and of many under that number, respecting which I felt any doubt. As for the remaining small entries, still very numerous, we again went through them one by one in committee; but this time I accepted any opinion based on reasonable grounds. But even so the entries which we had in the end to leave unclassified were still very numerous, though their totals formed a quite insignificant fraction of the population. In the cases referred to above, where people had returned themselves under a tribal name belonging to several castes, they were assumed, in the absence of any indication or information to the contrary, to belong to the landowning caste to which the clan belonged. For instance, almost every menial caste has a Bhatti clan. But while a Bhatti Rájpút would ordinarily describe himself as a Bhatti, leaving you to supply the conclusion that he was also a Rájpút, no menial would ever dream of so returning himself. Consequently, we were fairly safe in assuming that persons returned as Bhattis were also Rájpúts, and in classing them as such.

It will be seen that I spared no personal pains in classing my castes. The amount of labour involved in reading through and making notes of some sixty printed Settlement Reports, in comparing and digesting some 100 replies, many of them very lengthy, in referring to standard works on the subject, and in then coming to a decision in each case, was greater than will readily be believed. And a perusal of the Chapter on Castes will show that I am still only too doubtful as to the justness of many of my conclusions. My guiding principle here, as throughout the Census, has been to exhibit separately all items regarding the propriety of uniting which under a common heading there was room for reasonable doubt. Anyone who cares to do so can at any time add the figures together; but once joined in my tables, it would have been impossible to separate them again.

928. Tribal sub-divisions.—In section 914 I have described the method in which the figures for tribal sub-divisions were tabulated. The number of such sub-divisions for Jats and Rájpúts respectively were 52,380 and 23,925. Of course it was wholly out of the question to exhibit even a small portion of these clans in our clan tables. Nor indeed was it in any way advisable; for a large proportion of them included a quite insignificant number of persons. The first thing to be done, therefore, was to select the clans which were worthy of separate tabulation. Clerks went through the clan tables for each caste in each district, and prepared lists showing all clans which reached the number of 1,000 souls in any one tahsíl of the Province, and opposite each of these the total number for each district in which this condition was satisfied. Clans not attaining the above standard were neglected. I went through the tables thus prepared, with a list in my hand, drawn out by myself, of all clans mentioned as being important in any Panjáb Settlement Report; and selected the clans for which I thought it was worth while to obtain com-

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plete figures, rejecting those, where I rejected any, which appeared in not more than two districts. The clans thus rejected I had intended to mention in their proper places in the Chapter on Castes, and to give the figures for each; but lack of time has prevented my doing so, and I must now wait for the detailed clan tables which I hope some day to publish. In some cases, where it was doubtful whether castes returned under separate names were in reality separate or one and the same, I selected more unimportant clans than I otherwise should have done, and even reduced the standard from 1,000 to 500 in a tahsíl, in hopes that the results might throw light on the doubtful point.

The clans thus selected for tabulation were made over again to the compilers, who took a slip for each clan divided into columns for districts, went through the clan tables of all caste-headings included by me in the particular caste under treatment for each district in turn, and entered up all the numbers found opposite each clan in any of those tables, whether the clan appeared as a primary or as a secondary subdivision, the totals of the items so entered giving the figures of the final clan tables. This method led to double entries. For instance, Siál is a very important Rájpút clan belonging to the Punwár tribe. But Siáls as a rule have forgotten that they are Punwárs, and few have entered themselves as such. Thus it was necessary to take out the figures for Punwárs and Siáls separately. But in doing so the few individuals who had entered themselves as both Punwárs and Siáls were necessarily shown twice over, once under each heading. To meet this difficulty the clerks ticked off each item as it was entered in their tables; and if it had to be entered again under a different heading, made a note of the details in a memorandum form devised for the purpose. On examining these memoranda I found that the entries were very numerous, but as a rule included an exceedingly small number of people: and accordingly I have only noticed the more important cases in the text of my report. Another point which gave much trouble was the various spelling of clan names in different districts, or even in the same district. For instance, we find the Dháriwál clan of Jats spelt indifferently Dháriwál, Dhálíwál, or Dháníwál, and all three forms occurring without the "h," or with the long "a" shortened. On the other hand, some names which were perfectly distinct approached one another very nearly in spelling; as, for instance, Sindhú and Sidhú. Here everything had to be left to the local knowledge of the compilers, who were instructed to disregard mere local varieties of spelling and to refer all doubtful cases for orders. The tabulation of the figures for Pathans and Biloches in frontier districts has already been noticed in the section which treats of those castes.

929. Laborious nature of the work.—The above sketch will show the laborious nature of the task of compilation, and there was much additional work of a miscellaneous nature of which I have made no mention. The correction of proofs for the press of some 1,300 pages of tabular matter was in itself no light task; while I was continually called upon to furnish figures for the information of other Departments, which often had to be compiled from the village tables at the cost of considerable labour and delay. The compilation itself was most intricate and required the greatest exactitude. In compiling so many items under different heads errors naturally occurred, and the totals were often found not to agree. Now where this was found to be the case it was impossible to tell, without a re-examination of the whole work, whether the error lay in the figures we were compiling or in the process of compilation; and this latter had often to be gone through again, only to find that it was correct, and that the discrepancies must be looked for in the figures of the Divisional tables. When they were detected the village tables had to be examined, often the figures for individual blocks had to be checked, and sometimes the original schedules had to be referred to. Had it been possible to follow my original plan of making the tabulation keep pace with the abstracting, each set of tables would have been examined as it came in, and the errors would have been corrected in the Divisional offices. I have explained in section 918 how the changes made in the forms after the work had commenced not only largely increased the number of errors to be corrected, but transferred the labour of correcting them from the 1,000 men employed in the Divisional offices to the small staff of 30 clerks in the Central office.

The above remarks account in a great measure for the fact that we actually took 9 months instead of 3 over the compilation. But setting aside the disturbance of my plans just alluded to, there is no doubt that I had altogether under-estimated the amount and difficulty of the work. I had no conception that the entries to be classified would prove so numerous, or that their compilation would take so long: and I believe that others were no wiser than myself. Mr. Elliott, for instance, estimated provisionally that 30 clerks would compile the whole figures of the Province in a single month. The fact is that the work was new, we had no data to guide us, and it is hardly to be wondered at that our estimates proved to be inadequate. It must not be forgotten that, while the Native States abstracted and tabulated their own figures, we had to compile them; and that the population dealt with was thus increased from $18\frac{3}{4}$ to $22\frac{3}{4}$ millions, or by 20 per cent., directly compilation began. And there is another respect in which the task of compilation was heavier in the Panjáb than in any other province in India, and that is, the number of units for which separate figures had to be exhibited in the final tables. We have 32 districts, 11 divisions (including the Khaibar), 30 Native States, and 5 totals; or 84 complete sets of entries, of which 69 are independent and 15 are mere totals. The corresponding numbers for other provinces are shown in the margin.

Province.	Independent sets of entries.	Sets of entries including totals.
Assam	14	19
Burmah	19	25
Berar	22	29
Madras	25	29
Bombay	25	31
Central Provinces	33	53
N.-W. Provinces	51	63
Bengal	52	60
Panjab	69	84

Besides this we compiled the following information for each of our 84 items in addition to the information required for the Imperial tables:—

(1) details of urban and rural population for religion, sect, civil condition, birth-place, and education:

(2) details of all castes according to religion, instead of merely those shown in Imperial Table VIII without religion:

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(3) details of over and under 15 years of age, and of simple and combined with agriculture, for all occupations:
(4) details of resident families:
(5) details of occupied houses, of Jains and Sikhs, and of suburbs, for towns:
(6) tables of sub-divisions of tribes and castes.

The labour involved in Nos. (2) and (6) especially was immense. The sub-divisional tables occupied the whole compilation office for at the very least six weeks. I was, moreover, unfortunate in my selection of an Assistant to superintend the Central office. A man of exceptional mental power and possessing an unusually perfect knowledge of English, which was of the greatest possible use to me, he had had no experience in managing men, and was not well suited for the charge of an office. And this not only threw much work which I ought to have been spared upon my shoulders, but seriously hindered our general progress. At a future Census the compiling establishment should be trebled at least; and there should be two officers to assist the Deputy Superintendent, one a first-rate Tahsíldár or Superintendent of Settlement to superintend the vernacular office, and the other an English-speaking Extra Assistant Commissioner who would act as Personal Assistant and take charge of the translating department.

930. Classification at the next Census.—But at the next Census I hope that the whole work of both tabulation and compilation, so far at least as it depends upon classification of entries, will be rendered infinitely easier. The Panjáb Government has approved of a proposal that I should edit a Dictionary of Panjáb Occupations. This will include at least all the terms used to describe occupations in the Census schedules, and will explain the precise meaning of each. I understand that it is proposed to hold a committee after the reports on this Census are complete, in order to draw up a scheme for the classification of Indian occupations for use on future occasions. But whatever scheme be adopted, the Dictionary will enable the authorities to class all terms that may be entered in it at once, before the tabulation begins. All that will then be necessary will be for the tabulating clerk to look out in the classified list each occupation entered in the abstracts, and to enter against it the serial number of the main occupation-heading under which it is to be included. The columns of the village tables will be numbered to correspond; and the transfer of the figures will be easy. The entries not found in the classified list will be tabulated separately in supplementary tables, and the Deputy Superintendent will have to treat them as I had to treat the whole of the occupations of this Census. At each Census the unclassed residuum will grow smaller and smaller; while the successive editions of the Dictionary will form storehouses of valuable information on a subject about which there is at present very little available in print.

As for castes, Government has sanctioned a small expenditure on the complete tabulation of all the entries of tribal sub-divisions found in the schedules of the present Census. When the work is complete the results will be published, together with such information about the several castes and clans as I have meanwhile been able to collect from existing sources, and as I may have received in response to a letter asking for assistance in the matter which I have circulated throughout the Panjáb. The results can hardly fail to throw much light upon Panjáb Ethnology; and even if they do not afford material for an ethnical classification of the peoples of the Province, they will at least render easy the preparation of a classified list of clans showing to what caste each should be referred, which would be used by the tabulator exactly in the same manner as the list of occupations referred to in the last paragraph. Whatever form the caste columns may take in the schedule of future Censuses, it is quite certain that in numberless instances the clan will be returned instead of the caste, and that such a list will very largely diminish the labour and uncertainty which have attended the caste-classification of the present Census.

931. Progress and completion of the work.—The figures were received from the Divisional offices about the end of July; and after they had been translated and the Divisional tables issued, and the selected staff allowed a short rest, the work of compilation was fairly begun about the middle of August. In September and October I had to move the office to Amritsar, as the cholera then raging at Lahore so demoralised the men that they could not work. And here our misfortunes began. In escaping Scylla we had fallen into Charybdis, for the terrible fever epidemic which decimated the inhabitants of Amritsar laid the whole office prostrate, and during these two months very little way was made. The compilation in the vernacular was finished by the end of April, and all the tables save those for castes and occupations translated and checked. Thus the compilation itself took us more than six months of unremitting toil, besides two months of slack work. I have already explained in section 929 how enormous was the mass of work to be got through, how utterly everybody had under-estimated its amount, and how I was in a way unfortunate in the matter of officers. But here again our work was unexpectedly interrupted when nearly over. We had, as I have stated, received instructions that the English system of classification of occupations was to be followed as closely as possible in our tables; and I had accordingly followed it, completed my occupation tables in vernacular, and submitted Indices showing my classification in detail to the Census Commissioner. But just as those tables were about to be translated and checked, in the end of March 1882, came a completely new classification from the Census Commissioner, and I had to group the whole of my 2,075 different occupation headings anew. I spent a week's incessant labour on the new lists, and almost my whole office was occupied during all April (final orders were not received till 13th April, but we did not wait for them) in compiling the new Imperial table. Thus I was left at the beginning of May with the two heaviest tables, those for castes and occupations, untranslated, and unchecked in English. My vernacular office was composed of patwáris and qánúngos who were urgently needed in their districts, from which they had been absent for a year, and was superintended by an Extra Assistant whose services were required by Government. I could not keep them waiting till the tables were translated and the English copies checked, in order to correct any error that might be detected after the translation and check were complete; and I let them go. Thus I was left with only a small translating office, and with the whole translation and final check of the two largest tables on my hands. Moreover they were precisely the two tables in which I found the largest number of errors due to the uncertainty of vernacular writing. Shoe-makers had been entered as lime-

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burners (*jûta*, *chûna*), cotton scutchers as cotton merchants (*pínjna*, *bechna*), Rors as Aroras or Roras, Arabs as Banyas (*mahájan*, *muhájarín*), Maliárs as Maniárs, Barháis as Bharáis, and a thousand similar errors had been committed. The whole examination of the tables and detection of these errors had to be done by myself personally, since they were not errors which any mechanical checking of figures would bring to light, but required some knowledge of the subject to detect them. And when detected they had to be corrected by a small staff of six clerks. Thus some three months were spent over the examination and completion of these two tables, and till they were ready my clerks could make but little way with the preparation of the abstracts for the final Report, nor indeed could I make any satisfactory progress with it; and this because at the very last moment, when the work was done and the establishment about to be broken up, orders to recompile from the beginning our heaviest table were received from head-quarters.

932. Another reason why my Report has been so long delayed is the extraordinary dilatoriness of the Lahore Central Gaol Press in printing the final tables. These tables were begun in December 1881: a constant supply of copy has been kept up, proofs have never been delayed, and now at the end of February 1883, half the heaviest table, that of occupations, is still unprinted. With such bulky tables as those in question nothing can be done till they are in print. The number of separate items to be discussed, for which the figures are given in section 929, added much to my labours; while the various information which we collected in the Panjáb only for specially Provincial purposes, and which are detailed in the same section, greatly increased the whole work of the Census in every stage. During the whole time that I was engaged on the report I had other matters connected with the Gazetteer to attend to as well. And finally, one main cause of the delay in finishing my Report has been the utter absence of anything worthy the name of a Report on any previous Census in the Panjáb. Had there been a good Census Report to start with, such as exists in almost every other province in India, all I should have had to do would have been to compare my figures, add new facts, and modify previous conclusions. But there was nothing of the sort in existence, and I had to lay the very foundations as well as build the superstructure. It must not be forgotten in comparing the Panjáb Census with that of other provinces, that in hardly any other part of India was the record of caste made or tabulated in such great detail as here, while in no other province was any attempt made to collect information regarding tribal sub-divisions. The figures given on page 460 will show how greatly this added to the labour of tabulation and compilation. I do not think I exaggerate when I say that these figures alone added five months to the time which elapsed between the taking of the Census and the completion of the Report.

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CHAPTER XIII.
MINOR POINTS AND SUGGESTIONS.

PART I.—SUBSIDIARY MATTERS.

933. Legislation; and proceedings under the Act.—The Census was taken under Act XIV of 1880, the Indian Census Act, the third, fourth, and eleventh sections of which empowered all authorised by the Local Government in that behalf to appoint any person whomsoever to take the Census of any specified local area, and rendered him punishable if he refused or neglected to do so. The Act also provided for punishing householders who did not fill up correctly and return private schedules left at their houses for that purpose, and made persons, whether official or otherwise, who were in charge or command of or had control over bodies of men, or on whose premises considerable numbers were lodged or employed, responsible for their enumeration. Under this Act the Punjab Government issued notifications 149, 150, 151, 152, dated 11th January, and No. 396 of 31st January 1881.

These notifications extended the whole Act to the Panjáb, appointed myself, all Deputy Commissioners, and all officers in executive charge of sub-divisions and the Agents and Superintendents of the various Railways in the Province, to be Census officers, delegated to them the power of appointing other Census officers for the local areas under each respectively, and authorised them to sign declarations which should be conclusive proof of such appointments. They also empowered all Cantonment Magistrates and Tahsíldárs to sign similar declarations for enumerators, supervisors, and superintendents of Census. They further provided for the hearing of certain classes of cases before Magistrates of the 1st class, and that certain prosecutions under the Act should not be instituted without the sanction of specified authorities. The declarations of appointment for all subordinate Census officers usually took the form of a declaration made at the foot of the index-register under the signature of the Tahsíldár.

I think that some such legislative provisions were very necessary. No doubt the mass of the people did not even know that a Census Act had been passed; though in some Districts its main provisions appear to have been publicly notified, while in Gurgáon "its penal provisions were explained to the "people where necessary, and they were shown that they *must* answer." But the educated classes who alone were called on to assist, were probably very generally aware of its existence; and the Deputy Commissioner of Síálkot mentions its having had "a most salutary effect," while the Deputy Commissioner of Hushyárpur writes, "the fact of its existence was a salutary stimulant to the staff, though, "with regard to the outside public, I do not think there was a single instance in which it had even to be "warned." But it is with respect to the European and Eurasian population that such an Act was especially needed. During the discussion prior to the passing of the Act the Deputy Commissioner of Simla wrote:—

"In October 1878, when a Census of Simla was taken, the instances were not few in which the European residents filled "nonsense into their schedules, or where these, though frequently called for, were not filled in and returned to the Census officers, "who could only obtain the information after great delay and trouble. Again many European householders absolutely refused to "fill in the information for their own servants. There was also great difficulty in getting proper returns from personsliving in single "rooms in Clubs, Hotels, and Boarding Houses, who refused to give the necessary information to the Secretary or Manager filling "in the schedule. There were several instances in which enumerators were assaulted by European or Eurasian clerks and persons "of that class."

At the present Census a comparatively small number of private schedules were issued in Simla; but in no case was it found necessary to adopt coercive or punitive measures of any sort; a fact which seems to show that the provisions of the Act, which were prominently printed on the back of all private schedules, had a wholesome effect.

Prosecutions were instituted under the Act to the number of 25 in three districts only, as follows:—

Jahlam.—Five people in Talágang were fined Rs. 25 in all under section 5, for neglecting to fill in schedules for persons living in their premises or under their control.

Peshawar.—An enumerator in the city was found to have exacted a fee of two pice (three farthings) per house, and was sent to prison for three months.

Hazára.—Prosecutions for neglect of orders (section 11) were instituted, and Census officers fined as below:—

14	Patwáris fined	R68 12 0
4	Lambardárs fined	„ 26 12 0
1	Chaukídár fined	„ 1 0 0
19		R96 8 0

In Sháhpur Kandi (Gurdáspur) a supervisor tried to insist upon seeing the bride in a marriage-party. He was a Government servant, and was punished executively by the Deputy Commissioner. This I think was a great mistake. He should have been sent to gaol. In Sirsa there was only a single complaint made against a Census officer; and that a trivial one. In no other district was any instance of abuse of position or improper assumption of authority on the part of Census officers brought to notice. In Karnál

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one non-official Census officer neglected his duty, but it was a solitary instance, and it was allowed to pass. In Siálkot there were only a few objections to act as Census officers, and those were well founded. In several districts some officials were fined executively for neglect of duty. But on the whole coercive measures may be said to have been unknown; and the result speaks well for the readiness of the people to submit to and assist in the Census. As the Deputy Commissioner of Lahore says, "the enumerators were "not always men of tact and discretion; and no doubt people were occasionally questioned in an abrupt "and insolent manner. But," he adds, "I can only remember having heard of one single quarrel; that was "between an enumerator and a woman; and I believe the enumerator was in the wrong."

934. General attitude of the people.—All District officers, without exception, report that the attitude of the people towards the Census was all that could be desired. They constantly speak of "the cordial assistance" given by them, of "their perfect good humour and apparent desire that "the information they gave should be as correct as possible," of "information readily given and "assistance cheerfully and readily tendered, the people, with their lamps burning, waiting at "the door for the enumerator on the night of the Census." That foolish rumours should have obtained currency among the lower and more ignorant classes of the community was only to be expected; but it seems that they were almost wholly confined to those classes; and even their suspicions were soon allayed by the judicious action of the district staff, and by the pains taken to explain the real objects of the Census. The very elaborate and searching nature of the procedure as compared with that of last Census attracted attention; the recent campaign in Afghánistán suggested taxation, or that recruits, or even young wives for our soldiers were wanted; the care taken in defining and numbering houses and in recording families seemed to point to the probable revival of the old poll and hearth-taxes; while the minute enquiries as to occupation and source of income seemed to follow suspiciously on the recent imposition of a license-tax. But these ideas were often only mentioned to be laughed at. The grey-beards could remember two previous enumerations which no evil result had followed; and, when active interest was not felt, the feeling was generally one of amused indifference. Some of the rumours which were current in the more backward tracts are mentioned in section 40. On the frontier a little more difficulty was experienced, and some tact and discretion were needed on the part of the officers. The Pawindahs of Derah Ismáil Khán pleaded the fact that they had been counted on entering the Panjáb as a bar to further enumeration. In the far valleys of feudal Tanáwal, in some of which no European has ever set foot, some difficulty was experienced in persuading the people to allow their women to be enumerated; and in the Teri country of Kohát, where the Bárak sections were at feud with the Nawáb, they at first refused to allow him or his officials to take their Census.

935. The following extracts from the district reports are interesting. They show, among other things, that it is not difficult by judicious management, to give the people a personal interest in the work and cause them to feel pride in its success:—

Sirsa (Mr. Wilson).—"The attitude of the people was all that could be wished. They seemed sufficiently to understand the "object of the Census and to have no suspicion of any ulterior object. Not the slightest obstruction was offered. On the contrary "one and all helped as far as they could; volunteers readily undertook to act as enumerators, and the people generally gave every "information asked of them with readiness, scrupulously protected the numbers of their houses from obliteration by rain or accident, "kept lights burning on the night of the Census, and bore with great patience and without complaint the trouble to which they were "put."

Shahpur (Mr. Frizelle).—"No opposition to the work was met with, those who could assist were as a rule anxious to do so, and "glad of the opportunity of displaying their ability; not more than three or four persons of those selected asked to be excused, "and they were excused. Even those told off to blocks at a distance from their homes proceeded willingly, and yet no application for "pecuniary remuneration was made, although some slight mementoes of the services performed would be gladly accepted and have "been well earned.

"As to the general mass of the people their attitude was one of indifference, and not even curious as to the Census and its "cause; and this is the more noteworthy on account of the great prominence given to the work. Of course some absurd stories "were about, such as a search being made for a child of the Queen which had been lost, but they were raised only to be laughed at, "and most people had a very shrewd idea that no motive was immediately at work except, as they considered, a strange craze for "statistical information. No alarm whatever was felt. They are so accustomed to all sorts of enquiries being made at all times, "about outturn of crops, trade imports and exports, the number of cattle, carriage, &c., that such things come to them as a matter "of course. Any such enquiries, and even our recording of vital statistics, are likely to produce just the same effect in their minds as "a regular and entire Census."

Peshawar (Mr. Lawrence).—"It was fully expected that the Census in this district would be attended with certain difficulties. "Following quickly on the close of the war and preceded by three years' continuous drought, the Census found the people in a "very distressed and anxious condition. Tales were rife as to the objects of the Census; it was to collect a tax for paying the "expenses of the war, or it was to ascertain how many marriageable women there were for the army. The scanty Hindu popu-"lation in the villages and the large Hindu community in the city generally believed that a poll-tax was impending. These were "the views of the pessimists: others more sanguine went about saying that the Government intended to provide grain for its "subjects, and wanted to know how many mouths per family were to be fed.

"The minute nature of the house-lists and the maps of homesteads (which did not previously exist), and the careful numbering "of the houses, convinced all that something affecting their interests was about to be done. But when the preliminary record "commenced, the Pathán saw at once that no tax was to be levied. When they observed that the points most carefully gone into "were those of caste, clan, and family occupation, they argued that these details could have nothing to do with a tax. The Pathán "mind has a leaning to historical details of caste, clan, and family, and they soon realised some of the objects of the Census. No "Pathán ever gave the name of a woman, and great satisfaction was felt at the order forbidding enumerators to ask for women's "names. With the exception of the dislike felt by all for the entry of the zanána in the Census books, the Census was not unpopular. "One person renowned for piety in the Mardán tahsíl followed me for three days, begging that the members of his household might "all be entered as 'man' without distinction of sex."

Kangra (Colonel Jenkins).—"There was no disposition on the part of the people to obstruct the Census, but it was very "generally supposed that the intention of Government was to impose a poll-tax. The fact that such careful enquiries were made as "to the number of *chúlhas* lent colour to this supposition. The only argument that seemed to carry any weight to allaying "these suspicions was, that there had been two previous Censuses, and that neither of them had been followed by any increase of "taxation. Among the Gaddis a rumour was rife that Government wanted to know the number of unmarried girls, with a view "of course to their being carried off from their homes. This is reported to have caused a considerable number of marriages being "arranged before the Census took place, the month being fortunately considered a very favourable one for such ceremonies. This "report probably arose from the fact of such careful enquiries being made as to whether people were married or not."

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Kulu (Mr. Alex. Anderson).—"There was no obstruction to Census operations in Kúlu, where the people are quiet and law-"abiding. They have had little experience of such operations before, and though not afraid were very suspicious. Here as elsewhere "the first idea was that a tax was to be imposed to meet the expenses of the Kábul war, and newspaper articles as to the Patriotic "Fund and subscriptions given by Native States kept up the idea. They could not understand the careful enquiry into the number "of *chúlhas*, and some are said to have given out that a time much feared had come when there would be only one *chúlha* allowed "to a certain number of persons. These suspicions however did not give rise to even passive obstruction; they were willing to assist "in every way.

"On one point they took more than ordinary interest in the entries made. It is the custom for wives in Kúlu to live a consi-"derable part of the year with their parents; and husbands, fearing they might lose their rights if their wives were shown as living "in their parents' houses, tried to bring them to their own houses, and urged the enumerators and others to correct the error of enter-"ing the wife's name except in her husband's house. The Spiti people at first petitioned that they might be ordered to meet in one "place to be numbered."

Rawalpindi Division (Colonel Hall).—"Stories were of course told; but they were more in joke than in sober earnest. For "instance, the want of rain before the Census was attributed to it, as rain would have been fatal to the numbers of the houses. And, "as it rained a few days after the 17th February, I heard people say "Ah! It may rain now! The number may be washed off."

Rawalpindi (Mr. Steedman).—"The attitude of the people was after the first month of the preliminary enumeration apathetic. "When Census operations commenced they wondered a little, and asked one another why the Government was taking all this "trouble. Especially they were struck with the very great difference between the organization of the last and present Census. "They were puzzled at the extreme care with which they were questioned as to their occupation. This they considered proof "positive that a new tax was going to be imposed. When, however, they perceived that no measures were taken to find the actual "house of *every one*, they were somewhat comforted. Among other ideas as to the object of Government in taking the Census "were the following; that the Government wished to ascertain how many persons had been lost in the Afghán campaigns; that a "return giving the number of the warlike (*bahádur*) tribes was required to indicate whence recruits could be obtained to "fill up the gaps caused by the deaths in the late war; that the Government wanted to find somebody or other (this seems to be "peculiarly and intrinsically feeble); that it was only a part of the new settlement. In Ráwalpindi one idea was that the "population in Bengal and Madras being excessive, Government was anxious to ascertain where the surplus people could be located. "The general feeling was one apprehensive of new taxation, especially to defray the cost of the late campaigns. The people of "this district knew from personal experience what Government had been paying for carriage and grain and forage, and also for "earth-work on the Railways to Khushhálgarh and Attak. There was not the very least obstruction or hindrance whatever. "I am inclined to think that in many villages where the lambardárs or their sons had given assistance as enumerators, they were "rather proud of the accurate way in which the population was ascertained. It has now almost been forgotten that a Census was "taken only three months ago."

Jhang (Captain Bartholomew).—"At the outset of the Census operations much misunderstanding prevailed among the masses of "the community regarding its objects. The common people in towns thought that it had for its object the imposition of some new "tax on the people, others thought that Government was about to institute a conscription for the army in Afghánistán, and others "that some burden would be laid upon the people as in the old Sikh days. In villages and the jungle tracts, the ignorant people "thought they would be called upon to go to the frontier to fight with the Afgháns, or have some forced labour imposed on them "(*begár*). Many of the country people, and not few in towns, objected specially to mention the age of their females, and questions "regarding their occupation, &c., were taken as insults. People generally were disinclined to tell the numbers of the inmates of "their houses and other particulars. To remove all these difficulties and obstacles care was taken, in the first instance, to appoint "*such men as enumerators as had the confidence of the people, and they before setting to work explained the objects of the Census* "quietly. All Census officers were strictly enjoined to treat the people gently. The superior officers were constantly moving about "the district, inspecting, instructing, explaining and *encouraging*; and *while the work of preliminary record was in progress*, several "meetings were held at the headquarters of the district by the Deputy Commissioner, in which the objects of the Census were fully "explained to the headmen and notables. The result of this was that both during the preliminary enquiry and at the final enu-"meration, the demeanour of the people was all that could be desired; and this in a wild district, whose inhabitants a quarter of a "century ago lived chiefly by plunder, is specially remarkable. All suspicion soon ceased, and the people generally lent their aid to "furthering the objects of the Census."

Lahore Division (Mr. Elsmie).—"The Deputy Commissioner of Gújránwála states that the people were neither obstructive "nor suspicious, but they thought that Government was putting them to much trouble and annoyance for no adequate purpose. "Major Grey from Firozpur, on the other hand, says the people were interested in the Census and forwarded it in every way. "I was in camp myself in Lahore and Firozpur at the time of the Census arrangements, and had opportunities of talking to many "of the people on the subject. From what I heard I fully agree in Major Grey's view of the general attitude."

Hushyarpur (Mr. Coldstream).—"The general attitude of the people was most praiseworthy. No obstruction was encoun-"tered. Every one took his part in the scheme submissively; and I think I may even say there was a general alacrity to obey "the orders of Government in the matter. No cases of wilful concealment of persons from enumeration have to be reported. The "supervising officers only mention that the people were harassed by the constant verifications or *partáls*. It was not to be expected "that among an ignorant people, organised Census operations could be carried on without awaking suspicions and wonder as to "the object of Government in taking such immense trouble. One of the supervising officers notes an unwillingness on the part "of some to mention all the sources of their income. These suspicions, however, were not apparently universal, or very openly "expressed. Mr. Leigh, Assistant Commissioner, who was in charge of Dasúyah parganah, mentioned in his diary when he went "on tour supervising Census operations, that the people seemed to understand the object of the Census; and he had heard none of "the 'absurd reasoning given that had been quoted lately,' as representing native ideas on the subject.

"There appears also to have been an idea that the minute enquiry as to the occupation of those who held land portended some "increased *jama* at the approaching settlement. In a remote hilly part of the district there was among one tribe a curious misappre-"hension. It is thus mentioned by Mr. Kensington in his report:—

"'The objects of the Census can certainly not be said to have been understood, but on the other hand they were not, with few "'exceptions, misunderstood. The exceptions were some Gújar villages, the inhabitants of which refused to accept any assurance "'that the ultimate object was not increased taxation. These people got firmly into their heads that there was going to be a tax on "'hearths, and a few days before the Census went in for wholesale demolition of the offending *chúlhas*. The action was however not "'widely spread and is of course only worth a passing remark.'

"When I was on tour I heard in the hilly part of the district that the popular belief was that numbering men or cattle caused "them to die in numbers. Over some of the circles in that quarter some people died after the preliminary enumeration, and their "deaths were attributed by the hill people to the unluckiness of counting and re-counting individuals.

"The Commissioner of Jálandhar at the Kaisari Darbár held on 1st January 1881, and attended by the principal landholders "and traders of the Division, took occasion, opportunely, to explain the object of Government. The good effect of the assurance con-"veyed in Colonel Davies' speech on that occasion is mentioned in two of the reports of the supervising officers. Almost all that can "be said on this head as to the attitude of the people is satisfactory. The general feeling is well expressed by Sardár Gurdial "Singh, probationer Assistant Commissioner:—

"'It was very difficult for them to believe that Government spent so much money, neglected every other work even the settle-"'ment, simply to count its subjects, and had no selfish pecuniary object in view, especially after the expenses of the Afghán war. "'The *higher inhabitants* seemed to understand the unselfish motive of the Government.'"

936. Correctness of results.—There is an almost unanimous consensus of all concerned that the enumeration was performed with remarkable accuracy. Of course there were mistakes and omissions; there must be such in every Census. But I asked Deputy Commissioners to report especially upon the

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extent to which the work had been examined by European or gazetted officers, and from the result of the examination I find that an extraordinarily large proportion of the work was tested by superior officers, and that the general impression left was what I may almost call *surprise* at its unexpected accuracy. In some Districts the schedules of nearly every village were inspected by a gazetted officer; in many the Tahsíldárs examined nearly the whole of the work. Several Deputy Commissioners report that they examined the work in the most backward and difficult parts of their Districts, in which it had been least supervised, with especial care, and found it singularly good. Of course the errors in detailed entries, and especially in respect of age, caste, and occupation, were infinitely more numerous than errors in enumeration. But even in these respects Deputy Commissioners as a whole seem well satisfied with the work. Accuracy as to age is of course unattainable in India. In the matter of occupation the fault of the entries was want of precision, seldom absolute error: a term was used which was not wrong, but was more vague and indefinite than it need have been.

The entries of caste, and more especially of sub-divisions of caste, are the ones which are generally reported to be more unsatisfactory than any others. But I think that this is very largely because Deputy Commissioners misunderstood the object of the three caste columns. They seem not to have been satisfied unless all three columns were filled up correctly, whereas in a very large number of cases only two were intended to be used; and they apparently did not understand that all we really wanted or cared for was, the main sub-divisions of each caste, not the secondary branches of those sub-divisions; though to make sure of getting the one we had to provide for the entry of the other also. Thus, though there is no doubt that there are errors in the caste entries, yet I believe that, for our purpose, they were far more correct than Deputy Commissioners seem to think, and at any rate as correct as could be expected at a first experiment.

937. As for the mere numbering of the people, I believe it to be marvellously accurate; in the villages, which include some seven-eighths of the population, probably almost perfect. In the towns, where the population is more fluctuating, and often more difficult to deal with, errors were no doubt more frequent. But taking into consideration the fact that in England there is no preliminary record, that the schedules are in all cases filled in in the first instance by the householders, that more than half the entire population live in towns of more than 5,000 inhabitants, and that the Englishman is far less willing to submit to an inquiry of this sort, and far more likely to show an obstructive temper than the Indian, I believe it probable that the enumeration of population in the Panjáb is at the least as accurate as that of the English Census. To give in full the opinion of each Deputy Commissioner as to the correctness of the results would take too much space. I extract the opinions of a few officers well able to judge from personal inspection of the nature of the work done.

Mr. T. W. Smythe of Dehli, who takes one of the least sanguine views, writes:—"There is every reason to believe that the "returns are fairly accurate; and that the errors and omissions are comparatively few. The general impression of the inspecting "officers was, that the work was carefully done and that few names were omitted from the schedules. In the columns showing the "caste, *got*, clan or tribe, and the occupations, some mistakes and errors were discovered. These were owing more to defective or "erroneous information supplied by the heads of families than to neglect or carelessness on the part of the enumerators. Incorrect "replies were not as a rule given for the purpose of misleading, but chiefly through ignorance."

Mr. Benton of Karnal says: "I think that the Census results may be regarded as a close approximation to the truth. Much "trouble was taken for a length of time beforehand to impress on all engaged on the work the necessity for obtaining "correct results. There was hearty co-operation on all hands. The agency was throughout qualified for the task assigned to it, "and was satisfactorily supervised, admirably so I should say, by the tahsíl officials who had been employed in settlement. I "myself saw a great deal of the work in the three tahsíls, and more especially in Kaithal in which I had less confidence in the officers "than elsewhere, and I only found one circle in which the work had been inefficiently supervised and badly done. The work "throughout the Pánipat tahsíl and the Karnál parganah was almost all that could be desired. Behári Lál, Judicial Assistant, "took a strong interest in the work, as did also Mr. Anderson; and they both made favourable reports of what they had seen in "the Karnál tahsíl."

"Mr. Athim says that there might be two errors of sorts in a thousand entries in the town of Karnál; and he must have cor"rected half these, as he tested 50 per cent of the houses after the final enumeration. Mr. Turnbull tested over 200 houses before "the final Census, and as many after; and he found a few mistakes in the final enumeration arising from persons who were recorded "in the preliminary enumeration being struck out at the final, through being absent at their shops and the like, without being record"ed elsewhere. Such mistakes would probably have been corrected by the supervisors without his interference.

"There were a great many *baráts* moving about Pánipat during the time of the final enumeration. The Tahsíldár of Pánipat "tested 15 villages on the spot, and found that there was a very satisfactory correspondence between the erasures in one village and "the entries in another where the *barát* happened to be."

Mr. Wilson, Settlement Officer of Sirsa, writes as follows:—"The arrangements were carried out well, and the entries were "thoroughly tested by the supervisors and superintendents both during the preliminary enumeration and on and after the night of "the Census; the large number of supervisors made the work of supervision thorough. The Extra Assistant, the Tahsíldár, and the "Superintendent of Settlement were for some time in camp checking the entries. I was in camp in the tahsíl during the preliminary "enumeration and checked the entries in a number of villages. On the night of the Census I was in camp in the Sirsa tahsíl, and "on that night and the following days I checked the entries in a few villages there. In only one village I found mistakes, "a dead man's name not having been struck out and a newly-born child not having been entered; and in the jungle I found a small "encampment of gypsies, who had not been counted. In all important particulars, I believe throughout the tahsíl very few "mistakes were made; very few were omitted or counted twice over. The sex and condition (with regard to marriage) were correctly "stated. Age given is of course only a very rough approximation. An old man would give his age as '60 or 70' and when told that "he had been entered at 50, he would say '50 it is, let it stand.' Very few knew their age within 5 or 10 years. The entries giving "the age of children under a year old are not trustworthy. A child that was 8 months old at the preliminary enumeration in Decem"ber was 10 months old in February, and I do not think this change was noted in most cases; no doubt these entries were correct "within two months. The entries regarding religion and sect are correct regarding broad divisions, but there are great difficulties "in defining sects. Many Musalmáns hardly know whether they are Sunnis or Shíahs. Some entered the sweeper's religion as that "of Lálbeg, from the name of their deity; some called it simply "Sweepers' religion;" some called them Hindus. Where there are no "clear distinctions it is difficult to define. No doubt the entries given are trustworthy so far as they go, and approximately correct "returns can be compiled from them. So with the entries regarding tribe, clan" and *got*; many men cannot distinguish clearly "between these terms, and the entries have been various, but sufficiently correct returns can no doubt be compiled from them. "Difference of opinion exists regarding what makes a difference of dialect, and it is a question whether the Bágri spoken by many "in this district is a different dialect, or should be classed as Hindi. The place of birth was as a rule remembered and correctly "given. The question of occupation presents some difficulties, and it is impossible to lay down rules applying to every case, but the

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"general results are trustworthy. The entries regarding education are probably almost quite correct, and so I believe are those "regarding infirmities; the people did not seem to try to conceal such cases, which were well known in the village; of course what-"ever error there be is not on the side of exaggeration.

"On the whole I think that, allowing for the difficulties of many of the questions and the low order of intelligence of many of "the enumerators, and of the people generally, the entries are wonderfully correct; and that for all practical purposes the statistics "furnished by a compilation of the returns may be taken as quite trustworthy."

Mr. Walker, Settlement Officer of Ludhianah, "has no doubt as to the correctness of the results, and it would be strange if "they were not reliable, considering the effective nature of the supervision to which they were subjected. In **Simla**, where the "difficulties were exceptional, **Major Nisbet** 'has no reason whatever to doubt the accuracy of the returns.'"

From Hushyarpur Mr. Coldstream writes:—"Mr. Kensington, in charge of Garshankar pargauah, says, 'I believe the "'results to be as nearly approaching to perfect accuracy as can possibly be expected; the only exception is in the record of tribes "'and *gots*, in which there are probably some mistakes, chiefly owing to the utter vagueness of the mass of the people themselves on "'the subject. I was in camp in the Tahsil more or less from October to February, and was able to test the work at all its stages.'

Sardár Gurdiál Singh, C.S., who was *in charge of the lower part of the Unah pargauah*, says, 'I have personally examined "'and tested the entries in more than 130 villages and 7 ferries of the river with their water-mills and mooring places. I never "'found any omission. I believe that the result is correct, and that no persons have been left out uncounted. I found only one "'case in which a man had been counted twice on the night of the actual Census. He was an enumerator, and wrote his own name "'in his own book on that night, whilst it was not struck off in the house where he really resided.'

"In his diary this officer also stated that he one evening visited the villages in the hilly tract in the extreme south-east corner "of the district, and found that compared with the villages of the plains, there were very few mistakes in these parts. This is satis-"factory testimony to general correctness, for this tract is almost the most remote and inaccessible in the district, and probably "had least supervision.

"The officer in charge of the rural part of Hushyárpur Division is however not so sanguine. He thinks that a majority of the "enumerators were not sharp enough, and that mistakes were made in the numbering and arrangements of blocks. On the whole "however, even he thinks that the object of the Census has been attained.

"Sardár Gurdiál Singh found by enquiries from pilgrims, whom he met in numbers returning from Jawála Mukhi or proceeding "to the Ganges, that their names had been recorded on the Census night.

"I should doubt if many individual souls were that night left unnumbered; to assert that among the hundreds of thousands "none were omitted would be rash. I do not think, however, it would be possible to have attained on an average more correct results. "While on tour I was told (as quite a remarkable instance) of one man, a hermit, who lives in a hut on a hill summit on the confines "of three villages, having escaped enumeration in the preliminary stage. In one circle two children were born on the night of the "Census; one was entered as 2 hours, and the other as one 'watch' (*ek pahar*) old; one of them, a little girl, has been named by the "neighbours '*mardam shumári*' or 'Census.'

"I believe the result to be a fairly near approach to absolute correctness."

Mr. Tremlett, Commissioner of Jalandhar, who however saw none of the work himself, writes as follows of his Division. "The District Reports bear witness to the attention and interest the Deputy Commissioners have devoted to the work; and it "is satisfactory to find that in the plain districts, at any rate, full accuracy *for all practical purposes* may be considered to have "been attained. That there *are* errors here and there is from the nature of things certain; but I think we are justified in believing "they are so few and trifling in comparison with the large figures that have to be dealt with that they may fairly be regarded as in-"appreciable. About the **Kangra District** I feel less certainty. There the Deputy Commissioner had no settlement staff to call to "his assistance, while the persons whose services were available as enumerators are notoriously far more ignorant and backward in the "hills than those to be got hold of in the plains. In addition to this he found one of his Tahsildárs, Lachmi Sahái, most incompe-"tent and unsatisfactory, and he failed in enlisting the hearty cooperation of the hill Rájas. If to this we add the difficulties "arising from the mountainous character of the country, and the isolated struggling nature of many of the hamlets, I fear that a "fair amount of approximate accuracy is all that can be reasonably anticipated in the Kángra Returns." It is certain that the Census-officers in Kángra had exceptional difficulties to contend with; but on the other hand the population is practically stationary in the month of February; and Colonel Jenkins, the Deputy Commissioner, is of opinion that "so far as numbers go, the present Census "must have been very accurate;" and the Commissioner's opinion is perhaps partly based upon the length at which Colonel Jenkins dwells upon the difficulties he had to encounter, and especially upon the utter lack of interest which the jágírdárs or so called Rájas of Kángra displayed in the operations. Eventually their Census was taken by the Tahsíl officials. Of Kulu Mr. Alex **Anderson**, who examined the work most thoroughly, writes: "On the whole, and as the result of my own inspection, I am of opinion "that great accuracy has attended the work generally; and that the mistakes were more in the details than in the actual enumeration, "and that these mistakes were rather in the shape of omissions than of wrong entries." The omissions he explains to refer to the entry of one only of two occupations followed by the same man. In the more out-of-the way villages, where there was least supervision, he says: "In no case did I find that any one had been omitted from enumeration."

Mr. Hawkins of Amritsar says: "Although perfect accuracy is impossible, even in a civilised country like England, yet the "results are generally considered to be very fairly correct. I myself tested the figures in many villages and found them correct." **In Lahore, Mr. Stogdon**, who examined much work on the spot, found it done "intelligently and accurately, and believes the "results to be as correct as they can possibly be." **Mr. Bulman reports of Gujranwala** that the town Census "may be looked upon "as nearly perfectly accurate;" while the village entries "will generally speaking be found trustworthy in every respect save that of "minor sub-divisions of caste;" though he fears that the *bár* tract, the work in which he did not examine, "may not have been "done so thoroughly as other portions of the district," owing to the physical difficulties to be contended with. **Major Grey of Firozpur,** no lenient judge, reports: "I should say the figures were very correct. The enumerators took pains and the supervision "was most active and efficient. Nearly all the *enumerators were residents in or neighbours of the villages* where their work lay, "and personally knew the people therein. I tested the work for two months in every part of the district, and was thoroughly "satisfied with its general correctness. Of course there was some blundering over castes and *gots*, specially those of married "women."

Mr. Steedman, Settlement Officer of Rawalpindi, writes: "I believe that the Census returns of this District are on the whole "extremely accurate as far as the number of the population is concerned. On the Census night I checked the entries of between "15 and 20 enumerators in the city; and in one case only did I find a mistake in enumeration." **In Muzaffargarh Mr. O'Brien** is "confident of the general correctness of the Census." **Mr. Dames, in charge of the Derah Ismail Khan** *Census, who examined* the papers of more than 300 blocks, considers it "by far the most accurate Census yet taken." **From Peshawar Mr. Lawrence,** who was put on special duty for the Census of the District, writes:—"I am of opinion that the results are correct. The only point as "to the correctness of which I do not feel certain is the entries of age." **Major MacNeile of Hazara** will not commit himself further than to say "I can only say that the work was as correctly done as the means at my disposal would permit; and I consider the "returns fairly accurate as far as the Census itself is concerned, though there may be somewhat less accuracy in the return of caste "sub-divisions." The agency at his disposal, both official and non-official, appears to have been far below the average.

Of the officers whose opinion has not already been quoted, all agree in considering the results very correct except **Colonel Parsons of Jahlam, Colonel Gurdon of Jalandhar, and Mr. Fox of Gurgaon.** Of these officers Colonel Parsons is the only one who was in the District at the time of the Census. He thinks the results were "fairly accurate for all practical purposes." Colonel Gurdon, who returned from Europe after the Census, writes: "It may, I think, be assumed that upon the whole the results were fairly correct." The Commissioner's opinion on the subject has already been quoted. Mr. Fox, who has taken no part in the Census work, and who should not, as pointed out by the Commissioner, have been entrusted with the writing of the report, says:—"As I was not present "myself during the Census operations in this District, and the late Deputy Commissioner has left no notes whatever, I have only "been able to take a consensus of opinion from the various superior officers now present who were engaged in the work. Their "opinions would give 3 to 4 per cent. errors in enumeration, while in classification by *gots*, &c., the errors would be more numerous.

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"My own opinion is that generally speaking, considering the number of women perhaps not enumerated at all, those deceased or "emigrated not struck out, and those new names not in the preliminary record, errors in calculation of units of population must have "crept in to at least 3 per cent., having regard also to the agency procurable." Mr. Fox has not had much administrative experience; and I believe this estimate of error to be an absurdly high one. Excluding the hilly parts of Hazára and the Kángra *jágírs*, where the Census was probably less accurate than in other parts of the Province, I do not believe the error *in enumeration* approaches one per cent.; and I believe that even this error is in great measure confined to females. The errors in details such as age, caste, occupation, and the like, have already been discussed together with the results under each heading.

938. As for the subsequent operation of compiling the figures, I have already described at length the processes followed and the checks adopted. I believe the error to be considerably less than the error inherent in the material. The total numbers have been compiled with almost *absolute* accuracy. And in the details I believe the errors to be wholly insignificant, save in so far as my judgment may have erred in classifying the entries. At the same time I have only just come across an instance of how a very large and important error may be honestly made and yet escape detection. In correcting the proofs of the language table I was surprised to find that 1,507 females and only 397 males were shown in the Kángra district as speaking Tibeti. On examining the compilation slip, I found that the original entries were correct; but in adding up the Pahári-speaking males the total had been made 1,000 too large, evidently by a perfectly genuine mistake; while the 1 of the 1,358 males speaking Tibeti in Spiti had been written over the line of the column, and so escaped notice. Thus the total for Tibeti-speaking males was as much too small as that the Pahári-speaking males was too large; and the total males added up rightly, and so the error escaped detection till it caught my eye after it had actually got into type. I can only say that no labour has been spared to avoid such mistakes, and that I believe very few to have escaped uncorrected. There is one sort of error, however, which can hardly have been otherwise than frequent; and that is error due to the extraordinary uncertainty of the vernacular cursive. As far as was possible we compiled all our figures by men well acquainted with the district to which they referred; and they were always examined after compilation by such men. But there must have been hundreds of cases in which an unfamiliar term was mistaken for a familiar term and wrongly included under it. Some of these I have noticed in my discussion of the figures, and many of them I detected in my personal examination of the tables. At the same time this sort of error will have affected two tables only, those of caste and occupation; and the error will be almost wholly confined to small items, the larger and more important castes and occupations being sufficiently well known to the staff. In one case my Native Extra Assistant Commissioner, who was reading over a classified list of occupations with me, was obliged to confess with shame that by some gross carelessness boot-makers had been included in the priestly class; until a reference to the statement showed that the word that both he and I had read bootmaker in the vernacular, was really a local name for a class of Bráhmans. And even when the final occupation tables came up to me for examination before going to press, I found that many thousands of lime-burners had been returned for a certain district, who were on examination found to be entered as leather-workers and not lime-burners at all.

939. Progress reports.—Weekly progress reports were submitted to the Deputy Commissioners by the Census staff throughout each stage of the operations; and similar reports were sent to me from each district every fortnight. In order to enable them to prepare these reports, Census officials were instructed to keep up diaries in which the details of each day's work, notes of inspection, instructions given on the spot by supervising officers, and the like, were to be entered (see Appendix D, General Letter B).

On this subject Mr. Stogdon writes:—

"I think returns of the nature of Census diaries should not be required. They are of no practical value, and the probability is "that the majority of the entries in them are fudged. I do not see much use in the weekly progress reports. They are not trust-"worthy and do not conduce to the progress of the work. Every one knows that the Census has to be taken on a certain day, and all "parties concerned will do their utmost to have everything in readiness by that day, whether they keep Census diaries and submit "weekly progress reports or not."

I think perhaps the diaries were unnecessary. The idea was taken from settlement experience; but the Census work is of so temporary a nature that possibly the diaries give more trouble than they are worth. But the progress reports seem to me to be absolutely necessary. The Deputy Superintendent is responsible that the work is done up to time, and that all preparations are complete before the night of the final Census. If Government is of opinion that this may safely be left to the local officers, it would save the Deputy Superintendent an infinity of harassing anxiety. But I think that it would not be safe in all cases to leave matters so, and I believe that the necessity for reporting progress weekly stimulates subordinates by creating a rivalry among them; while I cannot conceive how a Deputy Commissioner is to know what is doing in his District, or to be answerable for the proper progress of the work, unless he receives something of the sort from all parts of his District. I would circulate to all tahsíls of the district a form showing the progress made in each, so as to encourage this rivalry. This plan I adopted with my Divisional offices, and found it had a wonderful effect on the work.

940. Inspection.—I have already noted (section 936) how very large a portion of the work was actually examined on the spot by superior officers. The gazetted staff of most districts was in camp almost throughout the operations, inspecting, instructing, explaining, and encouraging. I myself made tours of inspection through the Province, in the course of which I examined the work in the following Districts: Dehli, Gurgáon, Karnál, Ambála, Lúdhiánah, Jálandhar, Hushyárpur, Amritsar, Gurdáspur, Siálkot, Lahore, Gújránwála, Ráwal Pindi, Jahlam, Gújrát, Multán, Montgomery, Muzaffargarh, Pesháwar. Mr. Fanshawe, Revenue Under-Secretary to Government, while on tour with His Honour, inspected Firozpur, Sirsa, Hissár, and Rohtak. At each place I had the index-register and a few house-lists and completed schedules brought in from each tahsíl, examined them, and pointed out errors. A large part of the supervising staff was collected to meet me at head-quarters, and I briefly explained to them all points, and specially those regarding the filling-up of the schedules, on which it seemed to me that doubt was possible, and encouraged them to ask questions and propound difficulties for solution. I confess that I had been very sceptical as to

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any good result following upon these tours. I had much to attend to at Lahore; I knew that I should see only selected samples of work wherever selection was possible; and I only undertook the task at the express desire of Mr. Elliott. But the futher I went the more I became convinced of the value of my visit. In every place I found the staff ready with numberless doubts and difficulties which my growing experience enabled me to solve more and more satisfactorily as I went along; and my visit seemed to give a tangible reality to the Census Department, which had till then been represented by nothing more than orders on paper. Several Deputy Commissioners expressly refer to the assistance derived by them from my inspection.

And here I may refer to the importance of all instructions received in the Districts being verbally discussed in Committee by all the principal officials concerned. Nothing so conduces to uniformity of procedure and to a thorough understanding of the subject; and many of the difficulties experienced in Kángra, for instance, might have been avoided had this plan been adopted, as suggested in my instructions.

941. District and Divisional Reports.—A final Census Report was written by the Deputy Commissioner of each District; and it was intended that Divisional reports should be written by Commissioners. A memorandum detailing the principal points to be discussed in the reports was circulated in January 1881; and this plan should, I think, be followed on a future occasion. Some of the reports received were, as will have appeared from the extracts I have already given, exceedingly thoughtful and interesting. On a future occasion the Reports should be written on half sheets of paper in half margin, and *on one side only of the paper; no* two subjects should be treated of on the *same page; and* each page should bear the name of the District at its head. This would save an immensity of trouble in compiling the Final Report, as all the reports or parts of reports dealing with each subject or caste could thus be collected together. They should, moreover, be written for the District as a whole, and not in the form of a separate memorandum for each Tahsíl, as the latter form gives endless trouble, and sometimes consists of mere translations of reports by Tahsíldárs.

942. Commissioner's share in the work.—One defect of the plan of procedure followed at the late Census was, that it utterly failed to give Commissioners any necessary share in the work. Colonels Davies, Hall, and Minchin, and Major Young, have throughout taken a most active interest in the operations; but the other Commissioners did not concern themselves much about the Census, often not even taking the trouble to issue reminders to their Deputy Commissioners or to see that papers passing through their offices were drawn up in correspondence with the requisition to which they were an answer, and forwarding the final District reports with nothing more than a covering docket. It is very difficult to suggest any remedy for this state of things. Instructions are necessarily addressed to Deputy Commissioners, on whom the burden of the arrangements must fall; and it is perhaps not unnatural, though much to be regretted, that Commissioners should have taken so little interest in the Census.

943. Translation of instructions.—All circulars and instructions were translated in the Persian office of the Panjáb Secretariat, the recognised agency for this work. On this subject Mr. Gardiner of Gújránwála makes the following remarks:—

> "As regards the printing of the Census forms and instructions I have only to repeat what I have said above in para. 2, *viz.*, that "in the Census Department, as in other Departments in the Province, sufficient care is not taken in the preparation of the vernacular "forms and instructions. Translations which native officials were surprised to think they should be supposed to understand, mis- "translations and misprints, &c., in the headings of forms, which make the matter unintelligible, and which are not allowed to occur in "the English version, make the work of superintendence unnecessarily difficult."

I have received no similar complaints from any other quarter; and several of the Deputy Commissioners, while rightly objecting to the use of *ghar* to represent "family" and one or two similar expressions already commented on, say that the translations were in other respects satisfactory. My own impression is that, while they hardly justified Mr. Gardiner's language, they were not in all respects as good as they should have been. But the task was a difficult one; and though I nominally superintended the translation, I did not feel competent to override the opinion of the Secretariat Persian Office where I differed from them on a question of the choice of expressions. In fact, had I done so, I should probably have only made matters worse. As for misprints, I believe they were of very rare occurrence.

One error I most certainly fell into. I did not at first issue any translations of my general letters, as distinguished from the detailed instructions for the use of supervisors and enumerators. They were addressed solely to Deputy Commissioners, and dealt with either general principles or questions which arose only in certain special cases; I thought that they would only confuse subordinates if communicated to them; and I expected that Deputy Commissioners would embody so much (not a great part) of their substance as was necessary for the guidance of their vernacular staff, in orders which they would issue on their own authority. Mr. Benton remarks on this subject: "I have no doubt the intention was that Deputy "Commissioners should summarise the instructions received and issue orders to the Tahsíldár. But most "of them, I believe, preferred hurrying on the work of translation to taking the responsibility of issuing "their own orders with regard to a scheme which required a considerable amount of thought and study for "its adequate comprehension." I still think that such summaries would have been far better and safer than full translations; but there is justice in Mr. Benton's remarks. At any rate summaries were *not* prepared in many cases; and I think that at a future Census translations should accompany all instructions issued, whatever their nature. Eventually I issued translations of all my general letters, but not till some time after the work had begun, and I find that the delay is very generally complained of as having caused considerable inconvenience. That translations are not received till a few days after the English instructions, merely means that the latter are issued as soon as ready, instead of waiting for the translation to be made and printed. On this occasion time was so limited that it was necessary to do this; but when time allows of it, it would doubtless be better to despatch both together.

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944. Printing of Forms and Instructions.—The number of forms and instructions printed was as follows:—

Forms printed for the Census of 1881.

		No. of Copies.
1.	Schedules, Persian Character	1,130,200
2.	Ditto Nágri and Gurmukhi do.	198,700
3.	Ditto English do.	20,700
4.	Private Schedules, Urdu	3,500
5.	Ditto English	12,000
6.	Labels for Enumerators' Books	72,000
7.	Boat Tickets	7,000
8.	Railway Check Tickets	27,000
9.	Translation of General Letters (complete)	2,200
10.	Instructions to Supervisors (complete)	12,000
11.	Ditto Enumerators (complete)	72,000
12.	Instructions for Boat Census	2,000
13.	Ditto Military Census	1,000
14.	Sample Village Abstracts	10,500
15.	Blank Ditto	77,500
16.	Registers for Abstracting	900,000
17.	Check Sheets for do.	65,000
18.	Caste and Clan Abstracts and Slips	300,000
19.	Miscellaneous Forms and Slips (say)	10,100
	TOTAL FORMS PRINTED	2,930,000

Of these the only ones of which we had any considerable balance left unused, were No. 2 of which 78,600 were left over, the first indents having been exaggerated, No. 11 of which 5,750 too many were printed, and No. 4, which was hardly used at all. Out of these forms I supplied the Hill States, and also the Railway Census of Sindh and Rájpútána. The Persian and Nágri schedules were printed by steam at Calcutta, the other papers were lithographed at Lahore and Lúdhiána.

Major Grey of Ffrozpur suggests that the headings of the schedules should be lithographed instead of printed, as "few natives can read the printed character." But this would be quite impossible. Nothing but steam printing could have supplied the enormous number of schedules required within a reasonable time. Even were the suggestion that one heading only should be used for each book of schedules adopted (see section 957), the number of headings would still amount to something like 60 000. The heading is soon learnt off by the enumerator, and the printed character is at any rate sufficiently legible by him to save him from mistakes.

The text of the instructions was necessarily lithographed, as printed copies would have been practically useless. The greater portion of this work was done at the Lahore Central Gaol under the superintendence of Dr. Dickson, to whom I express my obligations in another place. But I was frequently compelled to have recourse to local presses, who were under no restrictions as to the number of hands employed or the number of hours during which they could work. Most of the registers for abstracting were printed at these last. The Gurmukhi schedules and instructions were printed for me by the Revd. Mr. Wherry at the American Mission Press, Lúdhián, and to him also I have to tender my most sincere thanks.

The rates charged by the Calcutta press were extraordinarily low; but unfortunately we began work so late in the Panjáb that we were obliged to get all our schedules up from Calcutta by mail train, which enormously increased their cost. The Central Gaol rates were inordinately high; but the money went into the Government pocket, and the only ill effect of the unduly high charges was to make the apparent cost of the Census greater than it should have been.

The paper used throughout for all Census forms was yellow Royal paper from the Bally Paper Mills at Calcutta. It proved exceedingly suitable for the purpose.

The English printing was comparatively small in amount so far as Census operations were concerned. But the printing of the final tables was a very heavy job. It was executed at the Central Gaol Press, and though the delay was excessive and the cost inordinate, the work was fairly well done in the end. The text of the Report was printed at the Government of India Press at Calcutta, and the style of work speaks for itself. I am under the greatest obligations to the Superintendent of Government Printing in India and his staff for the manner in which the task has been executed. The work was difficult, the abstracts required great nicety, the spelling of the vernacular names gave much trouble, my handwriting is singularly illegible, and I am much indebted to the care and skill which have produced so exceedingly satisfactory a result. The printing has taken a very long time; but if I had had to depend upon local presses for the work, the issue of the Report would have been still longer delayed.

945. Distribution of forms and instructions.—The distribution of forms and instructions to the Districts of the Panjáb was undertaken by the Lahore Central Gaol; and I think that this is the best plan that can be suggested. If, however, at a future Census the schedules are printed, as they should be, some months beforehand, they would best be sent direct from Calcutta to the several Districts, the supply needed by each being calculated roughly on the basis of the present expenditure (see section 946), and a surplus being kept at Lahore from which to meet supplementary or emergent indents. We were *most* unfortunate in the matter of distribution. The supply of schedules from Calcutta was very fitful, and not as rapid as I had hoped from the promises made me, and they had consequently to be doled out to Districts as they were received, instead of the whole supply for each being despatched at once. We naturally supplied the most inaccessible Districts first, leaving those on the line of rail to receive their schedules later on. But just as we had sent off the schedules for the Districts on the Panjáb Northern State Railway that line stopped through-booking of goods or parcels, and not only did so, but kept our parcels lying for a fortnight at various stations along the line without giving us notice of the fact. Some of the parcels for the most inaccessible districts miscarried; and the result was that in one or two Districts they were obliged to supplement the supply in a small measure by schedules ruled by hand. So, too, with covering boards for the books, which, as I have already pointed out, should be made in the Districts on a future occasion. Instructions were sent by post, and no difficulty was experienced in regard to them.

946. Consumption of schedules.—The number of schedules supplied to the various Districts is shown below[1]:—

[1] By "schedule" I mean a paper with one top heading on each side of it, forming when doubled across the middle, 4 pages or 2 leaves in the enumerator's book.

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Part I.—Subsidiary Matters.

Consumption of Schedules in each District.

Districts.	No. of Schedules.	Total Population.	No. of Schedules per 1,000 Souls.	Districts.	No. of Schedules.	Total Population.	No. of Schedules per 1,000 Souls.
Dehli	39,500	643,515	61	Ráwalpindi	45,750	820,512	56
Gurgáon	35,000	641,848	55	Jehlam	38,000	589,373	64
Karnál	45,000	622,621	72	Gujrat	42,000	689,115	61
Hissar	32,350	504,183	64	Sháhpur	29,000	421,508	69
Rohtak	37,400	553,609	68	Multán	41,620	551,964	75
Sirsa	17,200	253,275	68	Jhang	21,500	395,296	54
Ambála	83,500	1,067,263	78	Montgomery	21,500	426,529	50
Ludhiána	45,000	618,835	73	Muzaffargarh	20,000	338,605	59
Simla and Simla States	5,500	493,314	112	Derah Ismáíl Khán	24,150	441,649	55
Jalandhar	55,100	789,555	70	Derah Gházi Khan	27,520	363,346	76
Hushyárpur	69,300	901,381	77	Bannu	28,180	332,577	85
Kángra	69,230	730,845	95	Pesháwar	34,200	592,674	58
Amritsar	55,540	893,266	62	Hazára	35,180	407,075	86
Gurdáspur	54,520	823,695	66	Kohát	16,200	181,540	89
Siálkot	56,100	1,012,148	55				
Lahore	47,700	924,106	52				
Gujránwála	43,500	616,892	71				
Firozpur	33,540	650,519	52	Total, British Territory	1,249,780	19,292,633	66

It will be seen that the number used per 1,000 of population varies enormously in different districts. This is partly due to the fact that, while in most Districts the more incompetent enumerators made a rough preliminary record and had it corrected by the supervisors before they copied it into their books, in some Districts this was written on schedules, and in others on paper ruled by hand. But there is no doubt whatever that in some Districts the indents were absurdly large and there was great waste. Many of the enumerators' books had 10 or 20 unused blank schedules at the end, and thousands of spare schedules were in some cases sent to the Divisional office after the Census to be used as blank paper. In many Districts, however, the Deputy Commissioners complain that the supply was insufficient. I think that in future a very liberal margin should be allowed for waste; far more liberal than was generally provided by Deputy Commissioners in their indents for this Census. I think that the supply should be nearly if not quite double that which would be needed if all enumerators wrote closely and neatly, and there was no waste. It is better that rough copies, if they are to be made, should be made on the printed schedule, as the enumerators thus grow accustomed to its use, and are saved both time and trouble, while the cost of the extra schedules is quite insignificant; and above all things the schedules should be distributed as early as possible, and every effort should be made to estimate and supply the deficiency, if any, at as early date as possible. At this Census District officers seemed to think that schedules grew at Lahore; and I received emergent indents for thousands of schedules up to within a few days of the actual Census.

947. Mr. Steedman has the following remarks on the subject :—

"Native subordinates send in the wildest indents. The only way to keep them in check is to require explanations of how the "forms previously supplied have been expended. A short time before the final enumeration I sent a circular round the tahsíls "asking the superintendents to send me in all surplus forms, and pointing out that according to the number supplied, the population "according to the business returns, and the number of enumerators, there ought to be a surplus. I believe this went a long way towards stopping unreasonable emergent indents at the last."

This is very true; but care must be taken not to make the officers feel that they may not use schedules for such purposes as rough copies and the like, or that there need be any hesitation about replacing a schedule which has been badly cut up by corrections. The relative competence of enumerators is almost as important a factor in deciding the number of schedules required as is the population to be enumerated. Where other characters than Persian are used, two copies, or an interlinear biglot copy occupying double space, must be made, and the practice of preparing a rough copy seems to have been very general. Mr. Tolbort of Ambála says: "At the commencement of this stage of the Census proceed-"ings I endeavoured to check the practice of preparing a rough record first and fairing it afterwards, but "the general consensus of opinion in favour of the said practice was so strong that I considered it unwise "to persist in opposing it. I do not think that it led to '*farsi*' procedure;" and most Deputy Commissioners agree with him, though Mr. Wilson of Sirsa prefers the original as being more trustworthy.

Part I.—Subsidiary Matters.

948. Expenditure.—The total expenditure on Census operations was as follows:—

Expenditure on the Census of 1881.

Enumeration of the People—	R
1. Pay and allowances of Deputy Superintendent	3,430
2. Office establishment	900
3. Telegrams	800
4. Paid agency and allowances	1,160
5. Paper for schedules and instructions	5,810
6. Printing schedules, typographed at Calcutta	750
7. Printing instructions, &c., lithographed at Lahore	3,650
8. Carriage of schedules from Calcutta by mail train	5,980
9. Boards for enumerators' books	7,800
10. Packing and distributing forms	2,520
11. District contingencies—Paper	4,090
12. Do. do. Carriage, &c.	2,180
13. Do. do. Postage and telegrams	1,200
14. Rewards to non-officials who had worked well	16,000
TOTAL ENUMERATION	56,370

Preparation of the Figures—	R
15. Paper for forms	4,850
16. Carriage of do.	1,230
17. Printing forms (lithography at Lahore)	6,540
18. Central office—Pay and allowances of Deputy Superintendent	10,040
19. Do. do. Office establishment	7, 50
20. Compiling office—Assistant in charge	5,200
21. Do. do. Establishment	7,320
22. Do. do. Contingencies	650
23. Divisional offices—Officers in charge	9,430
24. Do. do. Patwaris' substitutes	16,050
25. Do. do. Patwaris' allowances	18,870
26. Do. do. Clerks	1,830
27. Rent	730
28. Printing and Contingencies	3,300
TOTAL COMPILATION	93,090
TOTAL ENUMERATION	56,370
„ COMPILATION	93,090
29. Printing Report (estimated)	5,640
TOTAL EXPENDITURE	1,56,000

The total does not agree with that given on page 13, as after those figures were prepared an additional bonus equivalent to raising the monthly allowances of the patwáris employed in the Divisional offices from R2 to R4 was sanctioned, and the Deputy Commissioner of Bannu sent in a bill for expenditure nearly two years after date. From the latter cause, item 4, which is composed of R560 for supervision of the Teri Census in Kohát, R450 for paid enumerators in Bannu, and R150 travelling allowance paid to a European Superintendent of the School of Wards in Ambála, does not tally with the statement made in the body of the report about paid agency. Item 8 might have been reduced by three-quarters if we had begun printing early enough, and items 9, 10, 12, and 13 by nearly one-half had the binding boards for enumerators' books been made in the districts; while if items 20 and 21 had been a little larger, items 18 and 19 would have been much smaller. By having the printing done at private presses, items 7, 17, and 29 might be largely reduced on a future occasion.

949. Certificates and rewards.—After the operations were over, certificates to the following effect were distributed to those whose services deserved recognition: "This certificate is presented to_____by "order of the Hon'ble the Lieutenant-Governor of the Panjáb, in recognition of the good and willing service "he rendered in the Census operations of 1881, and as a mark of approbation." These certificates were handsomely printed in gold, red, and black on foolscap paper surmounted by the Royal Arms, and the three colours constituted three classes of certificates. Of these the first was granted to officials of the higher grades and natives of position who had done specially good work; the number issued was kept low so that a first-class certificate should be looked upon as a very special distinction. The second-class certificate was given to all persons who had distinguished themselves by zeal and activity; and the third class to all others whose services were considered worthy of recognition. The numbers distributed were as follows:—

1st Class	559
2nd Class	4,711
3rd Class	10,464
TOTAL	15,734

With these certificates small amounts from the grant of R16,000 made by the Government of India were distributed as rewards. None of this money was paid to any official or other person whose services were paid for by the State, or as a reward for work done in connection with the Census of a cantonment, or of a municipality liable for the cost of its own Census. The money was to be "expended in rewarding "selected men for especially good service, and not in general payment for work done." Thus the rewards were confined to non-officials who received second-class certificates, recipients of the first-class certificates being of too high standing to receive small money rewards, while those to whom third-class certificates were awarded were, *ex hypothesi*, not men selected for specially good service. The Government grant for rewards was in some districts supplemented by grants from Local and Municipal Funds.

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Part II.—Census in Native States.

PART II.—CENSUS IN NATIVE STATES.

950. Form of schedule for Native States.—The Government of India, when determining to extend the Census operations to the Feudatory States with the exception of Kashmír, proposed to "draw the "broadest distinction between the enumeration in British and Foreign Territory. The results in the "latter case will necessarily possess much less completeness and accuracy, and we should not expect "them to conform to any uniform standard." The Committee of 1877 had recommended that nothing more than an enumeration by sex should be attempted, and the Government of India proposed a simplified form of schedule.

But the Panjáb Government preferred to adopt the same schedule in the Native States subordinate to it as had been prescribed for British Territory. It was of opinion that they were probably more advanced than the majority of similar States in India; that they had always imitated and adopted our administrative system in its main outlines; that they were fully competent to conduct their Census on the same lines as our own; and that they would feel it as a slight if a doubt were cast upon their capacity to do so by the suggestion of a simpler schedule for their use. The schedule prescribed for British Territory was sent to the various Chiefs, and they were asked if they thought that difficulty would be experienced in collecting any of the information therein detailed. They one and all professed their readiness to take a Census of their territories in precisely the same manner as that of British India, the only doubt expressed being by the Baháwalpur Darbár, which questioned the advisability of recording the age of females. Accordingly the general schedule and all instructions issued by me for the guidance of British officers were supplied to the Governments of Native States, who took their Census precisely as we took ours.

951. Nature of work in Native States.—The result has most fully justified the decision of the Panjáb Government, and I believe that, as regards the greater number of the States at any rate, their results are quite as accurate as our own. In most of them a special officer was appointed to superintend the operations: in some of them, as Pataudi, Loháru, and Dujána, officials were sent to our Districts to study the system; and all the larger ones, with the single exception of Baháwalpur, remained in constant correspondence with me, sent me specimens of their papers to examine and criticise, referred doubtful points for my decision, constantly kept me informed of their progress, and even sent men to Lahore to consult me personally on difficulties which had arisen. Some peculiarities there doubtless were. In Patiála a third column for eunuchs was added to the columns for males and females; and in this and perhaps in other States the Mahárája and his immediate family were excepted altogether from enumeration. But on the whole I believe the work was almost, and in most of the States of the Eastern Plains quite as well done as in British Territory.

In Loháru and Dujána great difficulty was experienced in obtaining competent agency for the final enumeration; but the Commissioner Colonel Minchin, who took great interest in the work, satisfied himself that the preliminary record, which after all is the backbone of the Census, was thoroughly satisfactory. In Pataudi Major Young sent an official to be trained at Gurgáon, and exercised an active supervision throughout; and the Census papers which were sent to me to examine from this State were admirably prepared. In Faridkot Mr. Fanshawe examined the work on the spot, and the papers which were sent me were very well done: Mr. Macnabb saw some of the work in Kalsia, and expressed himself well satisfied with it. Of the work in Baháwalpur and Maler Kotla I know nothing. But of Patiála, Nábha, and Jínd I can say from personal knowledge that the very greatest interest was taken throughout in the operations, that no pains were spared to secure accuracy, and that, so far as it is possible to judge from outside, the results were in every way admirable. The officials to whom the Census of these States was entrusted were men of great intelligence and ability; they took the greatest interest in the work, spared themselves no labour, and were in constant communication with myself on all sorts of difficult or doubtful questions.

In the Hill States greater difficulties were experienced, as might indeed have been expected; and in some of the wilder tracts the same special license had to be allowed as was taken of necessity in similar portions of our own territory, and has been described in sections 884, 885. In Chamba the operations were superintended throughout by Major Marshall, who tested much of the work on the spot and found it thoroughly and accurately done. In Suket the earlier work was capitally done under the superintendence of the Superintendent Munshí Hardiál Singh, an official of our Government, Colonel Davies the Commissioner exercising an active supervision. In Mandi Colonel Davies had practically to take the whole matter into his own hands, as the Rája, while only too glad to have the work done for him, and offering no sort of obstruction, would not himself move in the matter. Accordingly Colonel Davies obtained a trained staff of supervisors from Jálandhar and Hushyárpur and put Múnshi Nathu Rám, one of our Superintendents of Settlement, in charge of the Mandi Census, with the result that the work was well done in the end But it was done quite at the last moment; and it is wholly due to Colonel Davies' energy that a Census was taken at all in Mandi. In the Simla Hill States the Superintendent Major Nisbet summoned an official from each State to be trained at Simla, and throughout exercised the most active supervision. It was found impossible, both here and in the higher portions of Chamba and Mandi, to divide the country into blocks, as indeed it had been found in Kúlu; but enumerators were appointed to each hamlet or village, and Major Nisbet, who marched through a great portion of the country under his superintendence while operations were in progress, found that the enumeration was accurately done.

There is no doubt that, in the Hill States, as in the hilly parts of British Territory, more than ordinary difficulty was experienced in obtaining competent agency; and if they alone were to be consi-

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Part II.—Census in Native States.

dered, a more simple schedule would, I think, be advisable. But they include but a small fraction of the population of the Panjáb Native States, and it would be a pity to destroy the uniformity of the whole Census by making an exception in their favour.

952. Kapurthala Census incomplete.—In one State only, and that a State under the superintendence of an English officer, was the scheme of enumeration which was followed without other exception throughout the length and breadth of the Panjáb abandoned. In Kapúrthala the Superintendent directed that no religious sects should be recorded in the column provided for that purpose, that one of the two columns for tribal sub-divisions should be omitted altogether from the schedule, and that the other should be filled up for four castes only. The consequence of this action, which was not discovered till long after the Census was over, when the final tables were sent to my office, has been to destroy the value and completeness of our Tables in two not unimportant points. In all other respects I believe the Kapúrthala Census work to have been admirable.

953. Tabulation in Native States.—The abstracting and tabulating the figures was done by each State independently, with the exception of the Simla Hill States, which sent clerks to Simla where they abstracted the figures under the supervision of the Superintendent. Náhan, however, was excepted from this arrangement, and allowed to tabulate its own figures. Immediately after the Census was over I invited the Native States to send their principal Census officers to meet me at Lahore. There I fully described to them in Committee the scheme to be followed in the tabulation of the figures. I think that, on a future occasion, it would be well to collect these officers at a similar meeting before work is commenced, to explain to them the general plan of Census operations. A little verbal explanation is better than much written precept; and I am sure that the States would respond cordially to such an invitation.

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PART III.—DIFFICULTIES AND SUGGESTIONS.

954. Difficulties and suggestions.—In writing this Report I have attempted throughout to keep clearly in view the fact that this is not the last Census which will be taken in the Panjáb, and to leave on record as far as possible all experience that we have gained on this occasion for guidance in future enumerations. It can hardly be conceived by those who have never attempted the task, how much careful thought was needed and what countless minute points had to be considered in order to make the different parts of the machinery work smoothly and without jar or hitch, and to secure the measure of success which has attended the present operations. So far as the Census itself and the subsequent abstracting of the figures are concerned, all went far more smoothly than I had ventured to hope for. But in the final compilation and tabulation we broke down lamentably, for reasons which I have already stated in full. Still, even where things went best the arrangements might have improved. Many suggestions for their improvement will be found scattered through the preceding chapters of the Report; and I wish here to add such further suggestions as may occur to me, and to collect the references to the scattered suggestions already made, so as to form a sort of index to the whole subject of improvement at a future Census. And I cannot do better than follow the order in which the several stages of the operations have been treated in Book II. of the Report.

955. Preliminary operations.—*The demarcation of blocks.*—I have recommended (section 848) that the size of the block be increased, and many officers not there quoted agree with me. The importance of fixing natural boundaries as the limits of blocks should be insisted on (section 834).

Index-register. (*Form B, General Letter B, Appendix D.*)—Two officers have attacked this form as at once perplexing and useless. Mr. Frizelle writes from Sháhpur: "The great difficulty was the Index-register (Form A) and fortu-"nately this was confined to supervisors and those above them. I do not know what has been the experience of other Deputy "Commissioners, but I know that in this District the Form in question was for a long time looked upon even by Tahsildárs in the "light of a rather intricate puzzle or complicated piece of machinery. When it was filled up properly this was done only after long "practice, however simple it may seem now from use, and many rough copies were made out both of the abstracts and the Tahsil "Register before a final result was arrived at. Columns 5 to 10 especially might have been simplified, although one is bound to say "that any omission would have detracted from its value as a whole, and now that the Census is over and the beauties have grown on "one, it is easy to see that anything less complete might not have answered so well." So Mr. Stogdon of Lahore says: "This (the "Index) register appears to me to be one of the least important of the Census records. It has to be prepared piece-meal, and some "of the entries in it cannot be made until the house-list has been prepared. It is in fact a sort of ledger of the house-list, almost "useless for the purposes of the actual Census, but perhaps of some value when the results have to be ascertained and tabulated. "As a matter of fact the Index-register was in some cases prepared from the house-list; notably, I believe, in the case of the city "of Lahore." I confess I do not see myself where the great difficulty lay, nor do I find that other Deputy Commissioners experienced it, so far at least as their reports are concerned. Mr. Tolbert, however, modified it in Ambála; and his objection was, that as it stood it had to be built up from the house-lists instead of being prepared in the first place, and this I think touches the real defect in the form. It essentially consists of two parts. The first, which sketches out the general division of the tahsil into blocks, circles, and charges, should be drawn up at the very first; but the second part which contains details can only be filled in after the arrangements are considerably advanced. I would separate these two parts at a future Census. I would first draw up a preliminary list for each charge or tahsil something like the form in the margin.

Then as the demarcation of blocks, the numbering of houses, the selection of agency progressed in each circle, the following form might be filled up, which is based upon the suggestions of Mr. Steedman:—

TAHSIL ABBOTTABAD, CHARGE NO. 5, SUPERINTENDENT, AHMAD BAKSH.

Circle and Supervisor.	Village.	Population in 1868.	No. of blocks.
3 Kirpa Ram	Waisri	200	1
	Barána	300	
	Chhajpur	3,500	7
	Latifpur	490	1
			9

TAHSIL ABBOTTABAD—CIRCLE No. 3—SUPERVISOR, KIRPA RAM.

Number Collectori of Mauzah.	Name of Mauzah.	Subsidiary parts. Number.	Subsidiary parts. Name.	Block letter.	Boundaries of block. From Number — To Number —	Houses included in the block.	Number of families in the block.	Name and occupation of enumerator with the character he writes.

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Part III.—Difficulties and Suggestions.

The importance of the accurate entry of subsidiary parts has already been insisted on (section 835). Wells, however, should not be entered (section 842). It is essential to get the column defining the subsidiary parts next to the columns defining the block, or it will never be known in which block each subsidiary part is included. When this form is complete, I still think copies should be sent to the superintendent and tahsíldár, and that a general index-register for the tahsíl should be drawn up in this modified form. Not only will the Tahsíldár require some such index to the arrangements, but, what is far more important, the whole preparation of the tahsíl abstracts (section 890), the whole grouping of the blocks in the village tables prepared by the Divisional offices, and the whole check upon the completeness of the Census schedules as sent to those offices depends upon the correctness of this index-register. The greatest difficulty was caused by alterations made at the last moment in the arrangement of the blocks not having always been incorporated in the index-register. This form *must* be sent in an accurate and complete state to the Divisional offices, for they have no other guide in tabulation. It might perhaps be worth while to print blank forms of index-register on a future occasion.

Preparation of house-maps.—The importance of those maps and the best way of preparing them are discussed in section 837. In hill tracts and also in the Multán and Deraját Divisions—wherever in fact scattered hamlets are common—they are absolutely essential. The following suggestion of Mr. Gardiner seems a good one: "I think every Municipality should be made to provide "itself with a map of its town and to sub-divide the whole area of the latter into muhallas or streets and give a name to each locality "and a number to each house. This would greatly facilitate any future enumeration, and is required for the purposes of *municipal* "arrangements, the delivery of letters, and the general administration. The country is quite civilised enough to appreciate such an "improvement, and the sooner it is put into force the better." (See also Mr. Stogdon's opinion quoted in section 837).

House-numbering—should be down-hill in hilly tracts (section 836). The numbers are best affixed as indicated in section 836. The series of numbers should be continuous throughout the circle or village, whichever is larger (section 836). Wells and outlying huts in the field should not be indiscriminately entered (sections 841, 842).

Definition of house and family.—This is fully discussed in sections 838, 839, and proposals made in section 840. The existence of a hearth is perhaps the best test of whether a building is a house or not (section 841). A family should be defined and translated as suggested in section 843. A difficulty was felt in Hushyárpur in what was said to be the not uncommon case of one family occupying two houses. Who then was to be shown as the head of the second house? I think the column might be filled up "Same as No.—" *ba sharah lambar fulán*, and all families so entered omitted in the statistics for families, which should (section 844) be taken from the house-lists, not from the schedules. The following suggestion by Mr. Gardiner will I think be met by my proposals in section 840: "The definitions of a house and of a family are now understood, and any alteration of these on a future occasion "would only complicate matters more. But what I think is required is the localisation of the families in a house; under the present "arrangement as the families in a house are entered in no sort of order, there is no way of testing readily whether a family has been "omitted. Rooms might be numbered in some system from left to right as a bullock walks at a Persian wheel, and from the bottom "to the top of a house; and enumerators should be encouraged to enter buildings of any sort in their schedules (as after all the "enumeration is by locality) instead of being discouraged from doing so as they were."

Demarcation of urban limits.—This should be made most carefully on the lines indicated in sections 840 and 911.

Forms for house-lists.—I rather think I would print and distribute blank forms for house-lists at a future Census. It is well to leave as little avoidable work to be done in the districts and to render the duties of the staff as simple and easy as possible.

956. Agency.—The matter of agency is the real difficulty in a Panjáb Census; and though with the spread of education the difficulty will gradually decrease, yet I fear several enumerations will be made before it wholly disappears. The greatest difficulty was experienced in the backward tracts, such as the Meo country in Gurgáon, and the Sirsa district, in the hills generally, on the frontier, and in the great scantily peopled areas of the Deraját and the *bár* districts. The question of agency is fully discussed in sections 849 to 855.

It is always questionable whether it is not better to work with an inferior agency which knows the people, than with an imported agency to whom they and their ways are unfamiliar. The increase in the average size of blocks suggested above will partly diminish the difficulty. I do not think the general scheme of agency could be improved. The supervisors are the backbone of the whole scheme, and should be most carefully selected from the very best men available. Many officers are of opinion that the superintendents are useless and unnecessary, while others attach the greatest importance to them. I quote below opinions from Peshawar and Lahore:—

Peshawar.—"The supervisor was the man who really did all the work. The superintendent was too often a mere name. "I would suggest that at a future Census, the duties of the superintendent be entrusted to the Tahsíldár and Qánúngo. The "patwáris of the district, aided by the lambardárs, could perform the duties of enumerators and of supervisors, as they have done "in this Census."

Lahore.—"In my opinion the post of superintendent is not a very important one. The important point is to secure good "supervisors with manageable charges. If sufficient time is given for Census operations, the Tahsíldár, Náib Tahsíldár, and Assistant "Commissioner or Extra Assistant Commissioner in charge of the Tahsíl can effectually control and check their operations. If the "best men are appointed superintendents, it is difficult to obtain a sufficient number of efficient supervisors, and a Census cannot be "properly conducted. I think that the post of superintendent might be considered an honorary one to which zaildárs and men "of note in the district should be appointed. They would always be able to render very effective aid in making arrangements and "collecting the people, and impressing upon them the necessity of giving correct information and of furthering the work of "supervisors and enumerators to the utmost of their power; but the actual control of the work of the supervisors can best be performed "by responsible officials of experience, who are more likely to work in an intelligent and uniform manner than inexperienced "landholders, however influential and well-meaning they may be. The duties of a supervisor are of a most important nature, and "it is on them that the success or otherwise of a Census principally depends."

On the other hand Mr. Coldstream in Hushyárpur added intermediate officers between supervisors and superintendents for the final Census, employing for the purpose every man of position and available person in the district, in both towns and villages. Some of them rode 20 or 30 miles on the Census night. The mere fact of their appointment stirred up the enumerators and made them afraid to shirk their work; and for this purpose a zaildár who could not read was almost as useful as one who could. I should be inclined at a future Census to appoint all the best educated men I could get as supervisors; and to appoint as superintendents all the leading headmen, whether illiterate or otherwise, making it their duty to constantly visit their circles, stir up the enumerators, listen to complaints, and smooth away difficulties, rather than to supervise the actual entries, which task might be left to the supervisors. Mr. Coldstream points out the admirable effect of associating the natural leaders of the people with the operations, both in making the people careful and willing to give correct information, and in allaying their fear and suspicions. I think the supervising staff should be slightly larger than it was at this Census, especially if the patwáris are to make the preliminary record; and most of all in the sparsely inhabited tracts, where the distances to be traversed are great and where the intelligence of the enumerators is generally small. But if illiterate men are utilised as superintendents this will increase the number of educated men available as supervisors. I think (sections 49 to 52 and 854) that non-official agency should not be paid; but *whenever* they are sent to a distance from their homes, I would give them a small sum to cover actual expenses. I would avoid the use of Nágri and Mahájani writers as much as possible, even if the only alternative were to utilise school-boys as amanuenses (sections 855, 830). Where their use cannot be avoided, the best plan of transliteration is indicated in section 856. I would issue a second Government circular to heads of departments shortly before the Census as suggested in section 850, and would impress upon Deputy Commissioners the importance of meeting the convenience of the staff lent them by arrangements such as are described in section 850. I would make the preliminary record of villages entirely by patwáris (sections 851, 852).

957. The form of and entries in the Schedules and the enumerators' books.—I would put the name of the village on the label (section 858), omit the name of the tahsíl from the schedule (section 858), and add a column for the number and kind of house instead of writing it across the page (section 859). The bindings for the books should be made in the district, sample books being sent them to copy (section 860). Each village or urban suburb should have a separate book, even if it form only part of a block (section 860). Many officers point out that the printed character used is almost illegible to the people, and that the headings should be lithographed. This is impossible if the headings are to be repeated on every schedule. It is suggested however that a single lithographed heading should be supplied for each book, and the schedules consist of blank paper ruled by hand by the enumer-

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Part III.—Difficulties and Suggestions.

ator. This would save expense. The only danger is the headings being lost or torn. I think perhaps the plan might be adopted if the headings were pasted into the cover of the book, which should be made to project beyond the leaves so as to leave the headings clear above the top of each page (see however section 944). It would perhaps be an improvement to rule lines across the schedule, giving one space for each person. The number of spaces on the page should be even, to allow for transliteration (section 856).

Column 2.—I would add the relationship of the woman or child to the head of the house where the name was not given (section 859).

Column 3.—I would give instructions that unmarried prostitutes are to be entered as single (*kwári*, section 859). The question whether marriage or cohabitation (*mukláwa*) should be taken as the point to be recorded is an important one (section 681).

Column 4.—I would have only one column and would make the enumerator *write* "male" or "female" (section 859); and I would direct that eunuchs be entered as male or *mard* (section 859).

Column 5.—I would enter the age at the *next*, not at the last birthday (section 859).

Column 6.—I would have a separate column for sect (section 859). *All* sects of *all* religions should be recorded (section 196).

Column 7.—This column is fully discussed in sections 364 to 370 and 859, and suggestions made for a future Census. The *caste* of Native Christians should be shown, not the religion (section 291).

Column 8.—I think the translation would be better *zibán-i-pidari, yáni báp kí zibán.* "Father-tongue" would probably be nearer the native idiom.

Column 9.—I would direct in so many words that the name of a village should never be entered (section 905).

Column 10.—I would have the heading of the column "Occupation" only (section 859), and would enter the occupation of nobody under 15 years old (section 718). I would direct that under no circumstances should the name of a *caste* be shown in this column, but that if for instance the man was a Teli by caste and an oilman by trade, the words *tel nikálna* should be entered. Also that a woman should not be entered as of the same occupation as her husband unless she actually helped him in his work. I would I think distinguish between proprietors who cultivated with their own hands and those who did not, and I would direct that the sons of landowners were to be shown as landowners, not merely as cultivators (sections 726 and 774).

Column 11.—Very many officers have suggested that the character which the man can read and write should be shown. If so, two columns would be needed, one for "learning" and the other for "educated," and the name of the character would be entered in the proper column. The information would certainly be valuable (section 807). If a man wrote two characters he might be left to select the one to be entered. The point that children who are learning and men who have learnt to read only and not to write are *not* to be entered, should be emphasized (section 807).

958. Preliminary Record and final enumeration.—The preliminary record of the villages should be made almost wholly by the patwáris (sections 851, 852). The date of beginning should be as late as is safe (section 862). I would forbid a rough Census being made beforehand as practice for the enumerators (section 861). In fixing the date of the Census I would avoid *lagans* (section 863), and if fairs cannot be avoided, would take the day of the fair instead of the day before or after (section 863). If statistics for families are to be taken from the schedules, and not from the house-lists, which I should prefer (section 844), I would separate families entered for the first time on the final night (section 864). In no case would I enter temporary visitors in the preliminary record (section 862), and I would perhaps show people who were temporarily absent in order to arrive at the permanent population (section 865). A more difficult point is the question where people temporarily sleeping out of their houses at night are to be enumerated. It was found that some men absent for the night only at their shops or in their fields were cut out at the final Census, though this was contrary to the instructions. I do not see how to improve the rule. Men absent at their shops could easily be enumerated there, but that would be impossible in the case of men watching in the fields; and yet we must I think have one general rule for all similar cases. I think the rule must be left as it is; but the point might be emphasized thus: "But you must *not* cut "out those who are merely absent, &c." And the special attention of enumerators and supervisors should be directed to this point before the final Census.

959. Special measures.—The responsibility of Cantonment Magistrates for the Census of cantonments should be not only defined but enforced (section 867). Military lines should be treated as blocks (section 869). The lists of houses to which private schedules are to be granted should be made early in the operations (section 872). But there seems no necessity for giving them to natives at all (section 873). The European form of schedule should be recast, and separate columns substituted for foot-notes (section 875); and they should all be examined by a capable official in the district, and made really correct and complete before being sent to the compiling offices (section 875). Railway premises should be treated as circles (section 876) and house-lists prepared for them in large Railway centres (section 876). River turnpikes should be instituted (section 879). Tickets for travellers appear to be unnecessary (section 881). It is *most* important to avoid transfer of officers, and especially of Tahsíldárs, when the Census work has once been begun, till it is completely finished. Some remarks on the difficulties of a synchronous Census in mountain tracts will be found in section 885.

960. Discretion to be allowed in enumeration.—One *great* difficulty is the degree of discretion to be allowed to the enumerators and supervising staff in rejecting the answers given by the people and recording what they believe to be the truth. The subject has been noticed already in sections 32 and 37 of Chapter I. About age there can I think be no doubt. Discretion must here be exercised, for the man himself often does not know his own age. The really difficult columns are those for caste and occupation. Here I think that where the enumerator is the village accountant, who is personally acquainted with every man, woman, and child in the village, he may be trusted to make the right entries. What is wanted is the caste to which the man is ordinarily reputed to belong; not any theory of ancestral origin which he or his fellow castemen may have formed. One thing I feel quite positive about, and that is that there must be no "aiming at uniformity." I have already quoted in section 37 an example of the pernicious effects of such attempts. The same officer, a highly educated Native Extra Assistant Commissioner, writes: "To avoid "a description of the same occupation by a variety of names, a general list of occupations was issued, and "the enumerators directed to make a special report of every other occupation they had to enter. Similar "lists of the tribes and clans met with in the tahsíl were issued, and everything practicable done to secure "uniformity of results." No possible action could have been so eminently adapted as this to render the results of the Census absolutely worthless. There is no officer in the Panjáb, English or Native, who could draw up even a tolerably complete list of the occupations, tribes, and clans of a district or even of a single tahsíl; and the result of issuing such a list authoritatively naturally was, as it always would be, that some one of the entries in the list was made to fit every soul enumerated. Besides, it is useless securing one kind of uniformity in one district and another kind of uniformity in the next. What we want is *variety*. The greater diversity of entry, the larger our stock of information. Variations and discrepancies will set us thinking and inquiring. Such uniformity as is possible must be arrived at by a careful classification of the entries on one system in the central office, and not by forcing the entries into a predetermined mould. Instructions on the subject of what is and what is not to be entered cannot be too general; for not only are we without even a hundredth part of the knowledge necessary to enable us to enter into details effectively, but by attempting to do so we deprive ourselves of the very possibility of acquiring that knowledge. If we show the slightest preference for one entry rather than another, the natives will give us that entry, right or wrong. We cannot

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Part III.—Difficulties and Suggestions.

have our facts *recorded* in too great detail: we can classify, select, and reject when we are compiling the figures. I wrote to a district officer, when he wished to issue detailed instructions for his preliminary record: "My object throughout has been to lay down only the most general rules, to leave those rules to "be applied in detail by each enumerator making enquiries of each person in his block throughout the "Province, to get the results, swarming with errors, omissions, and discrepancies as they will be, and to make what enquiries we can, *based upon those very discrepancies.* When we have done all this we shall know, not the truth, but how to set to work next time to get nearer the truth." Some little has been done, I hope, in this present report towards bringing our mistakes and difficulties to the light, and perhaps towards smoothing away some few of them. But the great subject of caste I have only had time to glance at here, and hope to deal with it more in detail during the next few years. By next Census it is hoped that we shall have, as explained in section 930, classified tables of castes and occupations; doubtless with many errors and many omissions, but still giving an enormous advance upon anything that has been done or even attempted before in any part of India. I think that we shall do much better next time than we have done now, and with much less trouble. But I am certain that we must always *record* in detail, and leave classification, uniformity, and the like, to the later stages of the work. The greatest pains should be taken to impress this upon half-educated officials, who are apt to be injuriously obstructive in the plenitude of their self-conceit (see further section 32).

961. Abstracting the figures.—*Village Abstracts.*—I would have the preliminary tabulation done wholly by the patwáris, in the villages at least; and I think I would add a column showing the numbers added and cut out respectively at final Census, as suggested in section 865. The red-ink entries could easily be counted. If, as proposed in section 844, the figures for houses or families are to be taken from the house-lists and not from the schedules, they might be omitted from the village abstracts. Other simplifications in Part I. are suggested in section 900. Part II. I would omit altogether for the reasons there given. Part III. is all-important.

Examination of Schedules.—I would have all schedules examined in the tahsíl before being sent on (section 890), and I would have them all sent through the district office, where a certain percentage of them should be re-examined (section 891).

Tahsíl Abstracts.—These ought to be *most* carefully compiled, as they are the basis of the whole tabulation. They should be sent through the district office, and not direct, and Deputy Commissioners should be held responsible that they are thoroughly checked and examined in their offices (section 891).

System of abstracting.—I would keep to the system of Divisional offices (section 893) but would deal with the Bannu figures at Pesháwar (section 892, foot-note). I would adopt the method of abstracting proposed in section 897 and trust for my check chiefly to the system suggested in sections 896 and 898. Further proposals as to what particulars should be abstracted will be found at the end of this part of the chapter under the head "Final Tables." I think the registers we used might have been improved; but as they depend wholly upon the form of schedule and of final tables decided upon, it seems useless to discuss the subject here. The great thing is to avoid elaborate registers. It is better to have two or even three simple than one complicated one.

962. The tabulation of the figures.—The subject is fully discussed in Part II. of Chapter XII. I do not think the general system could be improved, nor do I think the tables could have been made more simple, supposing the facts to be tabulated to have stood unaltered. The system of check proposed for future use is described in sections 902 to 904. Care should be taken to follow the suggestions conveyed in section 904, and to tabulate male and female occupations in the same table; for though their occupations differ somewhat, yet it is surprising how little difference there is on the whole, and the loss of the check afforded by the double columns for sex is irreparable. In deciding what places to class as towns I think I would follow the strict numerical limit; that is to say, I would class no places of less than 5,000 souls as towns unless they were either Municipalities or Civil or Military Stations (section 909). Of course I would exclude, as has been done in this Census, populations widely scattered over large village areas. Arrangements for urban areas are suggested in section 911. The question of the village units in the Kúlu sub-division should be considered (section 910); but I fear any further sub-division would be impracticable, as Mr. Anderson tells me that the *tíkas* and *phátis* are sometimes intermingled with one another (*chak bat*). If any plan can be devised for recording the number and size of hamlets the information would be most interesting. It would be quite impossible to tabulate throughout by hamlets. All that is wanted is the total population of each hamlet, and that perhaps might be taken out separately (see section 910). The English schedules should certainly be wholly dealt with at a central office (section 913). Indeed they should, after examination at the district head-quarters (section 875), be sent straight to that office. The Census once taken, the European figures should be kept quite distinct throughout. I think that the totals for each village should not be filled into the village tables till the check is complete. At the late Census we filled them in at once in red ink. I doubt if they are really necessary at all except for the towns; for few villages can contain more than five or six blocks, the total of which can easily be given when required. At any rate the having two independent sets of figures on the same page, one only of which is to be added up in order to get the page totals, is very confusing; and the second set should not be added until the table is completely checked, totalled, and corrected.

Too much importance cannot, in my opinion, be attached to making each Divisional office completely finish the tables of one district and send them into the central office before almost they begin another. I am not sure that I would not make them send in tables tahsíl by tahsíl, at any rate for the first district. Abstracting and tabulating should go on "*pari passú*," and the tables should be examined and checked in the central office as fast as they are ready, and returned at once for immediate correction. I should have saved quite four months to my compiling office if this had been possible (sections 918 and 931). The Government of India might perhaps be begged to fix the forms finally in good time. Changes in the middle of the work not only lead to great delay but make error and fudging much more probable. It takes the heart out of an office to have to do work over again from no fault of its own (section 918).

963. Compilation of the figures.—At a future Census the existence of the classified lists of castes and occupations mentioned in section 930 should immensely reduce the work of compilation, as the great mass of the entries will then be classified during tabulation instead of during compilation (section 930). Still, the great error I made was to take too small a staff for this work, and it should not be repeated. Above all, a really good active Superintendent of Settlement should be appointed to superintend the office, even if this prevents any English-speaking Assistant being given to the Deputy Superintendent. The real work is done in the vernacular office, and it is there that driving power and system are wanted (section 929). Above all things the vernacular office should never be finally broken up till every single table has been translated, checked, and *passed for the press in English.* For the consequences of neglecting this rule see section 931. I think the compiling staff *must* be patwáris selected as on this occasion (section 920). No other men can deal with the local terms.

964. Issue of instructions and schedules.—Suggestions regarding the supply of schedules will be found in section 946. The discussion of arrangements and instructions by Deputy Commissioners in Committee with all their Tahsíldárs, and as many of the supervising staff as they can collect, is *exceedingly* important (section 940). At the same time it should be explained that "uniformity" is not wished for in the entries (section 960). I think that the Deputy Superintendent's tour of inspection is most valuable (section 940). By next Census he will be able to reach almost every district in the Province by rail. I would issue translations of all instructions *with* the instructions on a future occasion (section 943). The schedules must I think be printed at Calcutta unless lithographed headings for each book be adopted (sections 944 and 957). If printed at Calcutta they might be distributed from there by rail (section 945). Above all things both schedules and instructions should be distributed at as early a date as possible. The complaints regarding late receipt of these papers at the present Census were numerous. But we did not fix on our form of schedule till the end of August, and did not receive the whole supply ordered from Calcutta till January. The schedule form should be fixed upon long before any sort of Census work is really begun in the districts, or even any instructions issued; and the schedules should be printed at once at Calcutta. The stock of schedules should be complete and ready for distribution to districts,

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Part III.—Difficulties and Suggestions.

if not already distributed, by the middle of November at the very latest. It must be remembered that it takes a long time to reach parts of the Province from Lahore. Parts of Kúlu are 16 stages from Dharmsála over passes 10,000 feet high and often closed in winter. So with the instructions. On the present occasion the great delay in the issue of orders by the Government of India regarding the Railway and Military Census, and the alteration of the instructions regarding occupations in the middle of November, caused much trouble, delay, and confusion. But on a future occasion it ought to be possible to issue complete instructions for the whole operations at once, before work is begun at all. This would be the greatest possible boon to district officers, for men work at once more willingly and more intelligently if they know exactly what their work is to lead to. The complete instructions ought to be in the hands of Deputy Commissioners by the beginning of September if possible; for the first arrangements for demarcation of circles, and especially the selection of agency, give a great deal of trouble and take a great deal of time. The house-maps and house-lists could be prepared in November and December, and the preliminary record in January. Done in this way at leisure, I do not see why the Census work should anywhere prevent the preparation of the usual annual papers; and it certainly would be possible to test and check it more effectually than if the work were all crowded into the last few weeks. The only difficulty that I see arises from the fact that the issue of complete instructions by the 1st of September, and the completion of the printing of the schedules by the beginning of November, involve the appointment of a Deputy Superintendent of Census from the beginning of July at the latest. But when he had once got his forms and instructions complete, that is to say from the end of August, he would have little or nothing to do till about the middle of November, when he would begin his tour of inspection. During these two and a half months however, I think he might be employed in preparing classified lists of clans, tribes, castes, and occupations for the use of his tabulating offices as suggested in section 930.

965. Census in Native States.—I think that at a future Census the same schedule as is used for British Territory should be, as now, adopted throughout the Native States, not even excepting Hill States; though active superintendence on the part of the Superintendent of Simla Hill States and the Commissioners in political charge of the other Hill States will again be needed. (section 951). And I would, when the first instructions are issued, and before any operations are actually commenced, invite the States to send representatives to meet the Deputy Superintendent at Lahore, and there to discuss verbally the whole scheme (section 953).

966. Form of Final Tables.—I would, I think, show figures for towns and villages separately in *all* the tables. The urban population is exceedingly different from the rural in all respects; and the comparison would be most instructive. If it be thought unwise to double the bulk of the tables by showing separate details for towns and villages in each district, those details might be given for totals only. Where separate figures are not given for urban, rural, and total population, it would I think be better to show figures for urban and total population, rather than for rural and total as I have done. It is true that the rural population is more important and more normal than urban; but it is also so much larger that the variations in urban population are quite lost in the total figures, which may almost be taken as identical with those for villages so far as proportions and percentages are concerned.

Table I.—I would omit at any rate unoccupied houses. The figures are even more meaningless than those for occupied houses (section 841).

Table II.—For "Increase and decrease per mille" I would substitute "number living in 1891 per mille living in 1881," and so save half the columns at the end of the table.

Table IIIA.—See discussions of these figures in section 291. The table as it stands is useless. The European schedule should be remodelled (section 875); and I would separate race from sect, giving Native Christians in a separate table if the details are required.

Table IIIB.—I hope, for the reasons given in section 196, that at the next Census *all* sects, both Mahomedan and Hindu, will be both recorded and tabulated.

Table VI.—I would tabulate the civil condition of the earlier ages in greater detail, thus 0—5, 5—10 (section 687).

Table VII.—I would take only ten-yearly periods after 35, thus 35—45, 45—55, 55—65, and so on (section 666).

Table VIIIA.—I hope the classification of castes will have been greatly facilitated before next Census (section 930). I think all castes and all important tribes should be shown as now. But the "total" column in VIIIA. should have "both sexes" last instead of first, so as to allow of more ready comparison with the details for religion. I would omit details of Christians. I had hoped to arrive at the composition of Native Christians by caste, but they frustrated my desire by very generally returning *Krístán* as their caste.

Table XII.—The whole system of classification must, I think, be recast. The English system is utterly unsuited to the state of society in India (sections 715, 716), and I would suggest the adoption of the proposals set forth in section 720. I hope the lists of occupations to be published in connection with this Census will greatly facilitate the work in 1891 (section 930).

Tables XIV to XVII.—I doubt whether the details by age are of any value, unless the Sanitary authorities want them.

Table XIX.—The population by the present Census should be put first, and then that by the preceding enumeration. As now arranged it is difficult to find a town of any given population in the table.

967. Final counsel to my successor.—I would commend to the careful consideration of the next Deputy Superintendent of Census the experience described in sections 918, 919 and 931. I hope that he will find his task facilitated by my labours; and especially I hope that the detailed tables of caste divisions and occupational terms which it is intended to publish will do an immense deal to render the classification of these two entries simpler, more speedy, and more accurate. But I would once more earnestly recommend him to insist upon a sufficient compiling establishment, to have a first-rate *driving* man to superintend his office, and not to let his establishment go till the last table is passed in English and *ready for press*. If I had known what was before me and had acted on these principles, I should have had my work finished six months earlier. I have left him a complete set of my circulars bound up; and this volume, together with all files that can possibly be of any use either to him or for purposes of office reference, have been made over to the record department of the Panjáb Civil Secretariat with my No. 3928, dated 18th April 1883, to the address of the Secretary to the Panjáb Government, Home (General) Proceedings No. 119B of April 1883. I had intended also to leave him a complete set of the registers and other forms used in abstracting, and the vernacular instructions issued. But the surplus copies have been sold as waste paper, and I cannot complete a set. With the details given in this report he will probably be able to devise far better ones for his own use.

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Part IV.—The Last.

PART IV.—THE LAST.

968. Notice of officers.—In the Census reports for their districts the several Deputy Commissioners have brought to favourable notice the good work done by certain officers under their orders. I had intended to extract all these notices and to print them in this place. If Government should wish me to do so I can still submit them separately. But I do not propose to do so without orders to that effect, and that for two reasons. In the first place I have no time. But a more important reason is that, while some Deputy Commissioners have singled out three or four officers for favourable mention who had showed especial zeal, others have included in their praise all the higher officials and clerks who did good work. Under such circumstances those who had done equally good work under the first-mentioned class of Deputy Commissioners might fairly feel aggrieved at not being noticed. As for my own personal experience of the work, I saw good work in some places and bad work in some others, though not in many. But it by no means follows that what I saw was a fair sample of the whole in either case. I know positively that the whole work was admirable in certain districts; but I do not mention them because it may have been just as good in others where I had no opportunity of judging. But I have directly and indirectly seen a very great deal of the work and of the way in which it was done. I saw much of it on my tours of inspection; I heard much of it from officials and private persons both Native and European; I was in constant correspondence with many of the officers both officially and demi-officially; and finally the whole results passed through my hands in the end. And I may be allowed to say that the work was done in almost every case with an enthusiasm that compelled success. The officers felt that Government meant to have the Census well taken, and they set to work in earnest: the native subordinates followed the lead of their superiors, and the whole staff took a real active practical interest in the operations which I could hardly have realized had I not been brought into contact with it. They grumbled, but they worked with a will. I believe the Panjáb Census of 1881 to have been a great success; and I would submit for the consideration of Government that that success was wholly due to the zeal of the official staff, and that some public recognition of the fact might fittingly be made.

969. Of one matter I can speak with complete knowledge, and that is the nature of the Census reports sent in to me. I would specially mention the reports of Messrs. Wilson, Benton, Coldstream, T. W. Smythe, Steedman, and Alex. Anderson as most able, suggestive, and complete; and those of Mr. Frizelle, Captain Bartholomew, and Colonel Jenkins as exceedingly careful and complete. Mr. Frizelle's report on operations was especially suggestive. Messrs. Stogdon, Gardiner, and Lawrence sent in capital reports on the operations, and Colonel Beadon and Major Beckett on the results, none of these gentlemen reporting on both points, owing to transfer of officers. Messrs Walker, Dames, and Drummond, and Sardar Gurdial Singh, C.S., supplied some most useful notes chiefly regarding castes. Only three of the Commissioners sent me reports on their divisions; among them Colonel Minchin's report was particularly interesting. I must here acknowledge the ability and industry of Rai Búta Mal, Extra Assistant Commissioner, who was my Assistant while the figures were being compiled. His unusual command of the English language saved me very much time and trouble. He never spared himself, and his knowledge of the people was of the greatest use to me.

970. Success of the Census-scheme.—The Census of 1881 was, as I have said before, practically a first experiment in the Panjáb, for if detailed instructions had been issued on previous occasions, no record of them existed for my guidance. Consequently the whole scheme had to be thought out from the beginning, and the greatest care was necessary to secure that each set of instructions as it issued should not only fit in with what had gone before, but, what was still more important, should pave the way for everything that was to follow after it. Consequently it is most gratifying to me to find that the district officers of the Province with one consent speak in the highest terms of the arrangements. I shall I hope be forgiven if I indulge in the egotism of quoting a few of their opinions. The one fault they find, and that very generally, is that the scheme was too complex. But the only complexity to which they take exception, and the only simplification which they would as a rule suggest, is with regard to the threefold division of the caste column. That division was deliberately adopted after the most careful consideration by Sir Robert Egerton, and defended by him against the criticism of the Government of India; and few men are so intimately acquainted with the people of the Panjáb as he was. The subject is discussed fully in sections 365 to 367, and I believe that the three columns were necessary for our purpose, though perhaps they may be safely reduced on a future occasion with the help of the knowledge we have gained by means of them. Indeed much if not most of the difficulty experienced was caused by the misunderstanding of the district officers themselves, who in the face of the precise words of the instructions, would insist upon considering the schedule incomplete till each of the three columns had an entry to itself. But, setting the caste columns aside, the scheme is spoken of as satisfactory without I think one single exception.

971. Mr. Coldstream of Hushyárpur writes:—"The general arrangements left little to be desired. Instructions from headquarters "were very detailed and explicit, and experience suggests little in which they could with advantage have been altered. A few sugges- "tions on minor points will be made further on." Colonel Minchin, the Commissioner of Hissár, describes the instructions as "so "full and clear that no difficulty was experienced." Major Grey of Firozpur found them "in all respects save the caste column simple "and practical." Mr. T. W. Smythe of Dehli writes: "The instructions issued were so full and clear that there was little room for "doubt." Mr. Bullock of Hushyárpur says: "I have no suggestion to make or difficulties to bring to notice. The instructions "issued by the Deputy Superintendent of Census were as simple as they were complete." Mr Stogdon writes from Lahore: "*Difficulties and suggestions.*—I have very little to say under this head. I have already touched upon most of the difficulties I "met with, and I think that the general scheme prescribed was most complete and lucid. Its chief fault, if it had any, was that it "was too complex and intricate for the intelligence of many of the supervisors and enumerators, and too long for the patience of "men who knew that they were not likely to receive any reward for their labours beyond the approbation of Government. It is "important that the work should be of as little harassing a nature as possible, and on that account" Mr. Wakefield of Ludhiana

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Part IV.—The Last.

says: "I remember feeling somewhat impatient at first of the elaboration of the machinery and checks upon checks which "were promulgated. But experience and actual working soon showed the excellence of the arrangements, and the positive necessity "for every portion of the system which has resulted in so great a success." Finally Mr. Frizelle writes from Sháhpur: "There can be no doubt that the Census was carried out on a well-conceived plan, and that the instructions were full and "complete. If there is any room for criticism it is on the ground that the work was made too elaborate and difficult; though on "looking over all the forms and instructions it is hard to see what could have been well omitted, if perfection was to be secured. "The instructions to enumerators were simple enough and left nothing to be desired. The great difficulty was the Index-register "(Form A), and fortunately this was confined to supervisors and those above them. Columns 5 to 10 especially might have been "simplified; although one is bound to say that any omission would have detracted from its value as a whole, and now that the Census "is over and the beauties have grown on one, it is easy to see that anything less complete might not have answered so well. As to "the rest of the scheme as drawn up by the Deputy Superintendent of Census, I think there is nothing to be found in it that could "have been improved upon in theory or that could have worked better in practice, and the result is I consider a very trustworthy "Census." And these are only a few of the opinions recorded to the same effect.

972. The great difficulty I felt throughout in framing my instructions was, where to draw the line so as to be neither too concise nor too diffuse. There is always a great danger of written instructions being misunderstood, and points that are omitted in them will often be overlooked altogether, or will be treated differently by different people. On the other hand too much detail disgusts and confuses the reader, and leaves no clear impression on his mind. The opinions I have quoted above lead me to hope that on the whole I succeeded in avoiding both dangers. That there were innumerable faults in the scheme, I have made it my special business to point out in the second Book of this Report: that I greatly under-estimated the amount of labour involved in the compilation of the figures, and that my doing so has delayed the publication of the Report and somewhat increased the expense of the Census, I am only too painfully aware. But I can honestly say that at no stage of the operations have I spared myself in any way, and that whatever may be the value of the final product, it is the result of toil and application on my part which have, from the day I took up the appointment to the day I laid it down, been absolutely unremitting and incessant.

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